Non-linear absorption of indirect band gap materials upon intense bi-color laser light resolved by TD-DFT

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Hilase RP4 seminar
Prague, Czech Republic
September 2016, 26th

Gender balance at MPSD

- Co-workers at MPSD

(a) N. Tancogne-Dejean

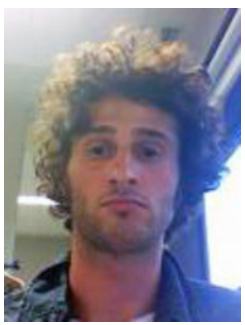
(b) Michael Ruggenthaler

(c) Heiko Appel

(d) Angel Rubio

- More octopus contributors

(e) Shunsuke Sato



(f) Umberto de Giovannini

(g) Micael Oliveira

Outline

1. Context
   - Motivations
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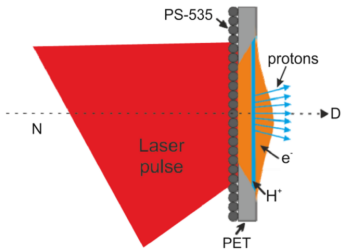
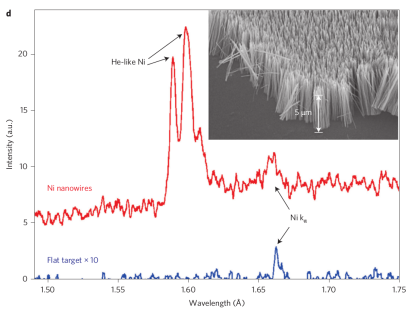
5. Conclusions

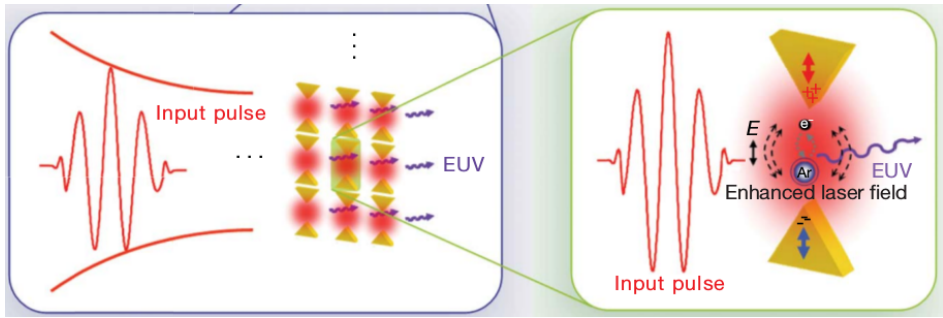
Electron dynamics in nanostructures

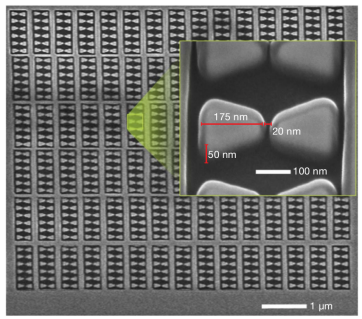


Margarone et al, *Laser-Driven Proton Acceleration Enhancement by Nanostructured Foils*, Phys. Rev. Lett. **109**, 234801 (2012) [1] (protons up to 7 MeV)



Purvis et al, *Relativistic plasma nanophotonics for ultrahigh energy density physics*, Nature Photonics **7**, 796 (2013) (e- @ 7 keV)





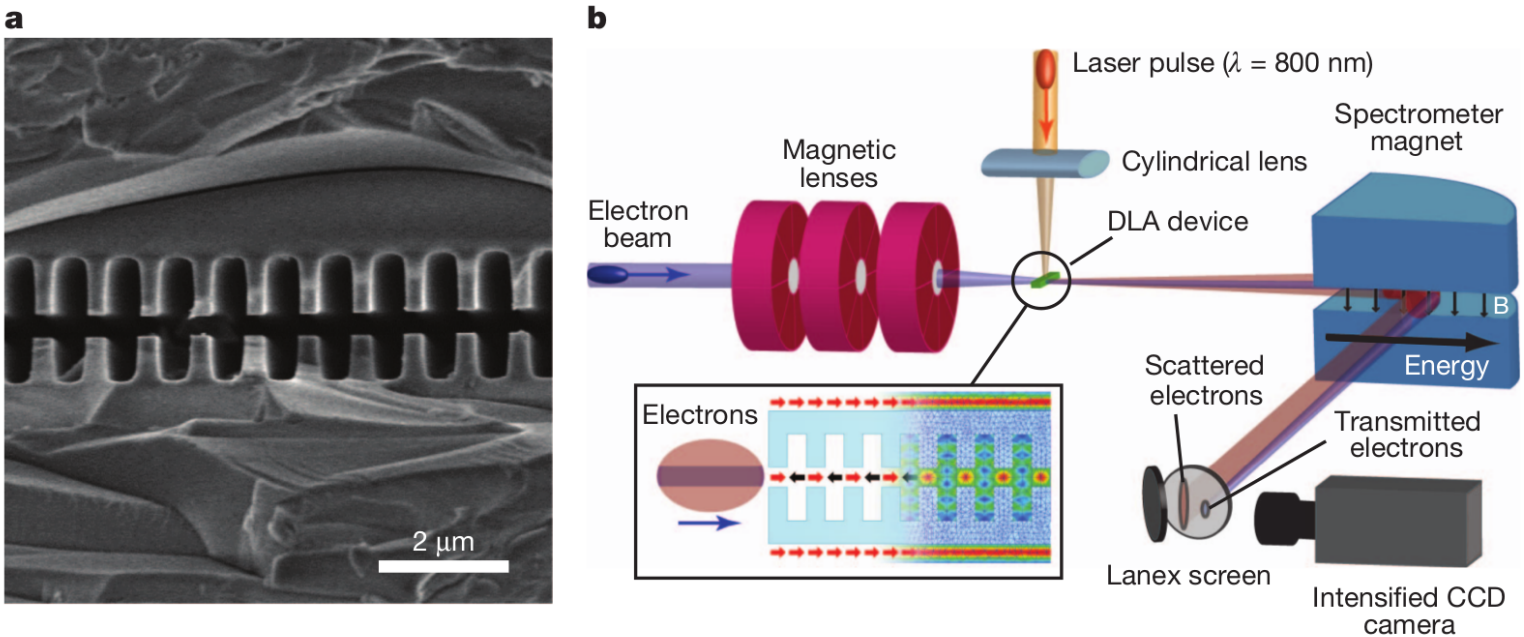
Kim et al, *High-harmonic generation by resonant plasmon field enhancement*, Nature **453**, 757 (2008) (~ 51st harmonic)

Laser + nanostructures = Future.

Accelerating charges within few cm$\sim 10^{-5}$ smaller than CERN.
High harmonic generation in solids: using 10^{-6} less energy than gas.

Acceleration using lasers

Peralta, E. A.; Byer, R. L. et al, *Demonstration of electron acceleration in a laser-driven dielectric microstructure,* Nature **503**, 91 (2013)



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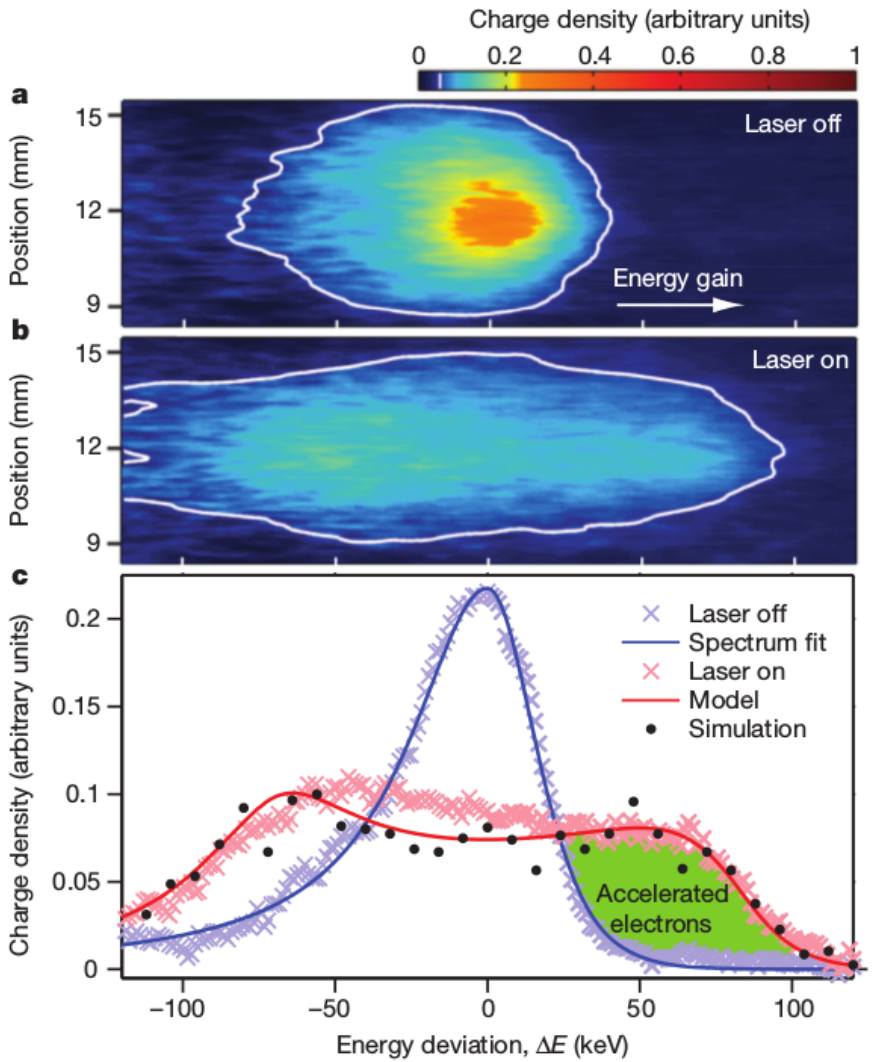
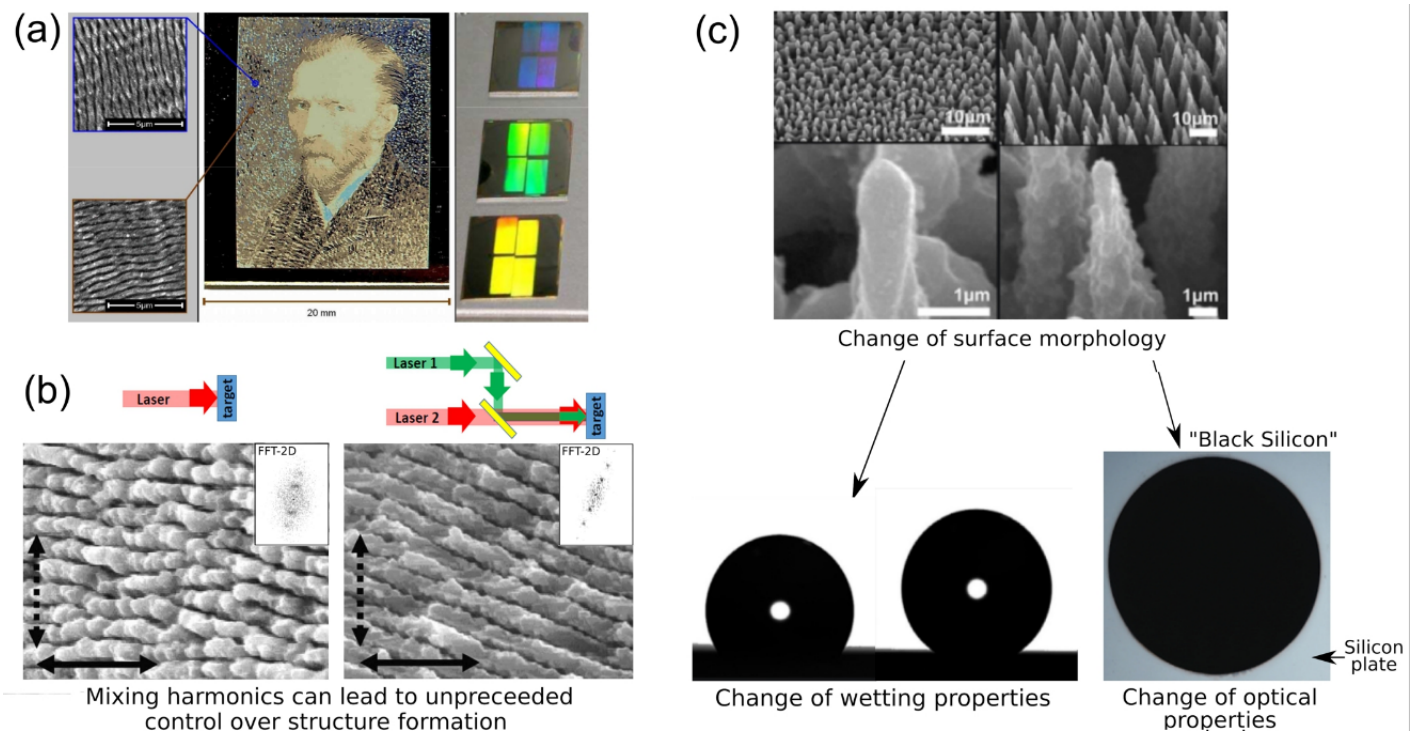
Peralta, E. A.; Byer, R. L. et al, *Demonstration of electron acceleration in a laser-driven dielectric microstructure,* Nature **503**, 91 (2013)



Currently based on SiO_2 (high damage threshold). They go now to Si

Surface functionalization with lasers



Functionalization of surfaces via laser nanostructuring

(a) Dusser et al, Opt. Express **18**, 3 (2010)
(b) Jia et al, Phys. Rev. B **72**, 12 (2005)
(c) A. Ranella et al., Acta Biomat. **6**, 2711 (2005)
(d) A.Y. Vorobyev, Ch. Guo, Laser Photon. Rev. **7**, 385 (2013)

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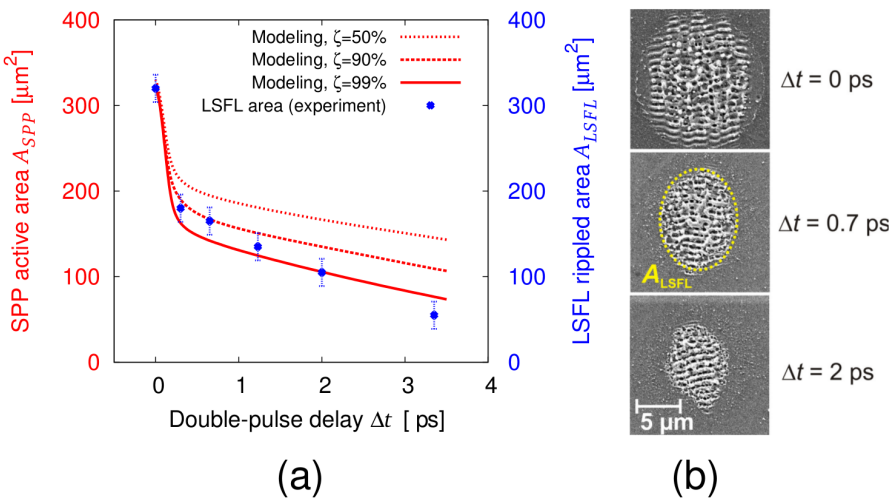
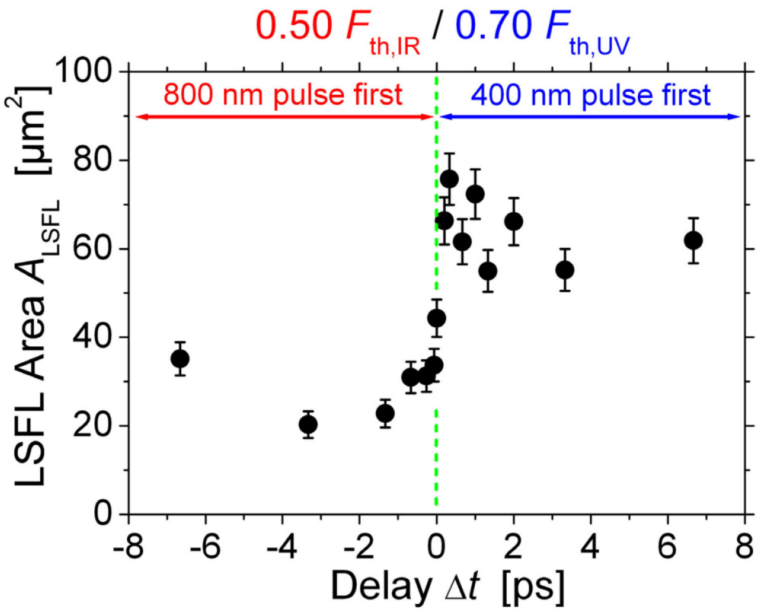
Laser nanostructuring of silicon

- Double-laser pulses give more control on the electron dynamics.



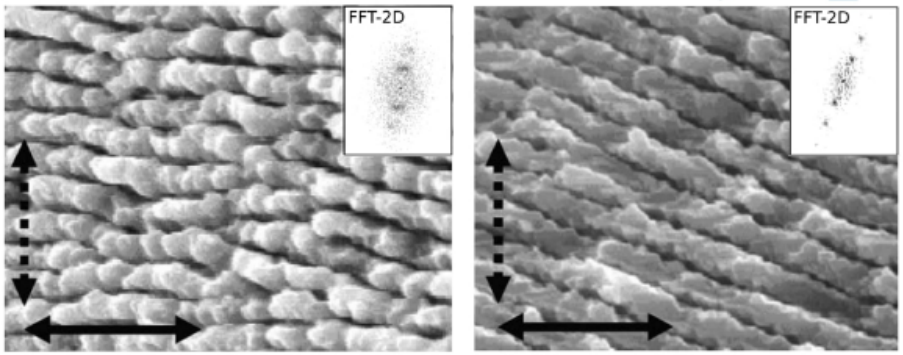
Optics Express **21**, 29643 (2013)



Höhm et al, Optics Express **23**, 61 (2015)

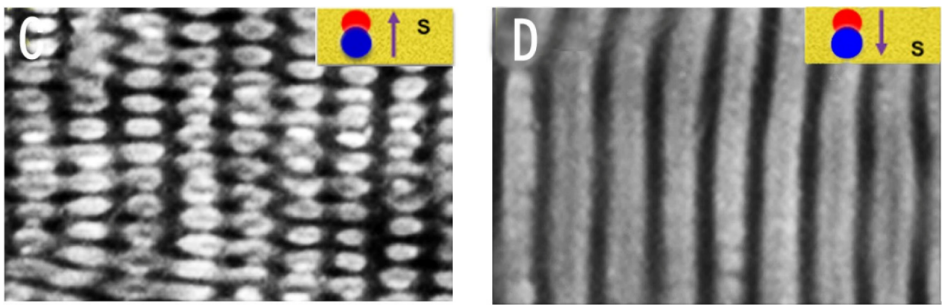
Open problems of electron dynamics

- Mixing harmonics improves quality dramatically.



Jia et al, Phys. Rev. B **72**, 12 (2005)

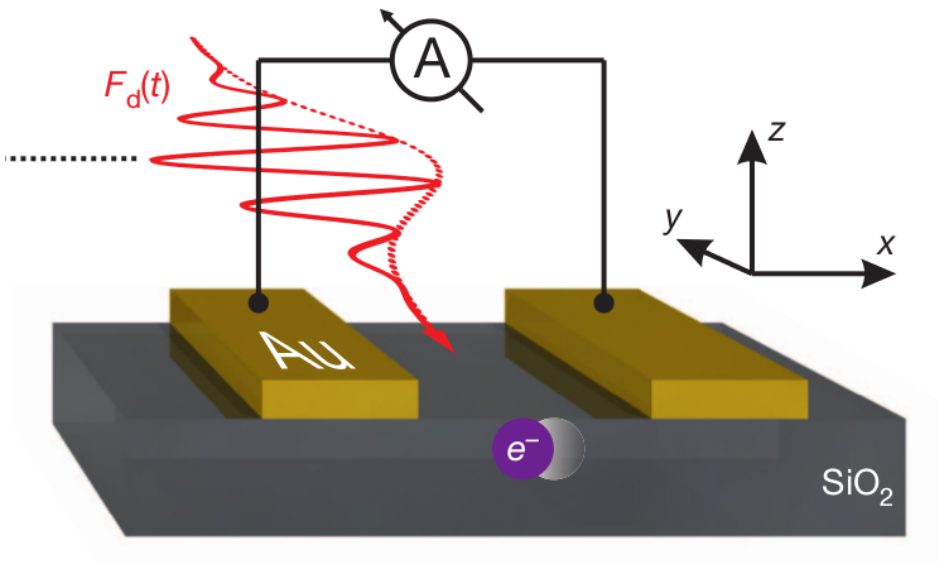
- Effect of scanning on nanostructure formation

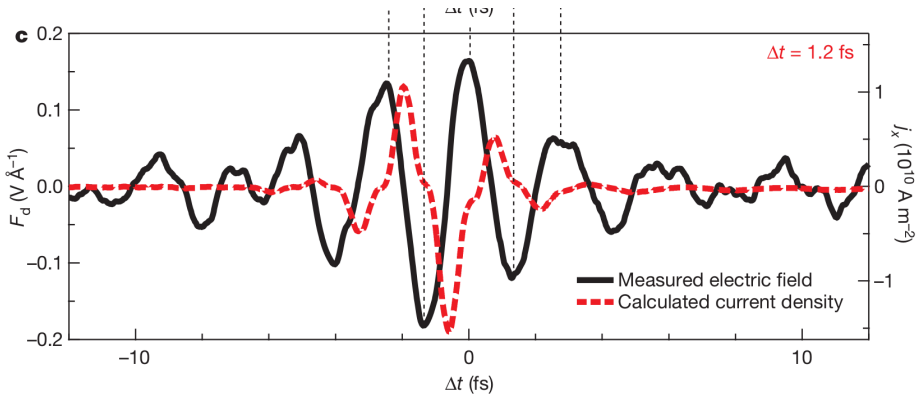


Cong et al, Optics Express **23**, 5357 (2015)

Current generation in band gap materials

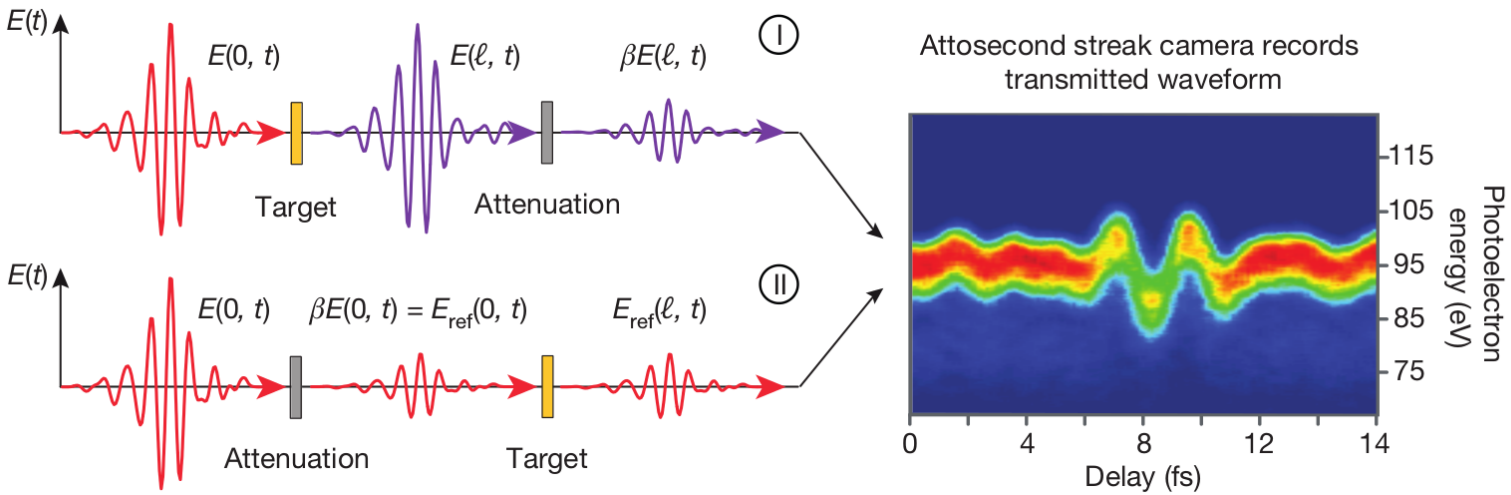
Schiffrin et al, *Optical-field-induced current in dielectrics*, Nature **493**, 70-74 (2012)





Use of bi-color pulses

Attosecond streak camera: [Sommer, A., Sato S.A. et al., *Attosecond nonlinear polarization and light-matter energy transfer in solids*, Nature **86**, 534 (2016)].



A lot remains to be understood.

Bi-color interactions look promising both for fundamental investigations and development of applications.

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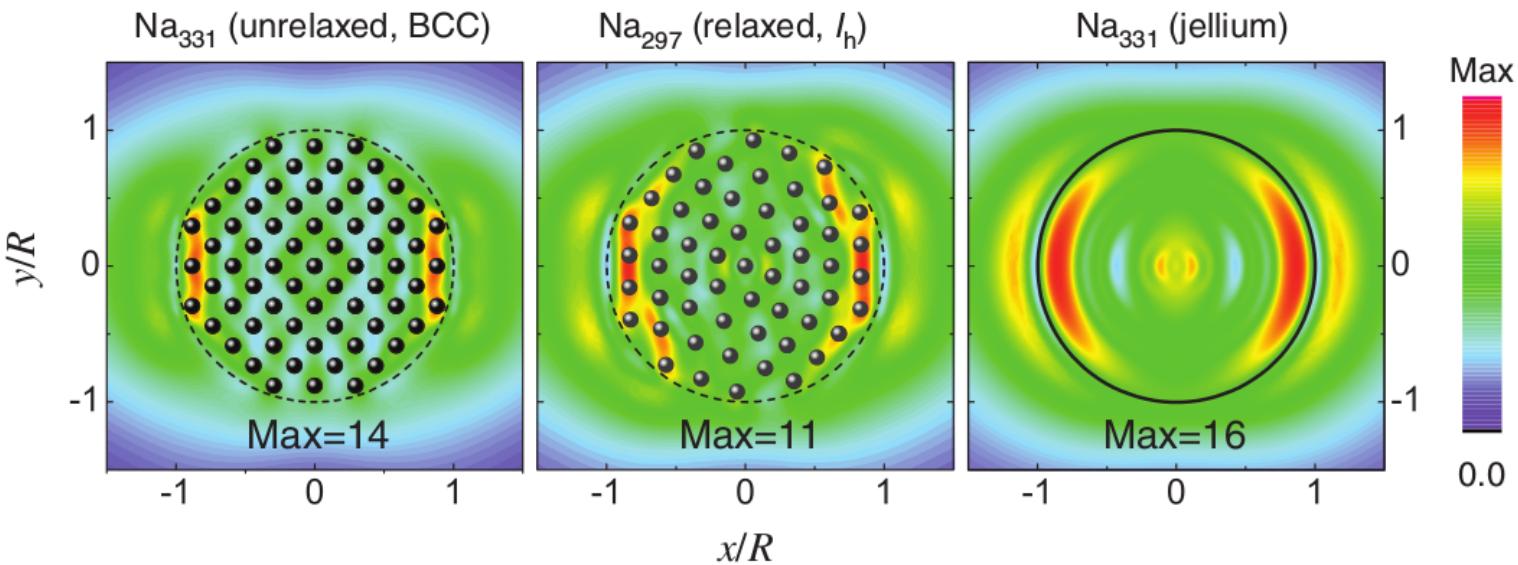
5. Conclusions

Simulate huge quantum systems

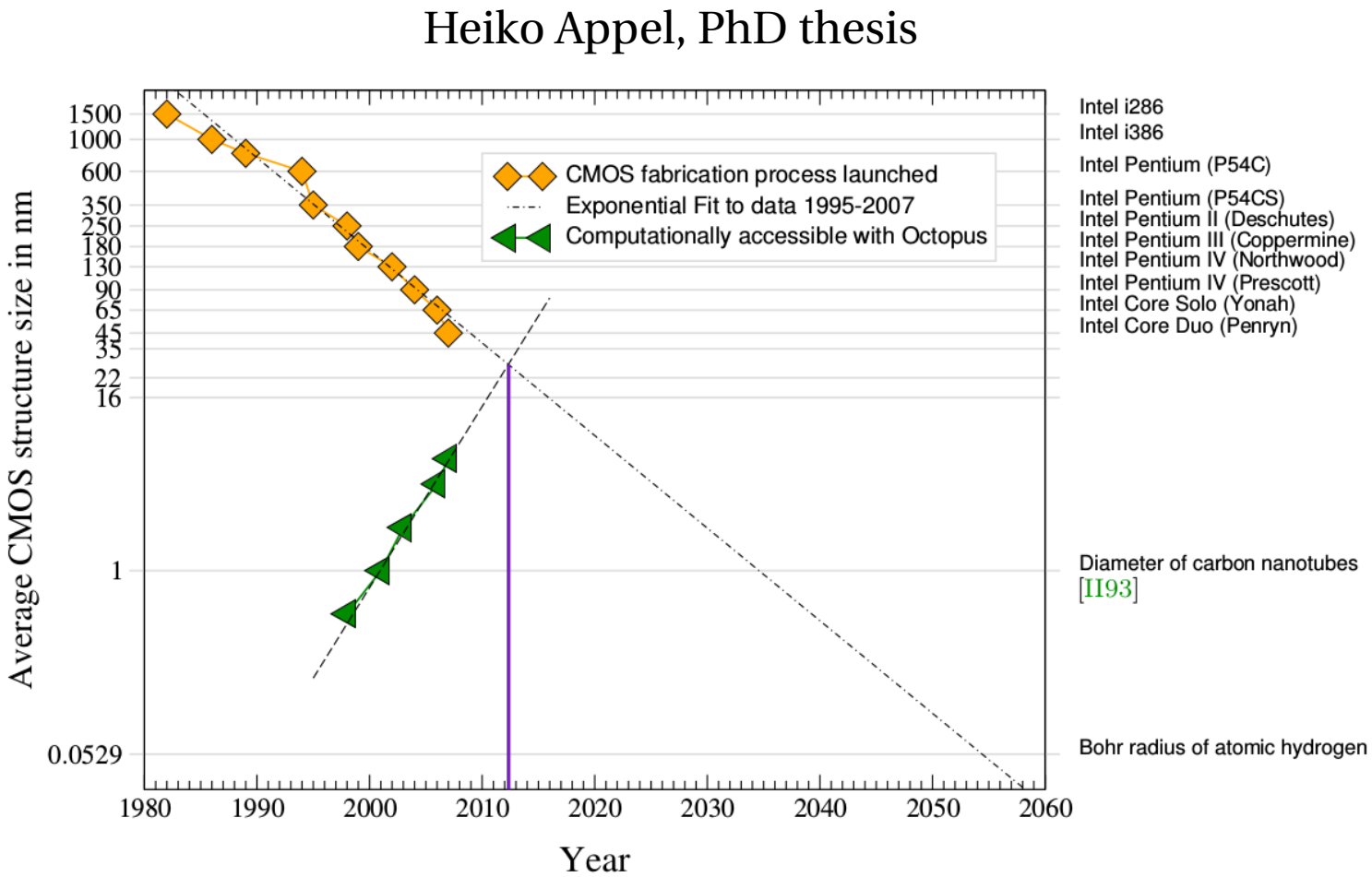
Varas, A., Garcia-Vidal, F.J. and Rubio, A., *Quantum plasmonics: from jellium models to ab initio calculations*, Nanophotonics (2016).



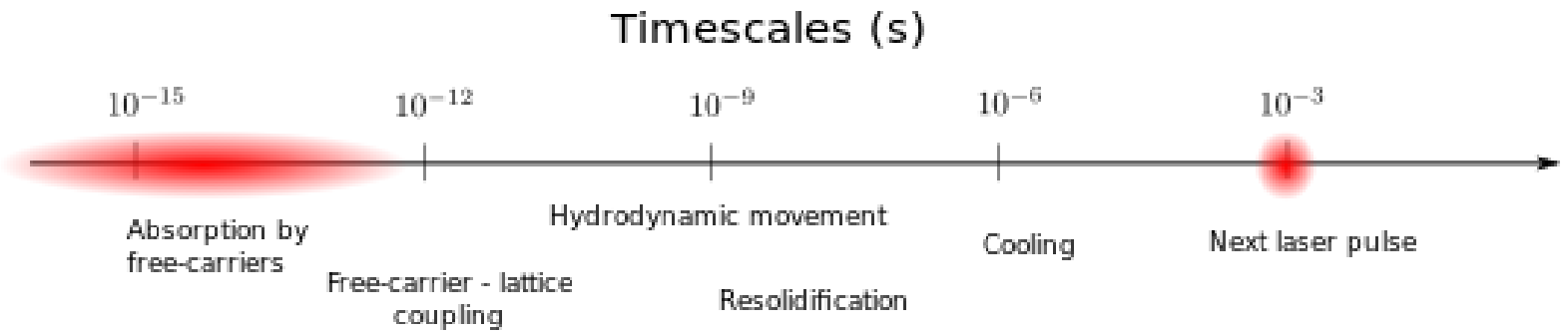
Limitations

Need huge computer facilities.
Not possible to address for small companies.

Maximum size ?



Continuum large-scale modeling



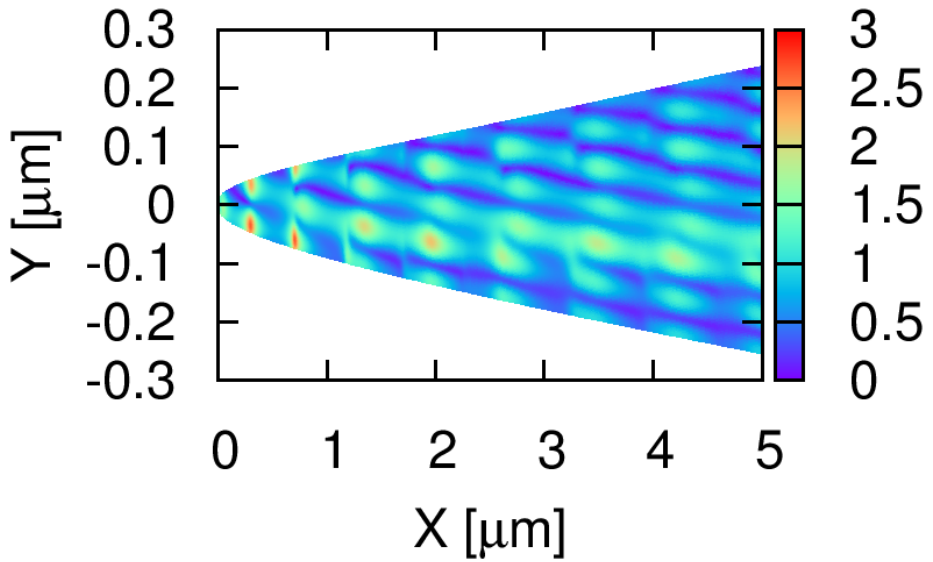
Electrodynamic processes ($t \sim \tau_{\text{laser}}$)

- Electron-hole dynamics in laser field.
- Surface Electromagnetic Waves generation (scattering on defects/roughness, excitation of Plasmon Polaritons, ...).
- Electron-phonon coupling.

Material reorganization ($t \gg \tau_{\text{laser}}$)

- Solid→Liquid →Gas/plasma phase transition (”ablation plume”).
- Liquid layer dynamics.
- Cooling.
- Recrystallization.

Features of large-scale modeling



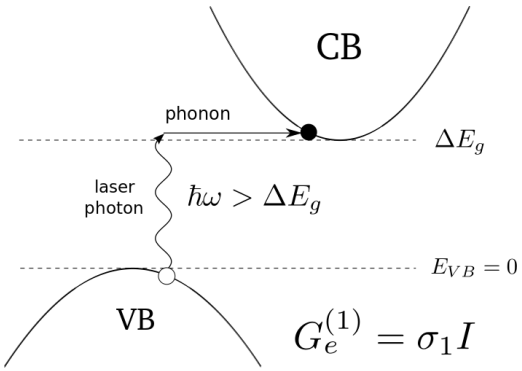
- Take into account the geometry and complex scattering of the laser source.
- Observe long range quantum effects, and the macroscopic transport of charges
- Correlate with real-size experiments: microstructures, etc.

Under construction

2D/3D continuum code : light propagation, matter excitation, and feedback.
Check http://www.QuantumLaP.eu/

Simplified model for Si

- One-photon absorption



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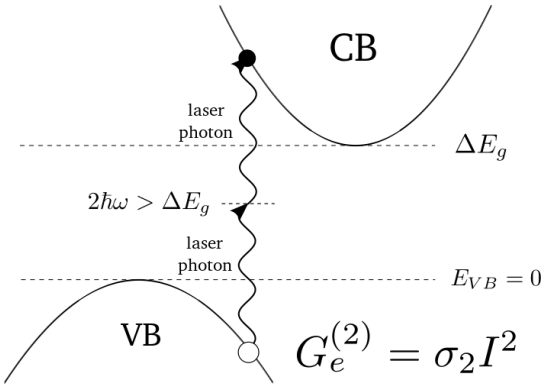
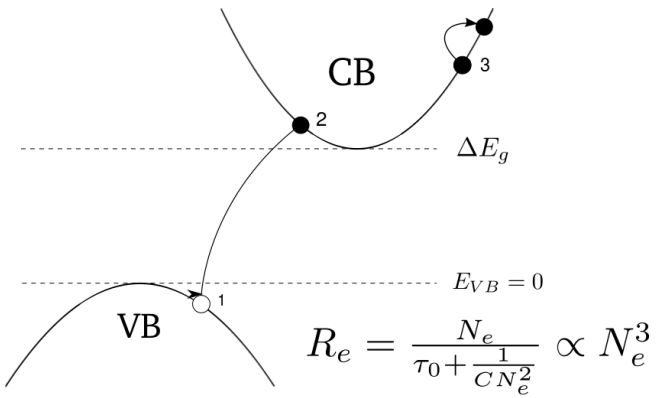
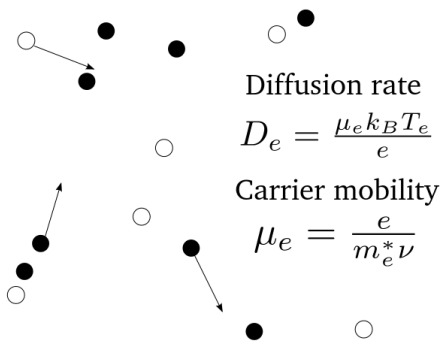


- Auger recombination

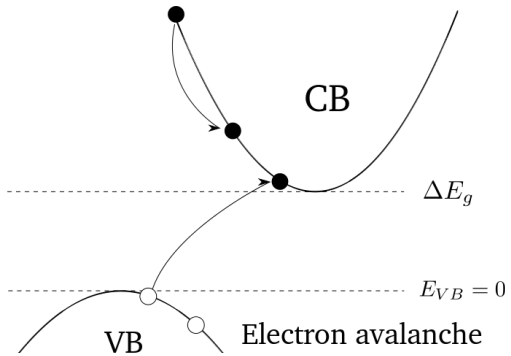


- Carrier diffusion

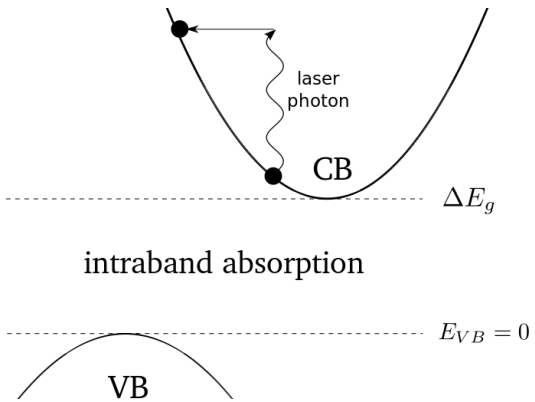


Simplified model for Si

- Impact ionization



- Free-carrier absorption



Number of free-parameters

Refs	2-γ absorption σ_2 [m.W^{-1}]	Effective mass m^* [-]	e-ph coupling τ_c [fs]	Collision time ν^{-1} [fs]
Sabbah et al [12]	6.8×10^{-11}	$0.15 - 0.20$	260 ± 30	32 ± 5
Sokolowski et al* [13]	45×10^{-11}	0.18	-	1.1
Sjodin et al [14]	10×10^{-11}	-	240 ± 30	-
Bulgakova et al [15]	$\uparrow$	1.0*	$\uparrow$	1.27
Bristow et al [16]	1.86×10^{-11}	-	-	-
Reitze et al [17]	34.6×10^{-11}	0.18	-	
Fischetti et al [18]	-	-	-	$\sim$ 10 fs

List of free parameters in several studies at $\lambda = 800$ nm.

Open problem

Discrepancy between fitted data in experimental literature.
$\Longrightarrow$ Physical ingredients are missing in models: ab-initio calculations are required.

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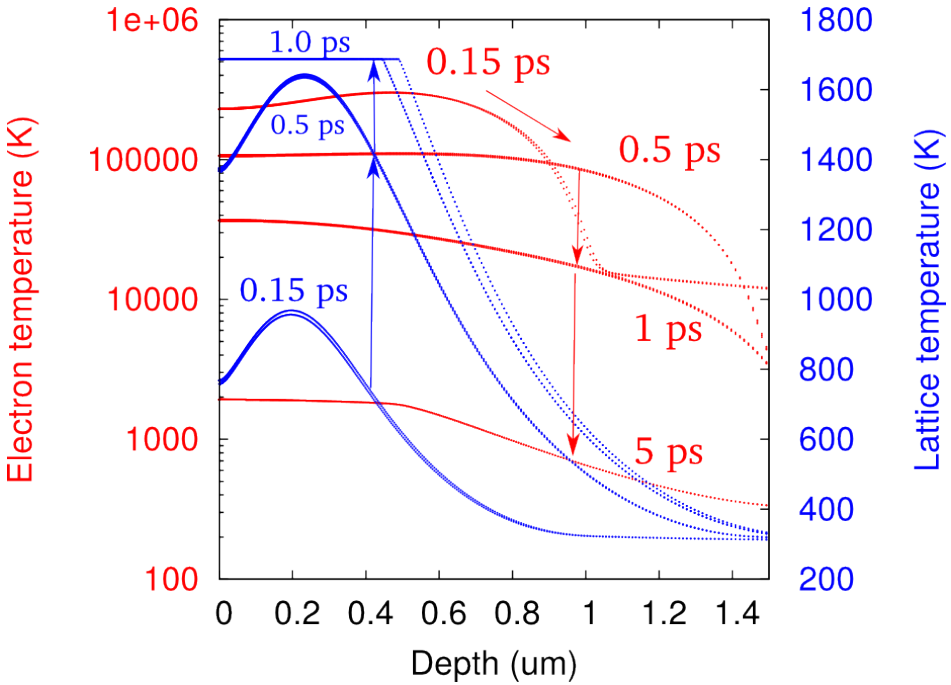
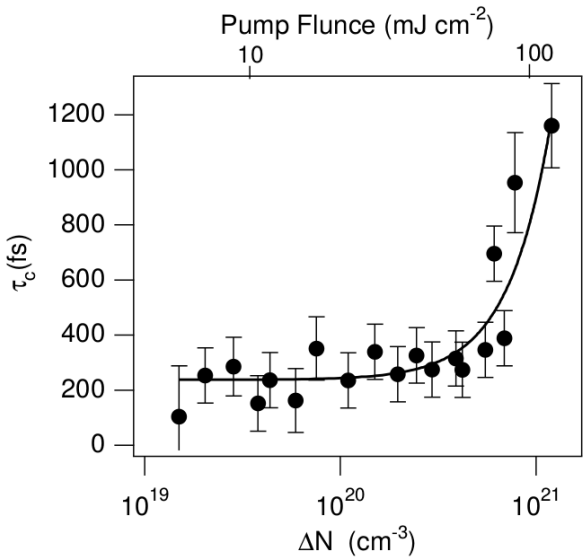
Quantum case →Large-scale effect

Electron-hole pair temperature - Lattice temperature



$F = 0.4\ \mathrm{J/cm^2}, \tau = 100\ \mathrm{fs}, \lambda = 800\ \mathrm{nm}.$



Sjodin et al, Phys. Rev. Lett. **81**, 5664 (1998)

Observation

Quantum electron-phonon dynamics are important to predict the location of energy-deposition.

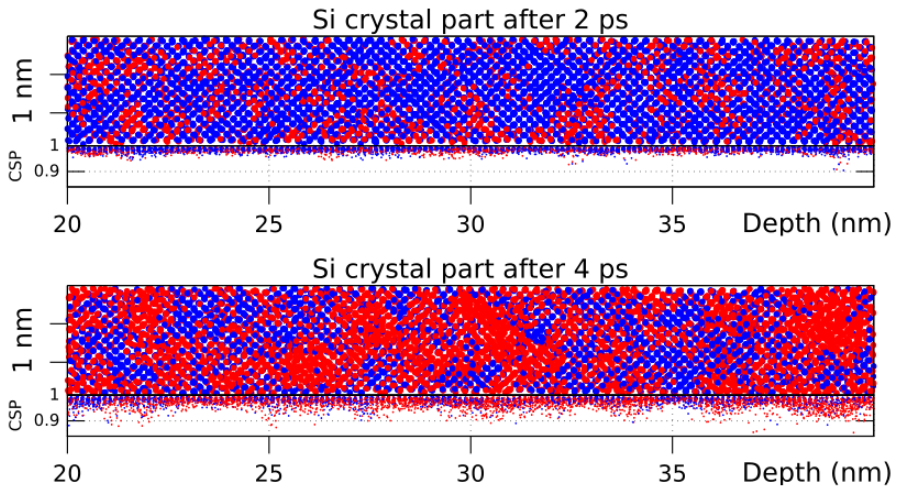
Homogeneous melting during the pulse

Wait: melting front velocity should be limited by the speed of sound!

- Laser energy is deposited deep in the material: overall $d \sim 1$ µm in $t \sim 1$ ps.
- Melting front velocity:

$$v_{\text{melt}} = \frac{d}{t} \sim 10^{-6+12} \sim 10^{6} \text{ m/s}$$

- Velocity of sound in c-Si: $c_s = 5843$ m/s.



V. Lipp et al, Phys. Rev. B **90**, 245306 (2014)

Problem

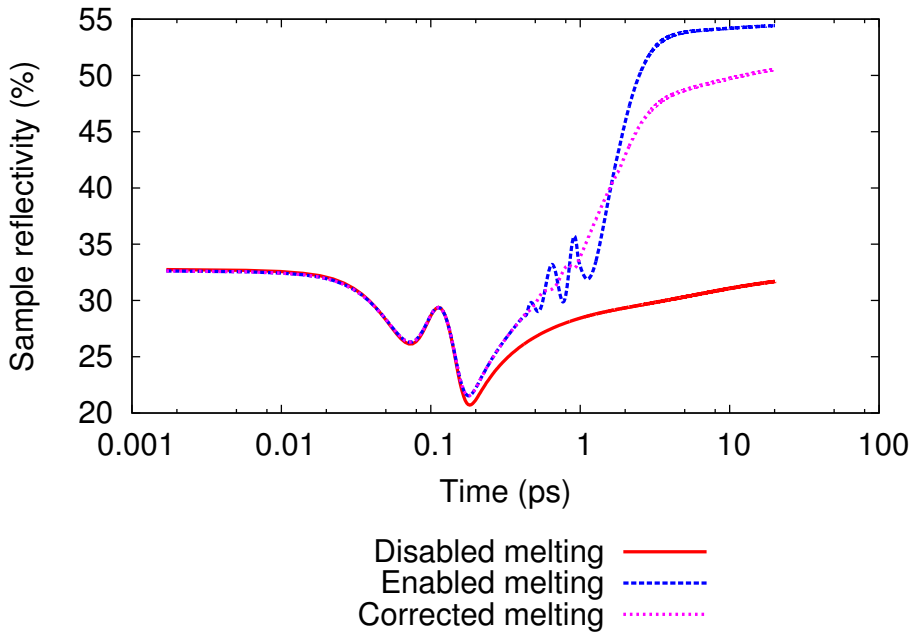
Melting front cannot be a single interface.

Molecular dynamics

Partial melting occurs via nucleation of molten centers.
Let's take into account !

Corrected surface reflectivity

$\lambda = 800\,\text{nm}$, $F = 0.8\,\text{J/cm}^2$, $\tau = 100\,\text{fs}$, $\theta = 0°$.



Neglecting optical properties of liquid Si

$$\frac{\varepsilon - 1}{\varepsilon + 2} = \frac{\varepsilon_{Si}(\lambda, T, N_e) - 1}{\varepsilon_{Si}(\lambda, T, N_e) + 2} \times (1 - \eta) + \frac{\varepsilon_{l-Si}(\lambda, T) - 1}{\varepsilon_{l-Si}(\lambda, T) + 2} \times \eta$$

Observation

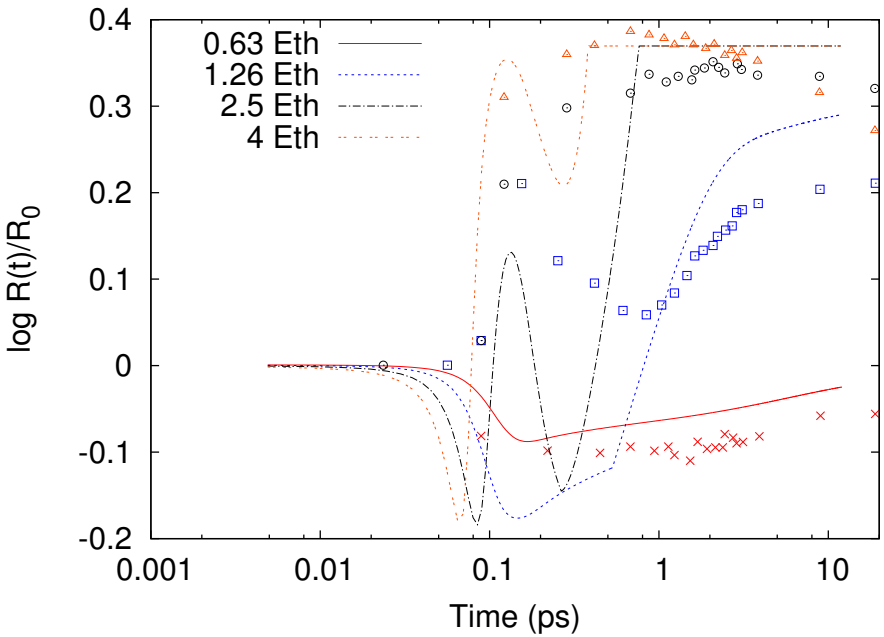
Oscillations have disappeared at $0.8\,\text{J/cm}^2$.
Effective medium theory helps but is not sufficient.

Scattering on homogeneously melting material

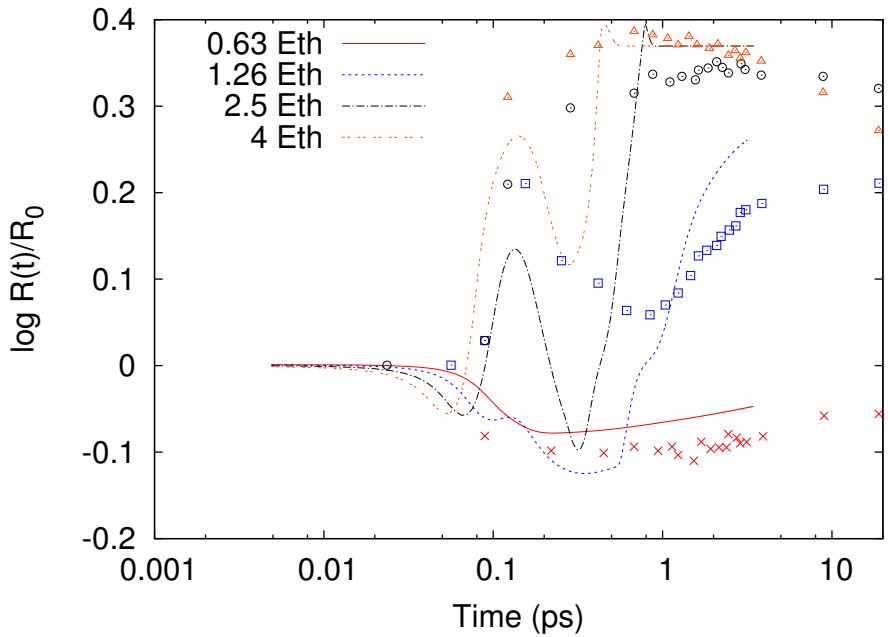
Parameters:

- $\mu_e = 1.5\ \mathrm{cm}^2/V/s$,
- $\nu^{-1} = 1.4$ fs.

Fresnel reflectivity



Mulitlayer reflectivity

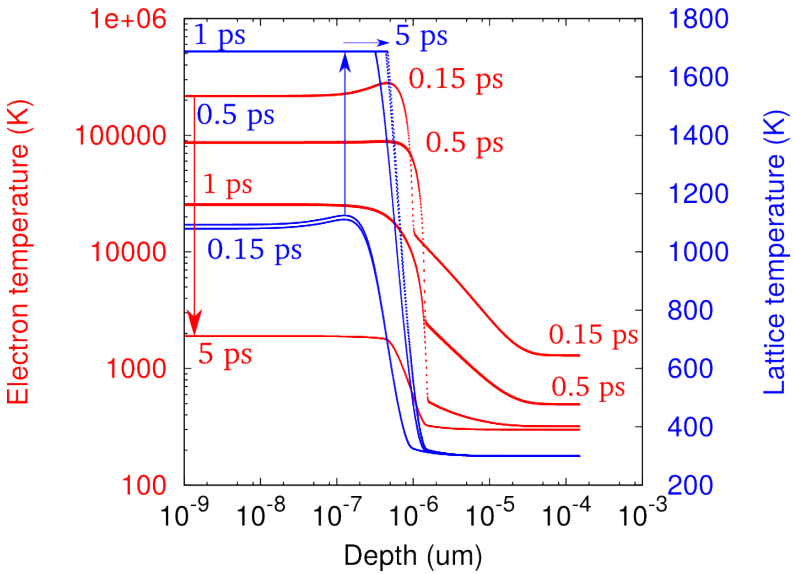


Observation

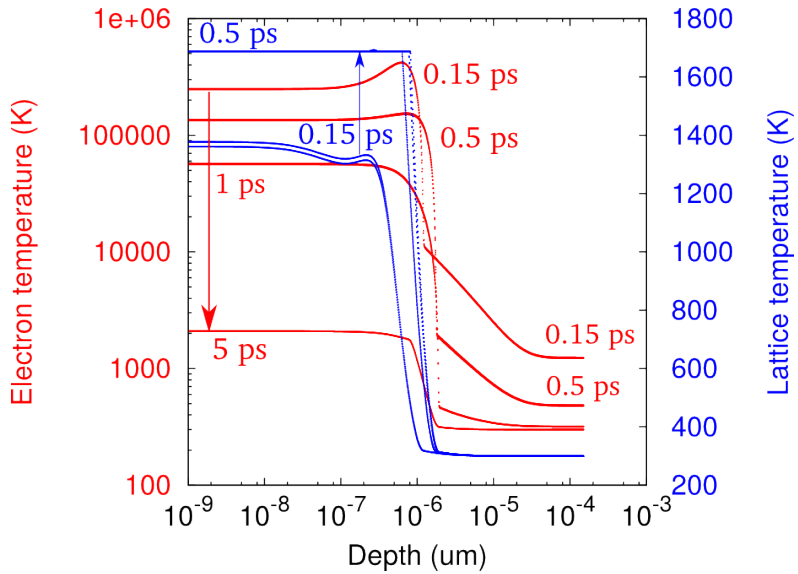
Effect of the probe light scattering on nucleation centers is important to describe the transient optical response.

Corrected energy absorption

Temperature profiles: Electrons and Lattice.



$F = 0.4\ \mathrm{J/cm^2}$



$F = 0.8\ \mathrm{J/cm^2}$

Competition between 3 processes ?

Non-thermal ultrafast melting (~ 300 fs)
VS electron-phonon screening (~240 fs),
VS scattering of probe light on the homogeneously melting silicon.

We need all levels of description

Each model can address effects that other methods can't.

Questions to address

TD-DFT as a theoretical description

How much TD-DFT is predictive for out-of-equilibrium situations?
How can TD-DFT address bulk effects? surface effects? intense laser?
Are there analytical tools for a non-equilibrium ab-initio approach?

Bi-color processing:

Are bi-color lasers good for laser processing?
Which new physical effects do they trigger?
Which experimental parameters are important to control?

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Many-body Schrödinger Equation

Solve the Schrödinger equation to resolve a N-body problem.

Stationnary Schrödinger equation

$$\hat{H}\left|\psi_0\right\rangle = E\left|\psi_0\right\rangle .$$

From the ground-state $\left|\psi\left(t=0\right)\right\rangle = \left|\psi_0\right\rangle$, we can solve

Time-dependent Schrödinger equation

$$\hat{H}\left(t\right)\left|\psi\left(t\right)\right\rangle = i\frac{\partial}{\partial t}\left|\psi\left(t\right)\right\rangle$$

using

$$\hat{H}(t) = \hat{T} + \hat{V}_{ee} + \hat{V}_{ext}(t)$$

Atomic Unit system

There exist several systems of "Atomic Units". See https://en.wikipedia.org/wiki/Natural_units for a great summary on units.

We use "Hartree atomic units."

- Distance are in Bohrs. 1 Bohr $= a_0 = 0.529$ Å $= 5.29 \times 10^{-11}$ m.
- Energies are in Hartree. 1 $H = 27.21$ eV.
- Time units are deduced: 1 time unit $t = \frac{a_0}{c} = \frac{1}{\alpha}$[u.a.] $\sim 2.419 \times 10^{-17}$ s [S.I.].
- $\hbar = e^2 = m_e = k_B = 1$.

Operators

In the Hartree system, operators read as:

- Kinetic energy: $\hat{T} = -\frac{1}{2}\sum_{i=1}^{N} \nabla_i^2$
- Coulomb potential: $\hat{V}_{ee} = \frac{1}{2}\sum_{i\neq j}^{N} \frac{1}{|r_i - r_j|}$
- External field potential: $\hat{V}_{ext} = \sum_{i=1}^{N} v_{ext}(\boldsymbol{r}_i, t)$, containing:
  - Nucleus energy: $v_{ext}(\boldsymbol{r}, t) = -\sum_{v=1}^{N_a} \frac{Z_v}{|r - R_v(t)|}$ (Z_v: charge of nucleus, R_v: position of nucleus)
  - Laser field: $v_{ext}(\boldsymbol{r}, t) = Ef(t)\sin(\omega t)\,\boldsymbol{r}\cdot\boldsymbol{\alpha}$ (dipolar approximation)

Why not solving Schrödinger?

Marques et al, *Fundamentals of time-dependent density functional theory* (Springer, 2011)

I want so solve electron levels of the oxygen atom (8 electrons)

$$\psi\left(\vec{r_1}, ..., \vec{r_8}\right)$$

- If taking a $10 \times 10 \times 10$ mesh, we obtain: $\left(10^3\right)^8 = 10^{24}$ entries.
- 1 DVD = 5×10^9 bytes.
- 10 g per DVD.
- We need 2×10^9 tonnes of DVD to store the ground state of Oxygen atom.

Existing quantum methods to avoid solving Schrödinger equation

1: Wave-function approaches
2: Functional theories.

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Kohn-Sham reduction

Kohn (Nobel prize: 1998) & Sham theorem

Access integrated quantities of quantum dynamics, avoiding to solve the Schrödinger equation.
Reduce the N-many-body problem to a single effective electron problem.

$$\psi\left(r_1, v_1; ...; r_N, v_N; t\right) \equiv \psi\left(r_{eff}, v_{eff}; t\right)$$

Theorem
"In a Kohn-Sham system, to a single density N_e corresponds a unique system state configuration described by the state density

$$N_e =< \psi^\dagger \psi >= N \sum_{\text{spin } \sigma_i} \int d^3 r_1 ... d^3 r_N \left|\psi\left(r_1, \sigma_1, ..., r_N, \sigma_N, t\right)\right|^2$$

In fine, N non-interacting fermions are evolving in a 1 body potential. Much simpler to solve.

Kohn-Sham equation

Notations. ψ: the exact wavefunction. ϕ: the Kohn-Sham wavefunction.

Kohn-Sham density

$$N(r,t) = \sum_{j=1}^{N} |\psi_j(r,t)|^2 = \sum_{j=1}^{N} |\phi_j(r,t)|^2$$

$$i\frac{\partial}{\partial t}\phi_j(r,t) = \left[-\frac{\nabla^2}{2} + v_{KS}[n;\phi_0](r,t)\right]\phi_j(r,t)$$

The solution are decomposed in 3 terms

$$v_{KS}[n;\phi_0](r,t) = v_{ext}[n;\psi_0](r,t) + \int d^3r \frac{n(r',t)}{|r-r'|} + v_{xc}[n;\psi_0,\phi_0](r,t)$$

Runge-Gross mapping theorem

Runge-Gross theorem

"To each single time-dependent density N_e corresponds a unique system state configuration."

Summary of TD-DFT.

Each state $\psi\left(\overrightarrow{r}, t\right)$ is described by one object: the total density $N_e(\overrightarrow{r}, t) = N_{VB} + N_{CB}$.

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Known issues of TD-DFT

Advantages

- Valid out of equilibrium: good to address high-field and intense laser physics.
- One main description for many materials.
- The theory has no free parameter: predictive state is reached for many problems.
- It exists many functionals allowing to describe a wide range of materials.
- Both surface effects and bulk effects can be addressed.

Current difficulties

- TD-DFT is not a black-box.
- (No energy dissipation.)
- (Temperature has no meaning at that scale.)
- Prediction of the band-gap are not always good.
- Where are the e-e and e-ph collisions?
- Missing magnetic effects.
- Wrong spontaneous emission rates.
- No memory effects yet.
- Charge transfer in molecules.

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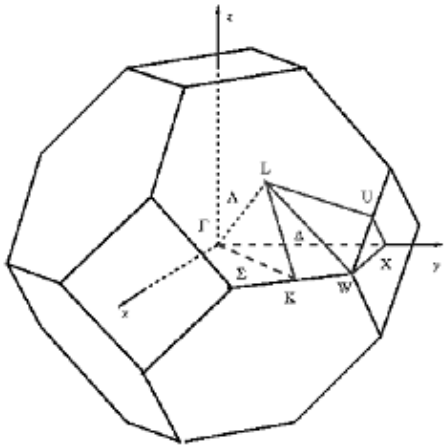
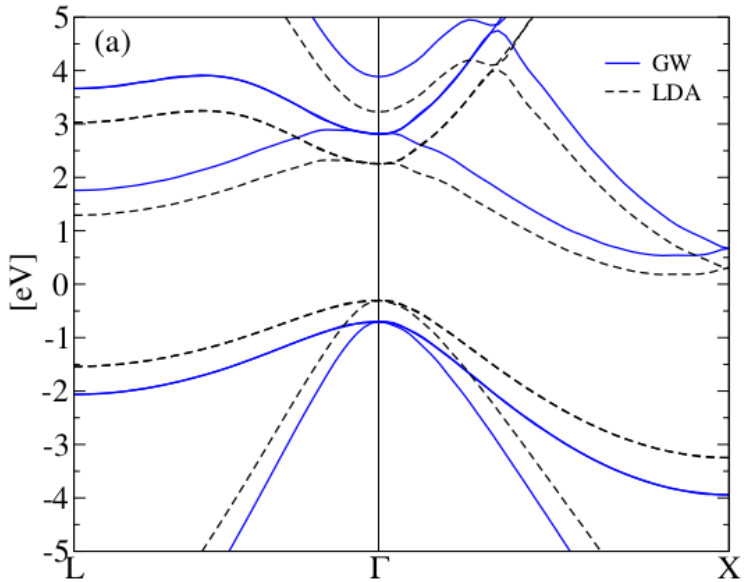
Quantum simulation of Si

Sangalli et al, *Ultra-fast carriers relaxation in bulk silicon following photo-excitation with a short and polarized laser pulse*, Europhysics Letters **110**, 47004 (2015)



Brillouin zone of Si



Quantum calculations of Si band structure

High symmetry points

Greek letters (Γ) belong to the inner part of the Brillouin zone.
Latin letters (L, X) belong to the edge of the Brillouin zone.

Band gap prediction

LDA: simplest approach (no e-e collision, no memory effects).
GW: state of art (Bethe-Salpeter Equation)

Equilibrium: the Si exciton peak

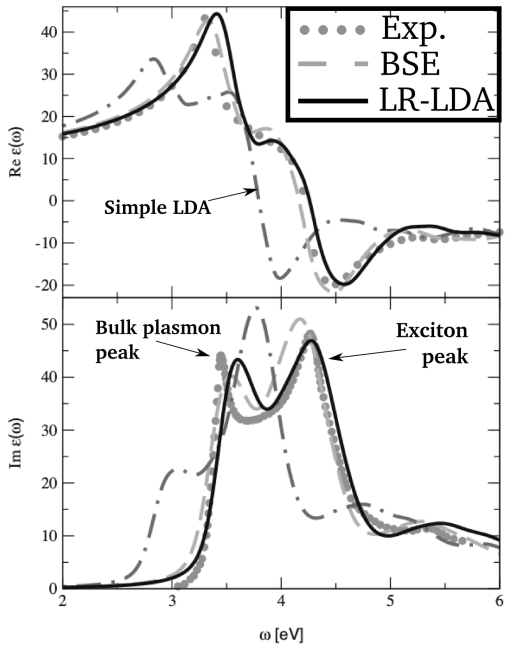


FIG. 1. Silicon, optical absorption (bottom), and refraction index (top panel) spectra. Dots: experiment. Dot-dashed curve: TDLDA result. Dashed curve: result obtained through the Bethe-Salpeter method. Continuous curve: TDDFT result using the long-range kernel derived in this work.

Reining, L.; Olevano, V.; Rubio, A. & Onida, G. *Excitonic Effects in Solids Described by Time-Dependent Density-Functional Theory,* Physical Review Letters **88**, 066404 (2002)

- Bethe-Salpeter: most accurate description (beyond TD-DFT).
- Good agreement was obtained by introducing long-range effects in DFT.

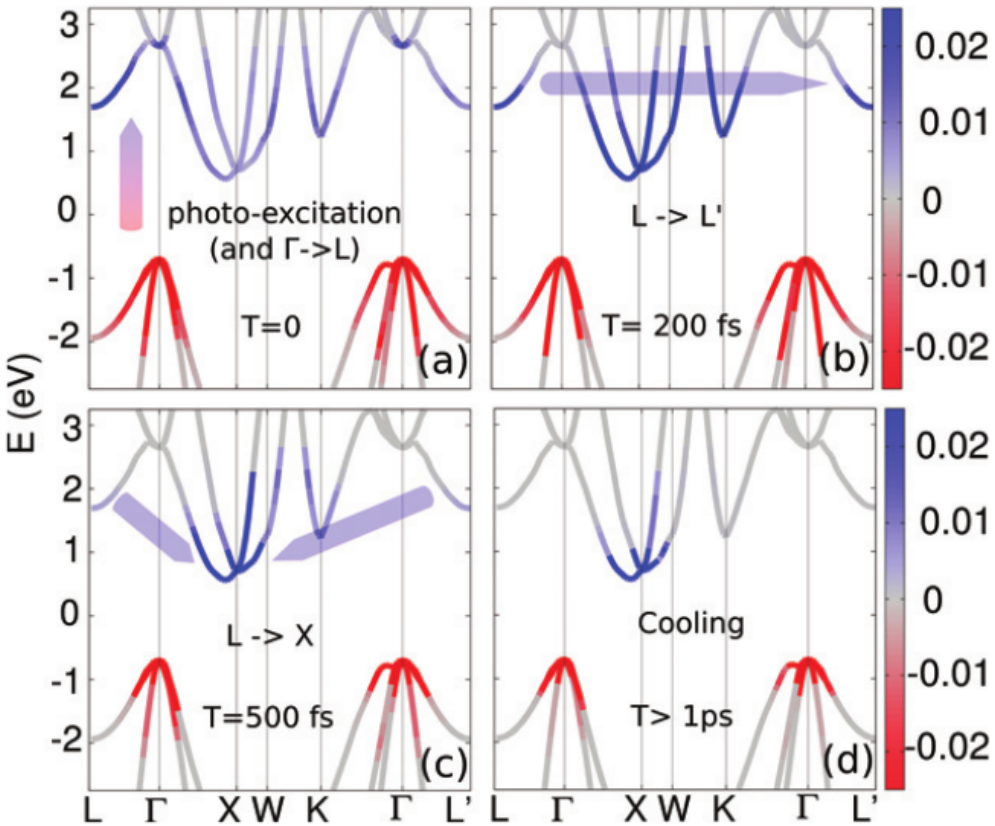
Open question

Does it work for intense laser irradiation?

Near-equilibrium linear response of Si

- UV light: 1-photon absorption.
- $I = 10^9$ W/cm^2, $\tau = 110$ fs, $\hbar\omega = 3.4$ eV.

Sangalli et al, Europhysics Letters **110**, 47004 (2015)



(a) Non-trival interband transitions.
(b) Excited electrons transfer between bands.
(c) Relaxation to lower energy levels.

Far from equilibrium: state of art

Sangalli, D.; Marini, A. et al, *Non equilibrium optical properties in semiconductors from first-principles: a combined theoretical and experimental study of bulk silicon,* Physical Review B **93**, 195205 (2016)

Brown dots: experiments. **Black line**: quantum electrodynamics theory (BSE).

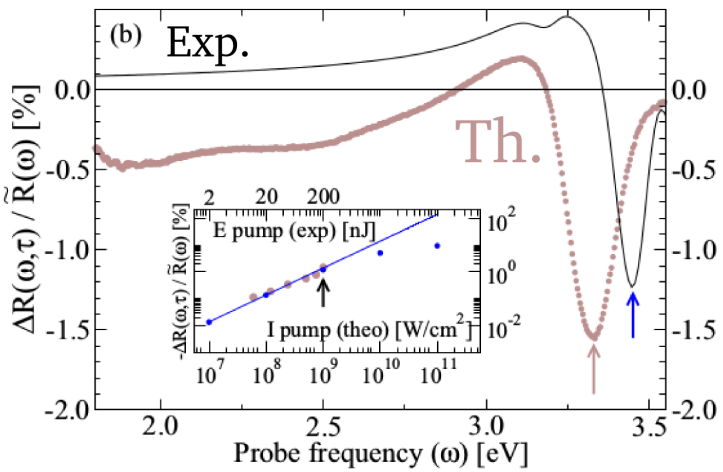


FIG. 7. (a) Renormalization of the quasiparticles energies from the COHSEX self-energy due to the screening of the nonequilibrium carriers density. The change induced in the reflectivity near the main peak is shown in the inset (blue arrow) together with the effect due to the renormalization of the electron-hole interaction (red arrow). (b) Transient reflectivity for a pump intensity of 10^9 W/cm^2 and $\tau = 200$ fs; the variation of the BSE poles due to the screening of the non-equilibrium carriers density is considered. In the insets the change in the gap (a) and the signal at 3.45 eV (b) are shown as a function of the pump peak intensity. Theoretical results are compared with experimental data (brown circles).

BSE: Bethe-Salpeter Equation (perturbative approach)

BSE is still a perturbative approach.

Non-perturbative BSE-like approach ?
"Non-equilibrium green functions."

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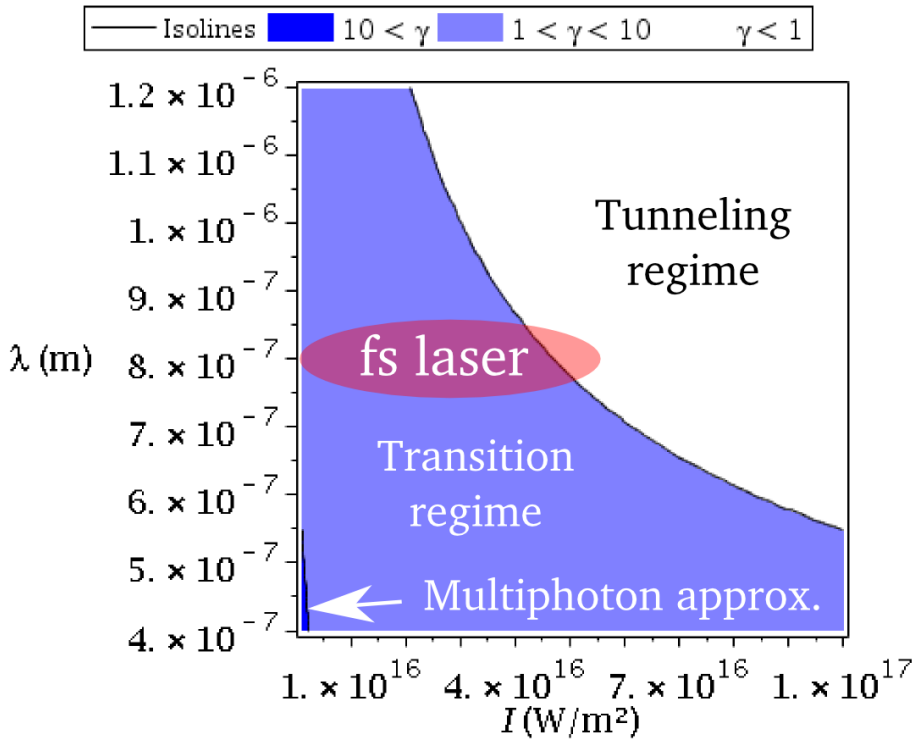
Non-linear absorption

Valence band (VB) electron may be excited to CB by intense light in different ways. **Adiabatic vision**:

- Multiphotonic absorption
- Tunneling effect

Two models can be used:

- Keldysh model (direct band gaps, parabolic band structure) [24]
- Multiphotonic rates (adjusted by experimentalists) [16]



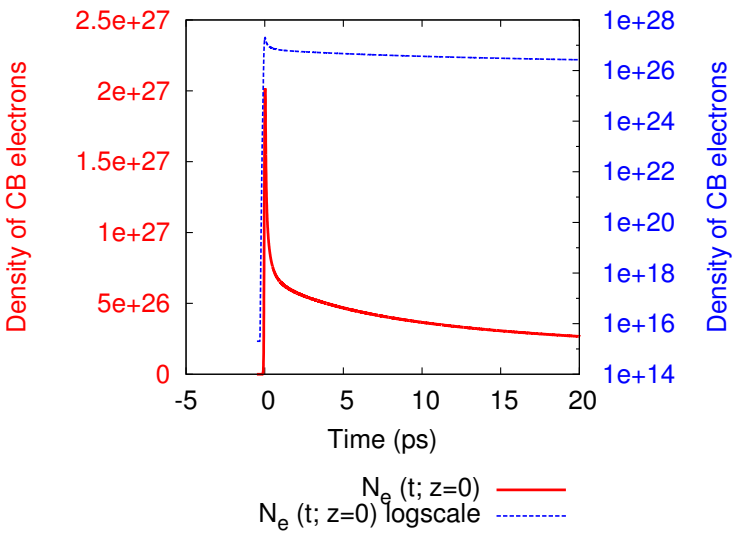
$E_g = 1.12$ eV (300 K), $m^* = 0.18$ [13]. Isolines: $\gamma = 10$ and $\gamma = 1$.

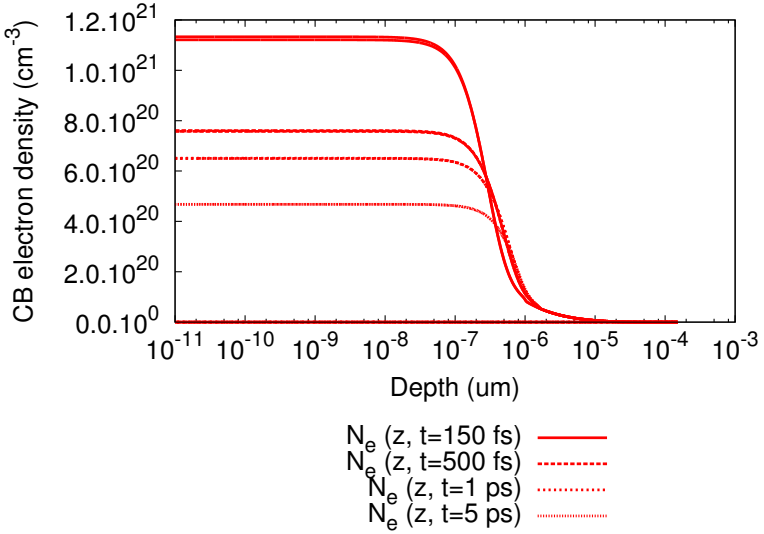
Limitations

Real band structure of materials is not parabolic.
fs laser ablation uses an intermediate regime
Monochromatic light

Evolution of excited electron density

Density of excited electron N_{e-h} (m^{-3}).
Derrien, T. J.-Y. et al, *Possible surface plasmon polariton excitation under femtosecond laser irradiation of Silicon,* J. Appl. Phys. **114**, 083104 (2013)





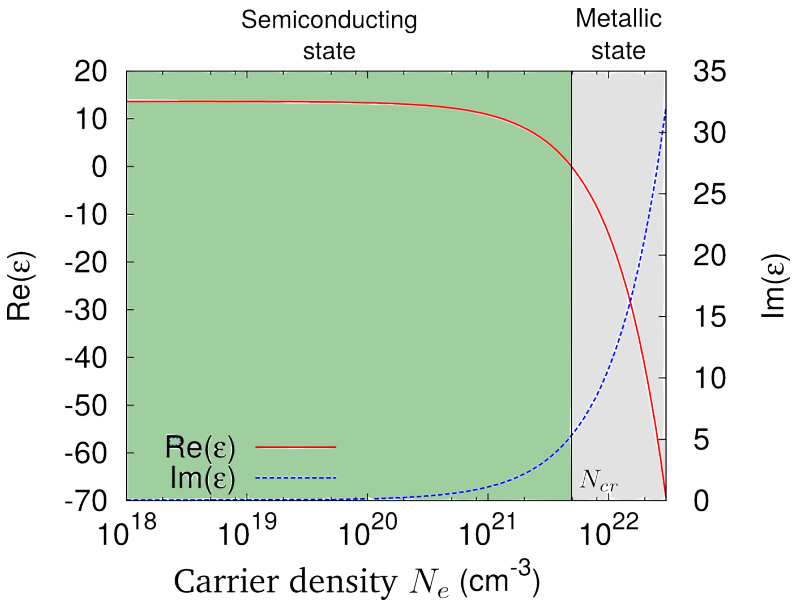
Missing:

Light propagation.
Better description of non-linear absorption.

Transient change of refractive index

- Modified Drude model
  [Sokolowski-Tinten et al, Phys. Rev. B **61**, 2643 (2000)]

$$\varepsilon^*_{Si}(n_e) = 1 + \underbrace{(\varepsilon_{Si}(\lambda, T) - 1) \times \frac{N_0 - n_e}{N_0}}_{\text{shift of potential: band-filling effect}} - \underbrace{\frac{\omega^2_{pe}(n_e)}{\omega^2\left(1 + i\frac{\nu}{\omega}\right)}}_{\text{response of conduction e-}}$$

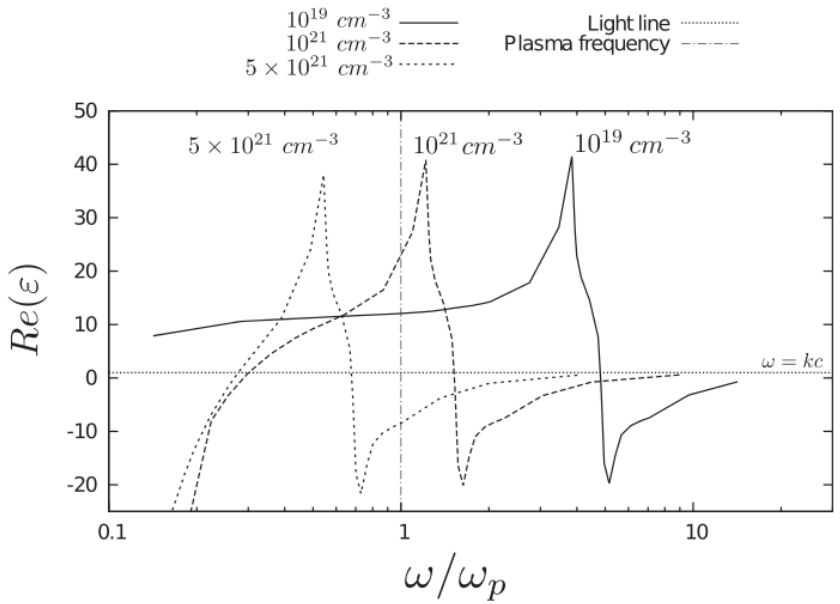


$$\omega^2_{pe} = \frac{n_e e^2}{m_e m^* \varepsilon_0}$$

m^* depends on the band curvature. Evolution of m^* and ν with T_e and T_l are unkonwn (average values are measured experimentally)

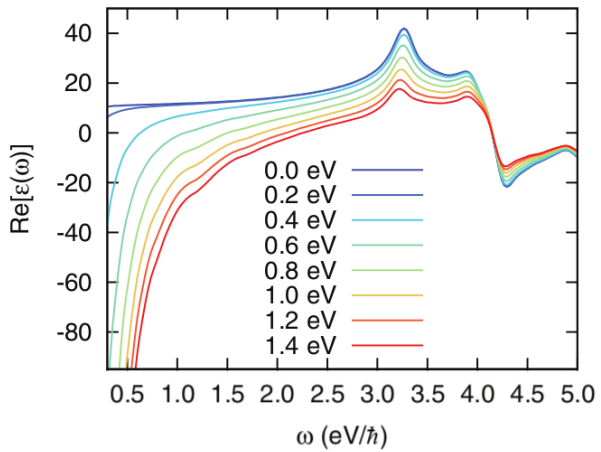
Si: validity of the Drude model

Classical Drude model



Derrien et al, J. Appl. Phys. **114**, 083104 (2013)

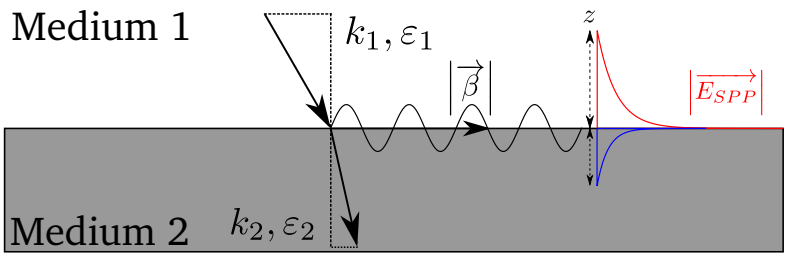
TD-DFT simulation



S.A. Sato and Yabana, Phys. Rev. B **90**, 174303 (2014)

Transient SPP excitation

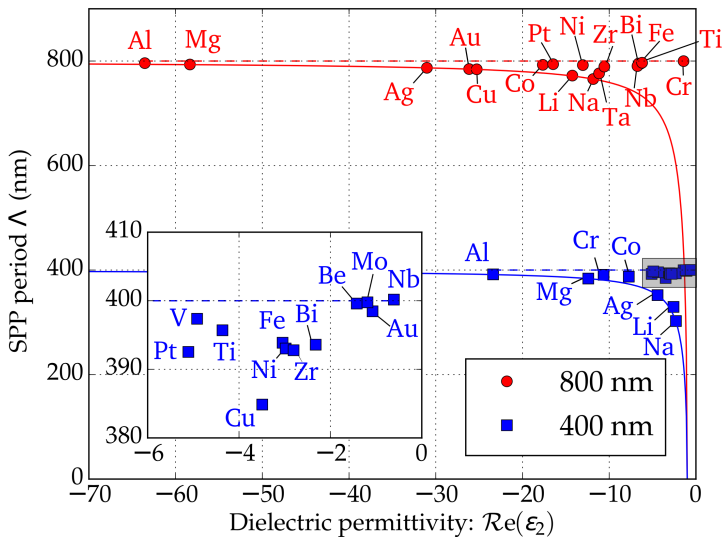
Derrien, T. J.-Y.; Krüger, J. & Bonse, J. *Properties of Surface Plasmon Polaritons on lossy materials: Lifetimes, periods and excitation conditions*, Journal of Optics (2016), Accepted for publication.



Prediction

SPP may not be directly visible in the quantum simulations.

- Large-scale oscillation of the SPP field: Λ is close to laser wavelength λ ($\gg$quantum simulation box).



xc functionals for Si

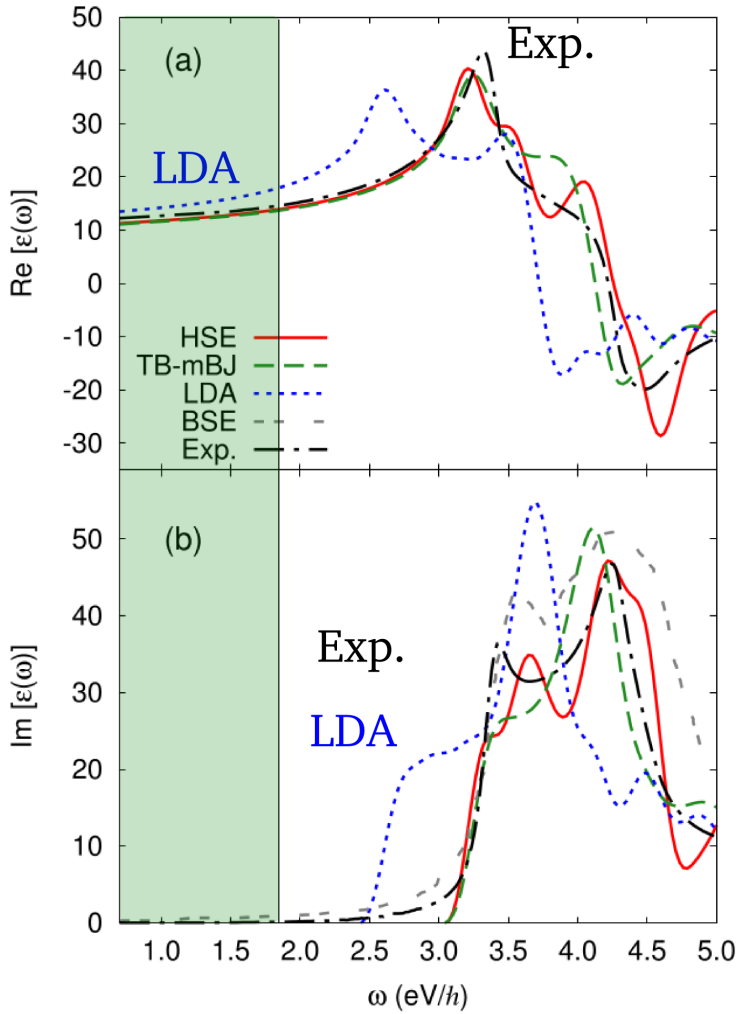
Sato, S. A.; Taniguchi, Y.; Shinohara, Y. & Yabana, K. *Nonlinear electronic excitations in crystalline solids using meta-generalized gradient approximation and hybrid functional in time-dependent density functional theory,* The Journal of Chemical Physics **143**, 224116 (2015)

Question

Hybrid functionals implemented in octopus?

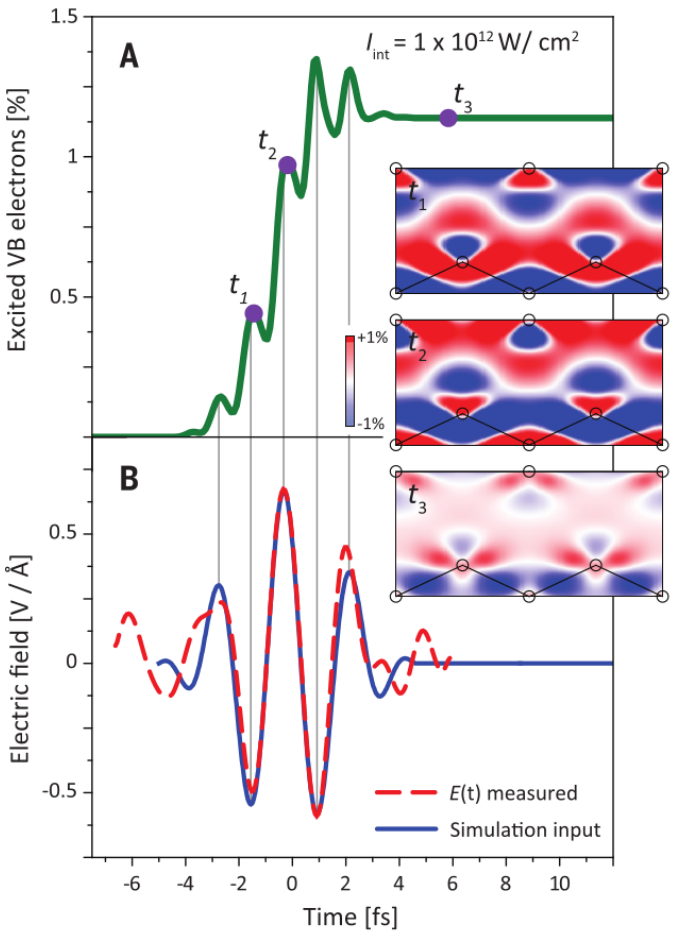
Accuracy in Si

Upon IR light, everything should work well.
Upon UV light, difficulties are expected.



Attosecond electron dynamics of Si

Schultze, M.; Sato, S. A.; Yabana, K. et al, *Attosecond band-gap dynamics in silicon,* Science **346**, 1348-1352 (2014)



Output of the study

- TD-DFT method is accurate to describe the transient dynamics of carriers in Si.
- Excitation is rather due to tunneling (10^{12} W/cm^2).

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Overview

Purposes

- Learn TD-DFT as a tool for bridging continuum descriptions with ab-initio approaches.
- Extend TD-DFT calculation of electron excitation rates from VB to CB for many laser/material parameters.
- Sample upon femtosecond, intense, non-linear excitation, as used by fs-processing community.
- With single- and mixed two-color beams.

Method

- Using a 2D-slab geometry: modifiable thickness, resolve surface effects.
- Reduce the k-points grid to $1 \times 1 \times 1$ to avoid Schrödinger / Heisenberg discrepancies.
- Irradiation slab with bi-color laser pulses (11-dimesional parameter space),
- Fully automatic using Bash and Python scripting.

Expected output

Expectations

- Input: numerous laser parameters as used by laser processing scientists (very high sensitivity).
- Time- and space-resolved surface and volume phenomena: generate the missing ingredients of large-scale models.
- Output: non-linear absorption rates, and other quantities (later).

Side effects

- Benchmarking Octopus (EOS, and the new born DRACO).
- Find the missing blocks of Octopus.

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Excited electron definitions

Otobe, T.; Yabana, K.; et al. *First-principles electron dynamics simulation for optical breakdown of dielectrics under an intense laser field* Physical Review B **77**, 165104 (2008)

Definition ?

- Heisenberg vision: projection of observables on the ground state (easy, fast, intuitive picture).

$$N_{el} = \frac{1}{V} \sum_{nn'k} \left(1 - \left| < \phi^{GS}_{n,k} | \phi_{n',k}(t) > \right|^2 \right)$$

- Schrödinger vision: accounting for the laser field A(t) [more rigorous, difficult to compute]

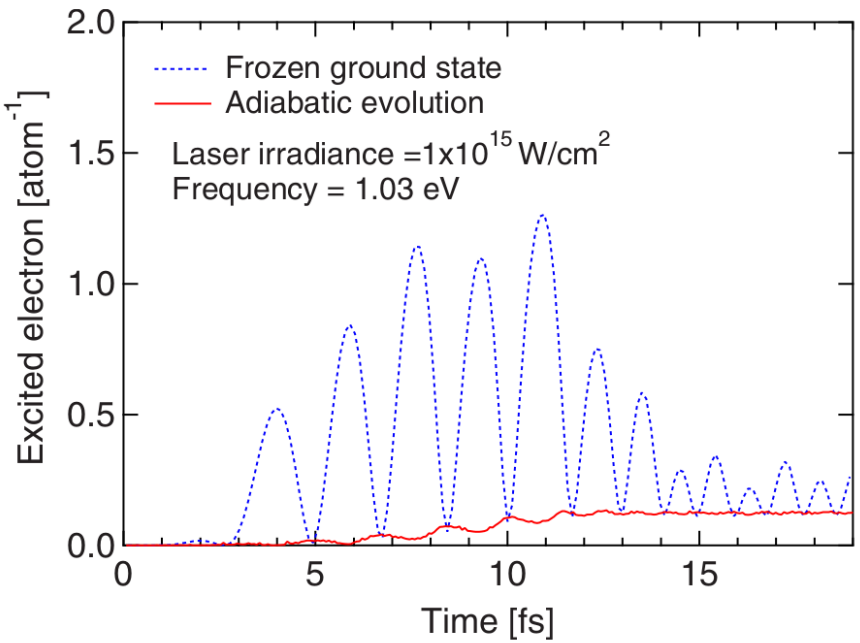
$$N_{el} = \frac{1}{V} \sum_{nn'k} \left(1 - \left| < \phi^{GS}_{n,k+A(t)} | \phi_{n',k}(t) > \right|^2 \right)$$

Which method?

- Method 1 runs faster (10 fs/day) with Octopus.
- Method 2 needs to re-converge a ground state for each time-step (1 day per timestep ?).

Definition of excited density

Definition of electron density in TD-DFT ?
Otobe, T.; Yabana, K.; et al. *First-principles electron dynamics simulation for optical breakdown of dielectrics under an intense laser field*, Physical Review B **77**, 165104 (2008)



Heiko's trick

If using only 1 k-point (Γ) to kill the k-space dispersion, both definitions must give the same results.

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Multiscale problem

- Small size sample (12×0.5 Å ~ 3.0 nm),
- Laser wavelength ($\lambda \in [100\text{ nm} - 1\ \mu\text{m}]$) is large compared to the simulation box.

Approximation

Laser field is described at a single point.
Focusing effects are too large in respect of the system.

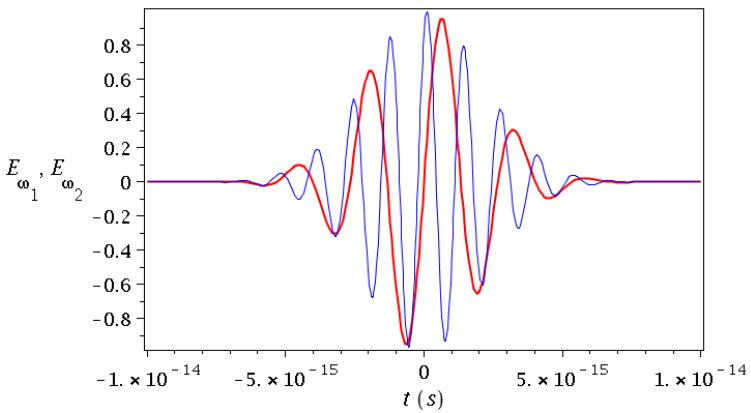
Bi-color laser pulses

Bi-color laser pulses generated through their electric fields

$$\boldsymbol{E}(t) = \boldsymbol{E}_{\lambda_1}(t) + \boldsymbol{E}_{\lambda_2}(t, \Delta t).$$

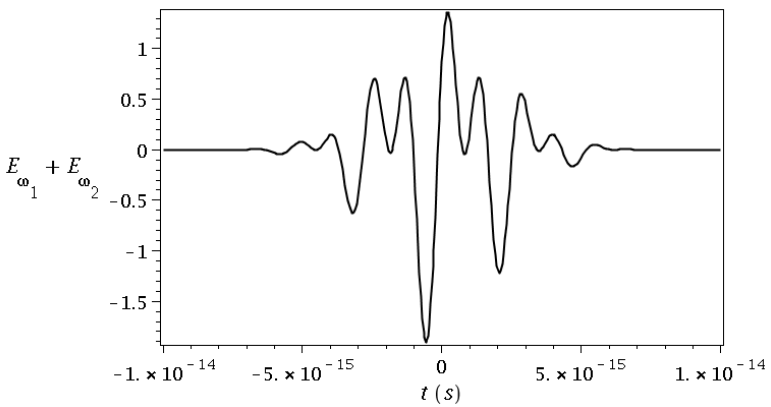
$$\boldsymbol{E}_{\lambda_1} = \boldsymbol{E}_0^{(1)} \sin(\omega_1 t + CEP_1) \cdot \exp\left[-\frac{1}{2}\left(\frac{t}{\sigma(\tau_1)}\right)^2\right]$$

$$\boldsymbol{E}_{\lambda_2} = \boldsymbol{E}_0^{(2)} \sin(\omega_2 t + CEP_2) \cdot \exp\left[-\frac{1}{2}\left(\frac{t - \Delta t}{\sigma(\tau_2)}\right)^2\right]$$



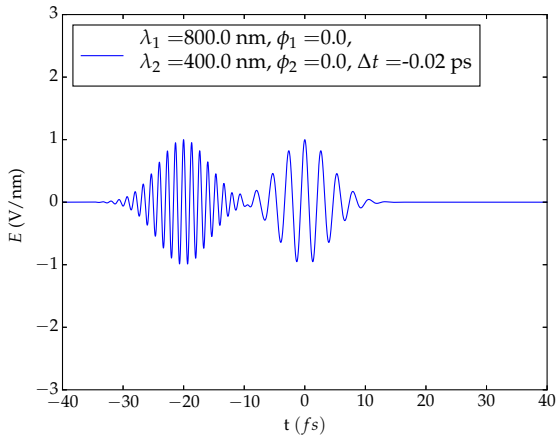
Main idea

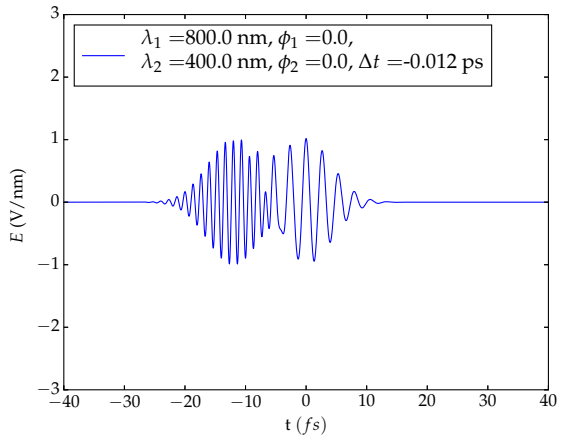
Electron dynamics must be strongly affected by the bi-color laser pulses.

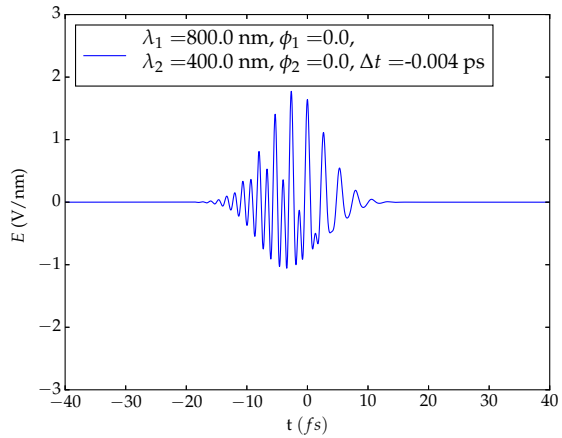


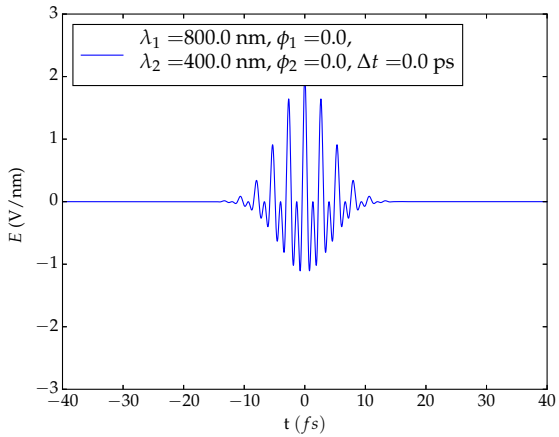
Effect of the pulse delay Δt

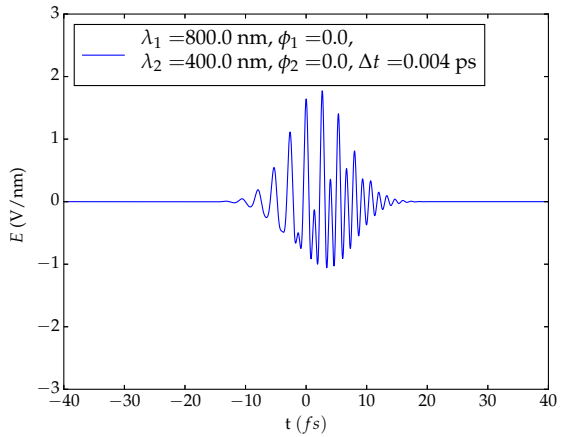
Wavelength: $\lambda_1 = 800$ nm, $\lambda_2 = 400$ nm.

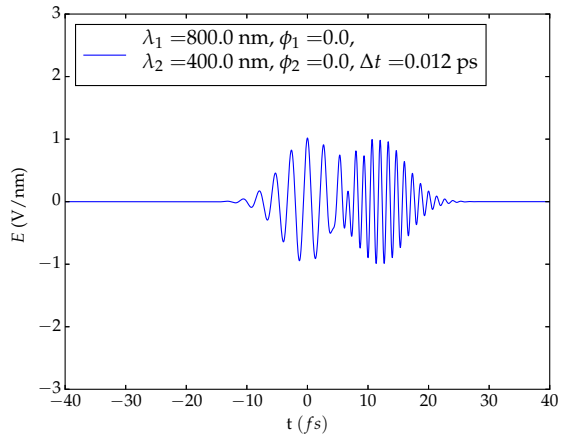






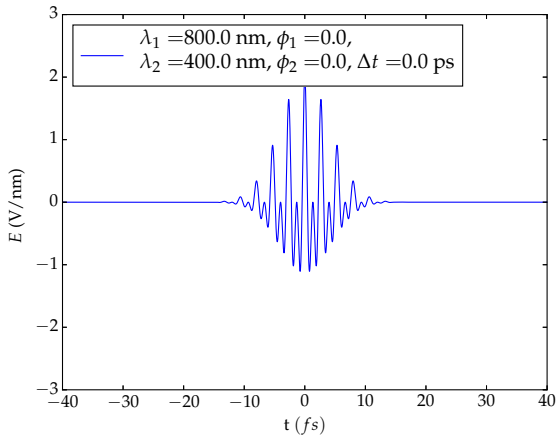


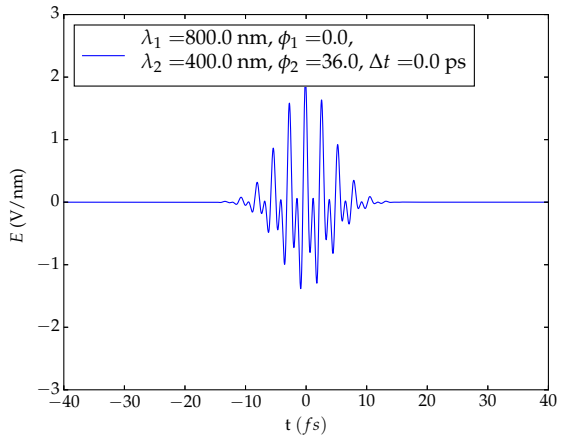


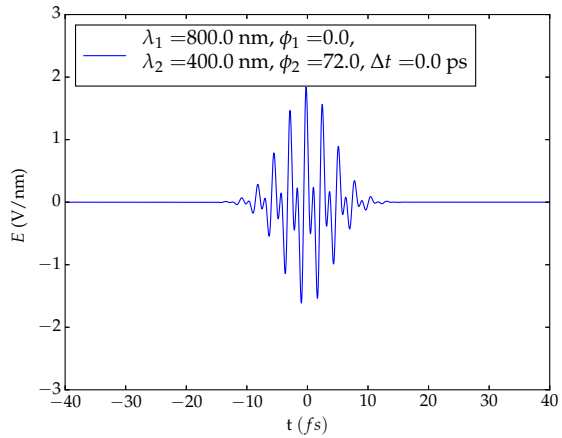


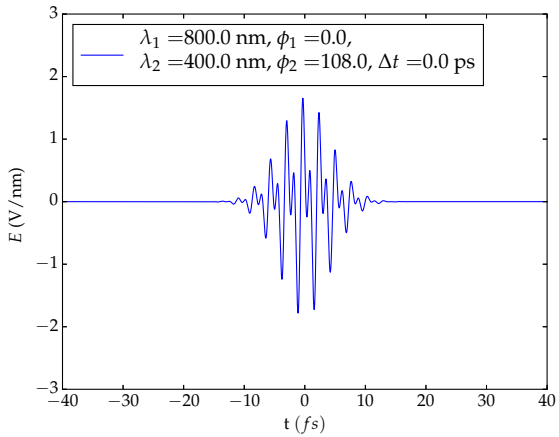
Commensurable wavelengths: CEP

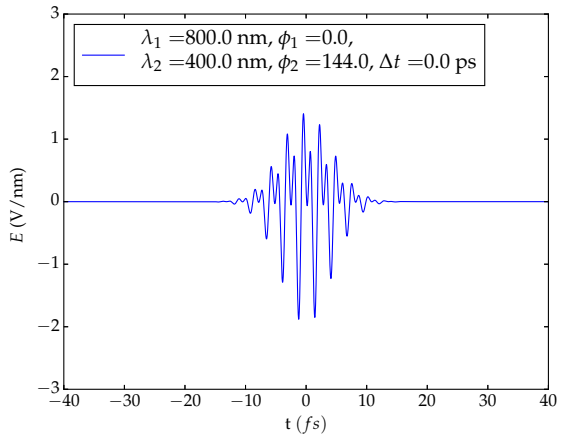
Mixed beams (800 nm and 400 nm) of various respective phases.

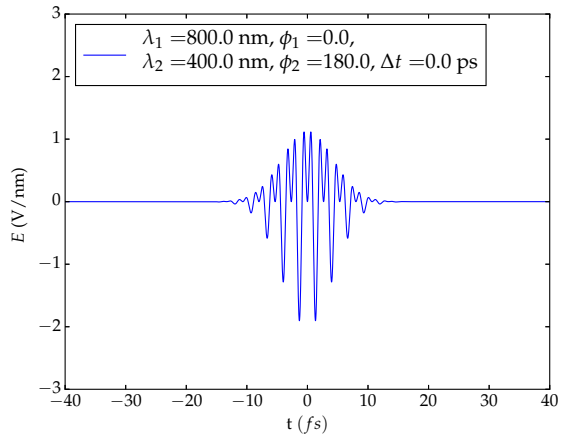






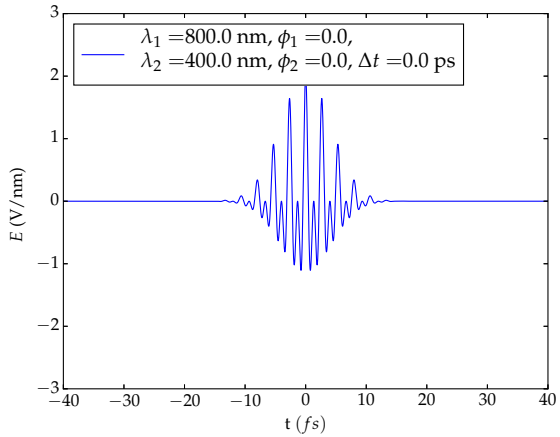


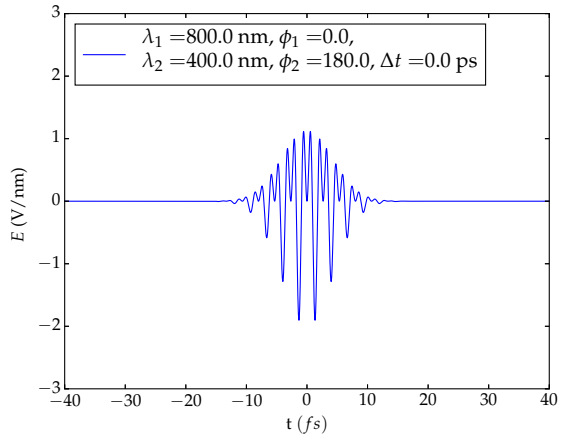




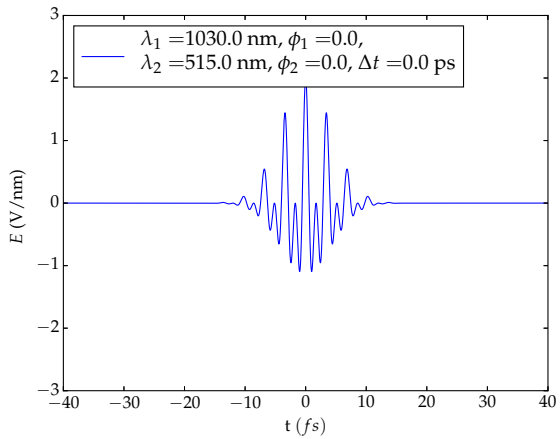
Commensurable wavelengths

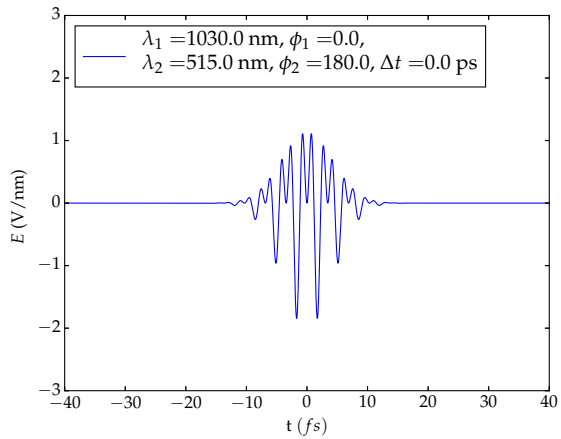
800/400 nm. $\phi = 0°$ and $\phi = 180°$.





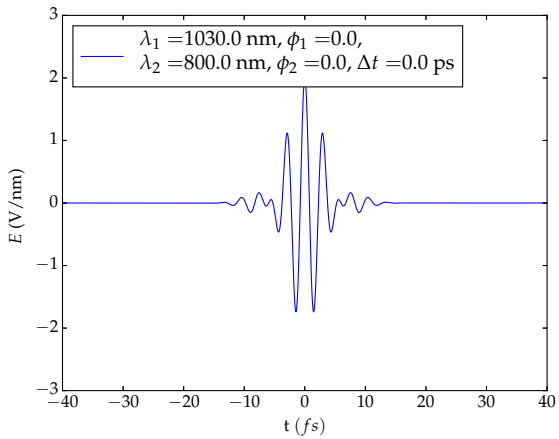
1030/515 nm

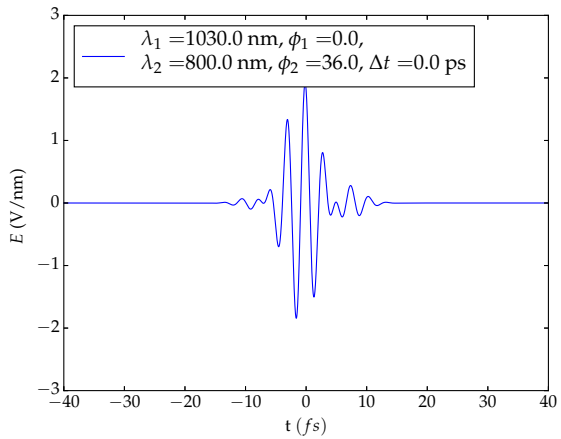


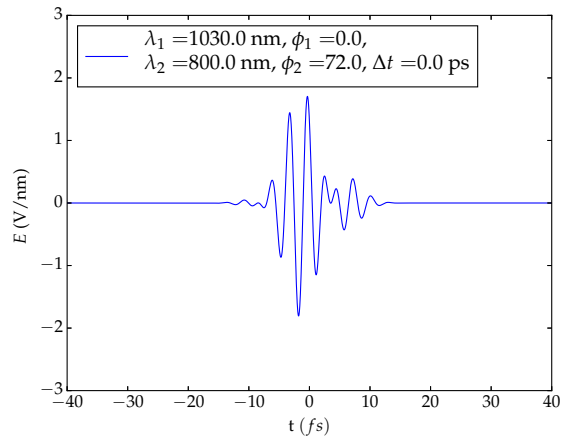


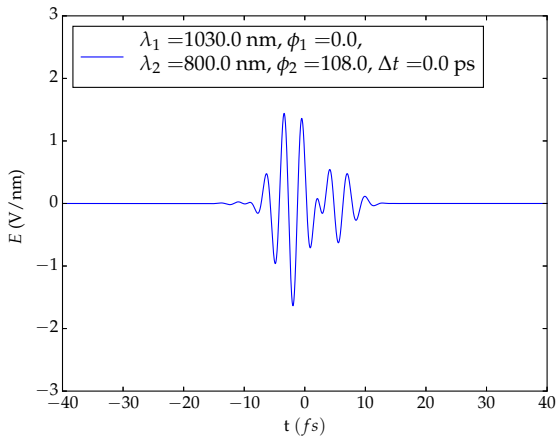
Incommensurable wavelengths: envelope

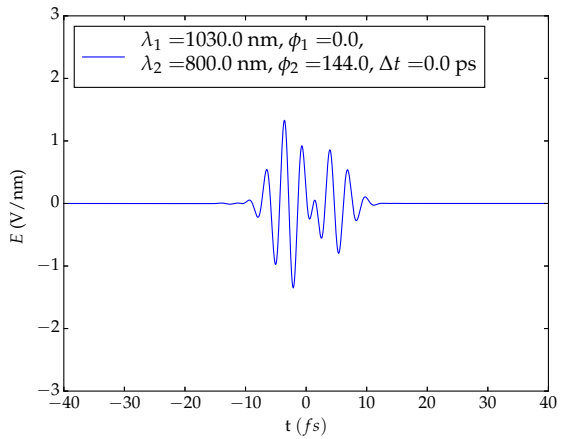
Wavelength 1030/800 nm. Various phases.

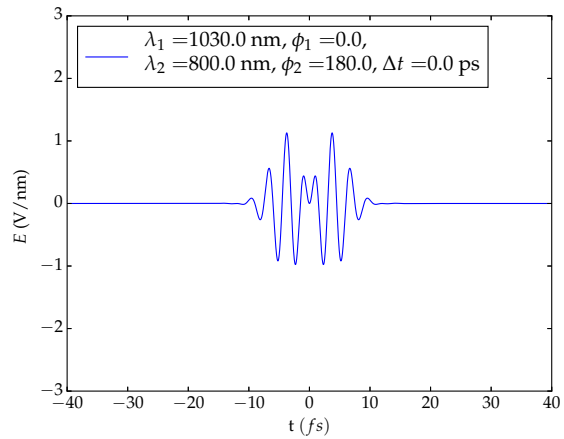












A collection of bi-color pulses

Set of parameters

Wavelengths λ_1, λ_2
Pulse durations: τ_1, τ_2
Envelope phases: CEP_1, CEP_2
Laser fluencies: F_1, F_2 (or intensities)
Polarization: linear, circular.

11 dimensions to explore.

- How to structure / classify them?
- Symmetries in the parameter set?
- Huge computational time is required to build a non-linear absorption database.

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5. Conclusions

Surface reconstruction of Si

Reconstruction of semicond. surfaces Si (100):H, Si(100):O, etc...

- Tancogne-Dejean, N. *Ab initio description of second-harmonic generation from crystal surfaces*, PhD thesis (2015)

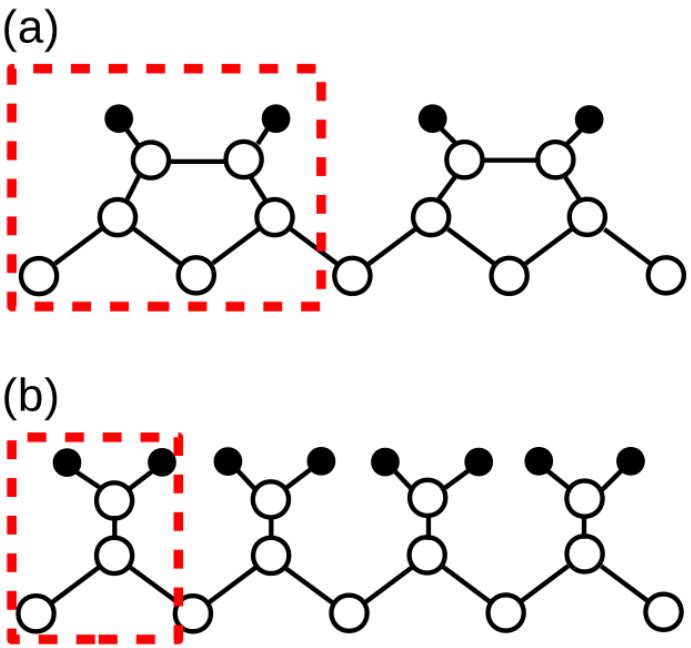
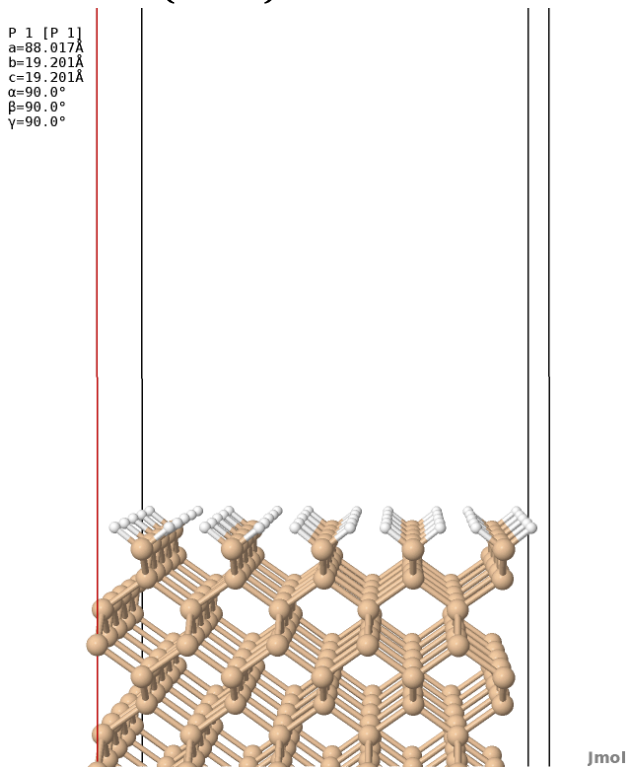


Fig. 4.5: Side view of different possible reconstructions of the hydrogenated Si(001) surface. (a) Monohydride or Si(001)2×1:H surface. (b) Dihydride or Si(001)1×1:2H surface. The H atoms are displayed in black. Red dashed lines correspond to the smallest cell needed for mapping the entire surface.

$Si(100)1 \times 1 : 2H$



Minimization of energy

Build a 2D slab with enough layers to obtain converged surface energy.

- Tancogne-Dejean, N. *Ab initio description of second-harmonic generation from crystal surfaces*, PhD thesis, Ecole polytechnique (2015)
- Bechstedt, Friedhelm, *Principles of Surface Physics*, Springer Berlin Heidelberg (2003).
- Stekolnikov, A. A.; Furthmüller, J. & Bechstedt, F. Physical Review B **65**, 115318 (2002)

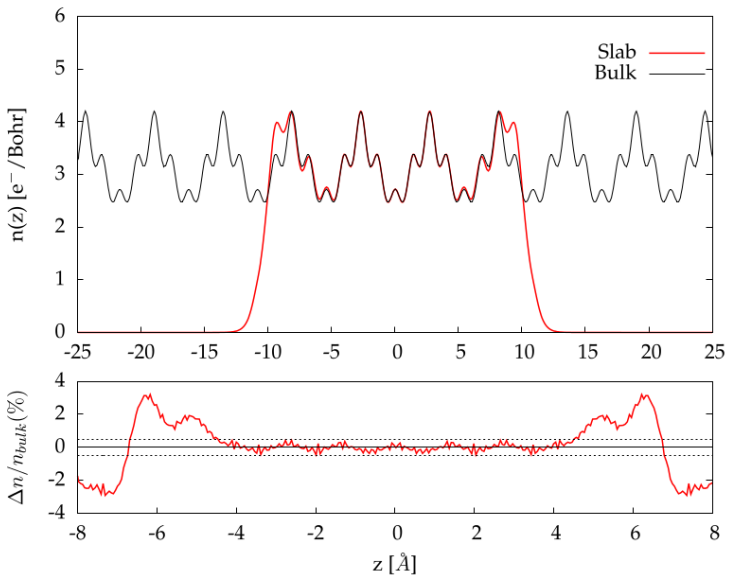


Fig. 5.9: Top panel: Variation of the number of electrons par Bohr along the z direction. This value is obtained by averaging the electronic density in the $x-y$ plan for the entire cell. Bottom panel: relative differences between the bulk and the slab densities of the above panel. Note the change in the scale. The slab is composed of 16 atomic layers in total, with asymmetric dimers as surface reconstruction on both sides.

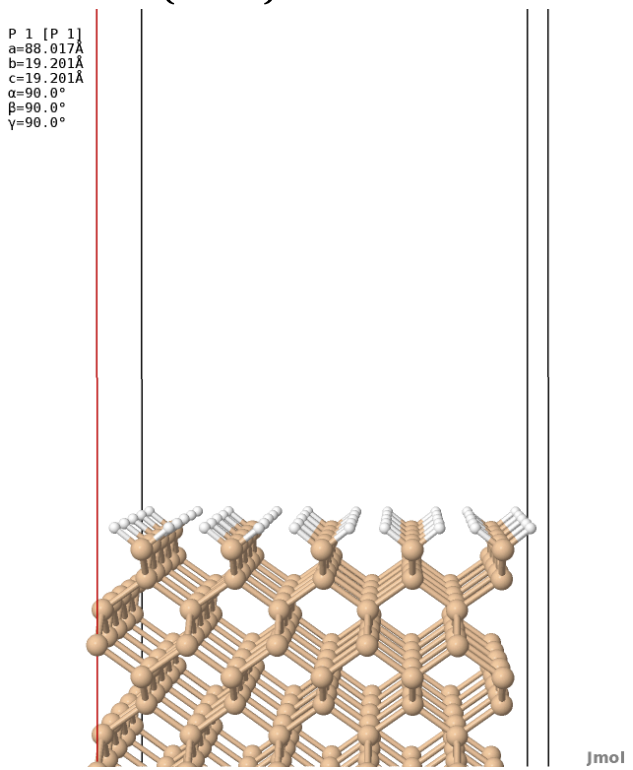
A converged sample

- Number of Si atoms:
  $$\underbrace{4 \times 4}_{\text{k-points=1} \times 1 \times 1} \times \underbrace{12}_{\text{atomic layers}} = 192.$$
- Number of H atoms:
  $$\underbrace{2}_{\text{sides}} \times \underbrace{2}_{\text{dimers}} \times \underbrace{(4+1)^2}_{\text{cells in XY}} = 100.$$
- Total number of atoms: 292 atoms.

Possible problem with $k = 1 \times 1 \times 1 = \Gamma$

We may require more atoms during TD-calculation out of equilibrium!

Si(100)$1 \times 1 : 2H$



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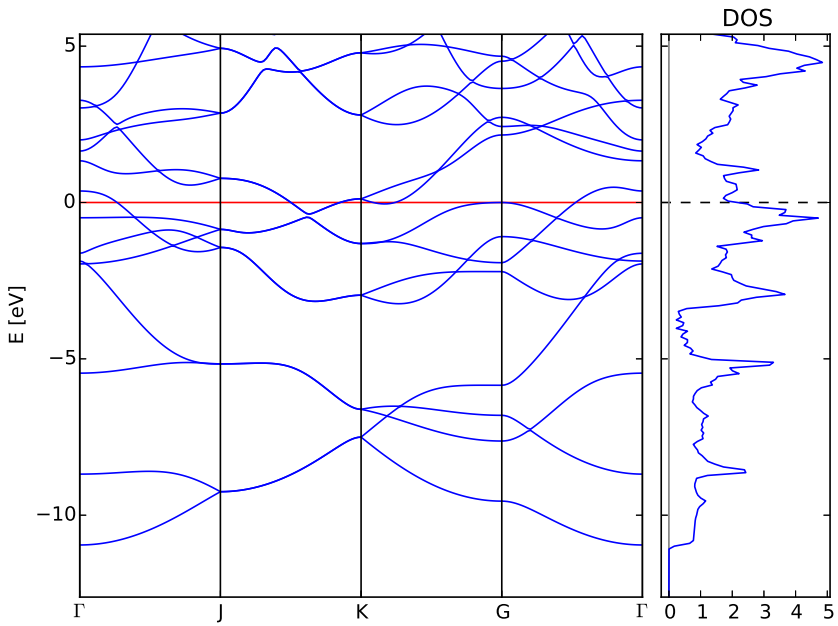
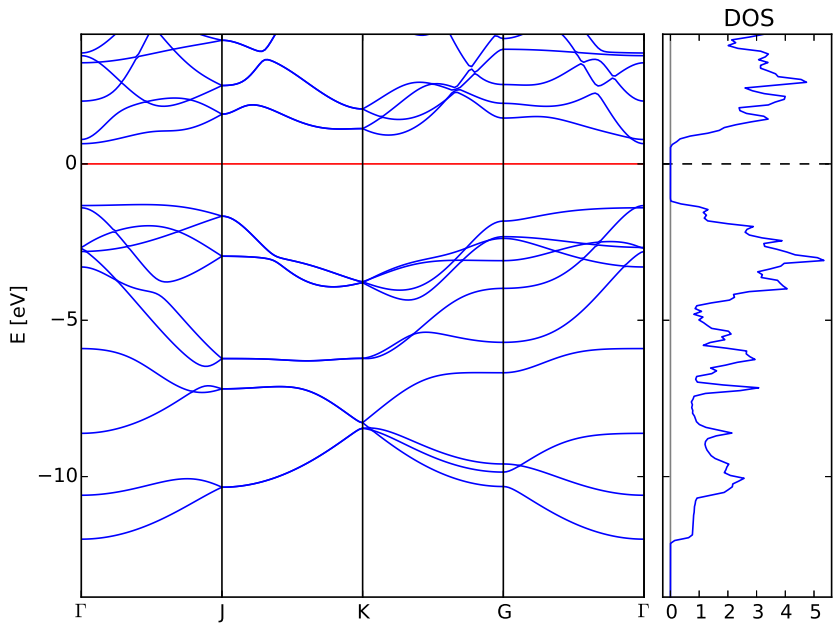
Latest results: $1 \times 1 \times 4$ atom

Without H terminations.

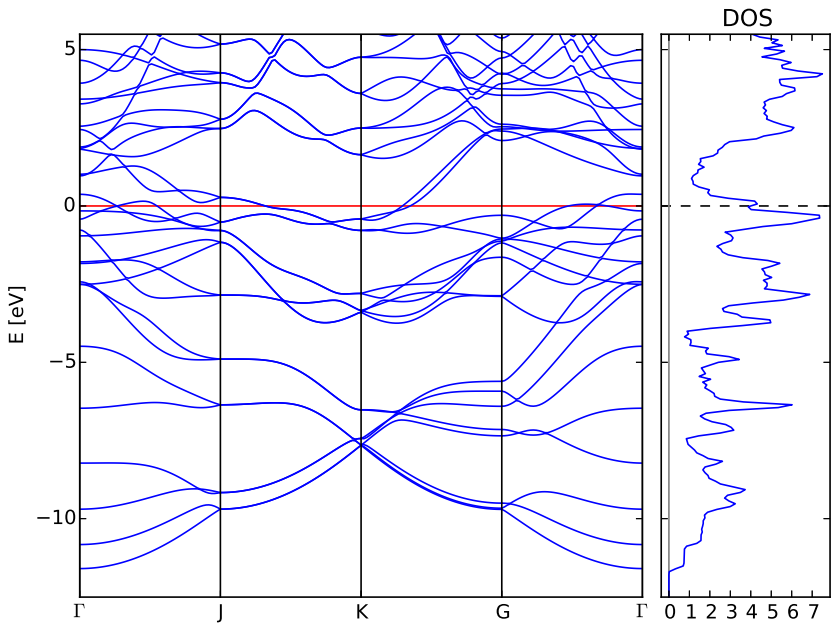


With H terminations.

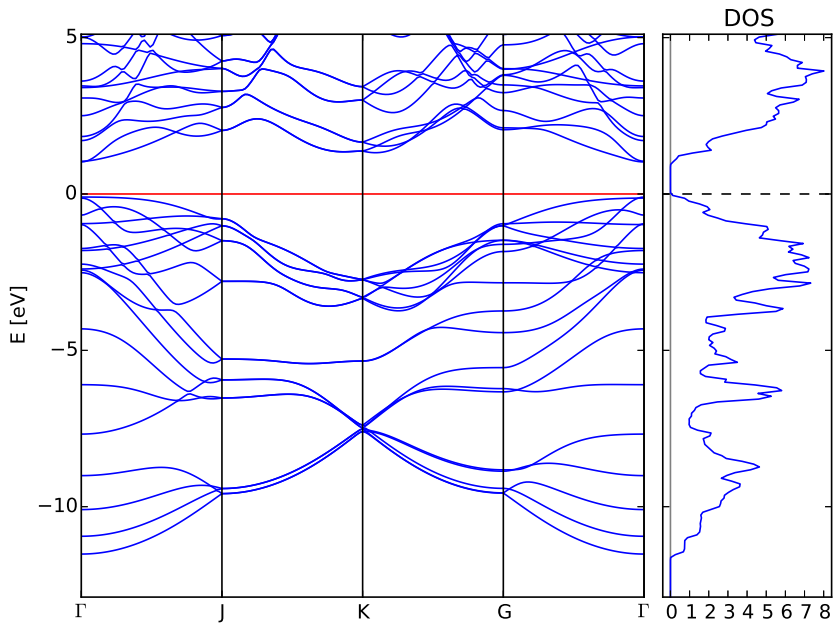


Latest results: $1 \times 1 \times 8$ atoms

Without H terminations.

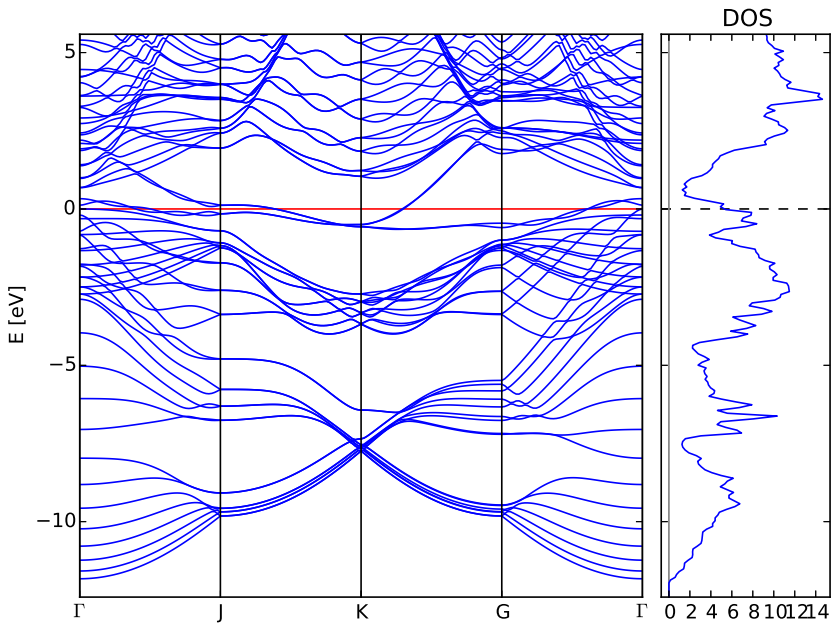


With H terminations.

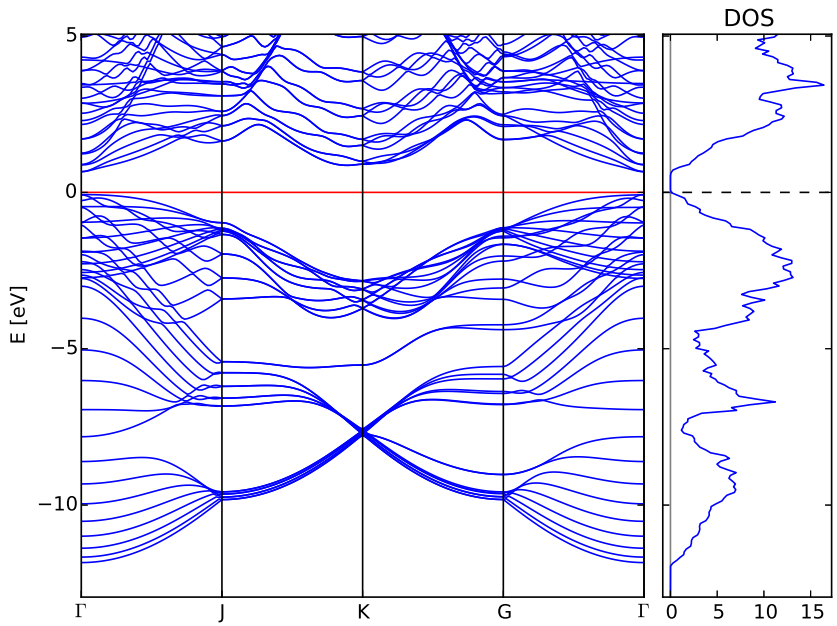


Latest results: $1 \times 1 \times 16$ atoms

Without H terminations.

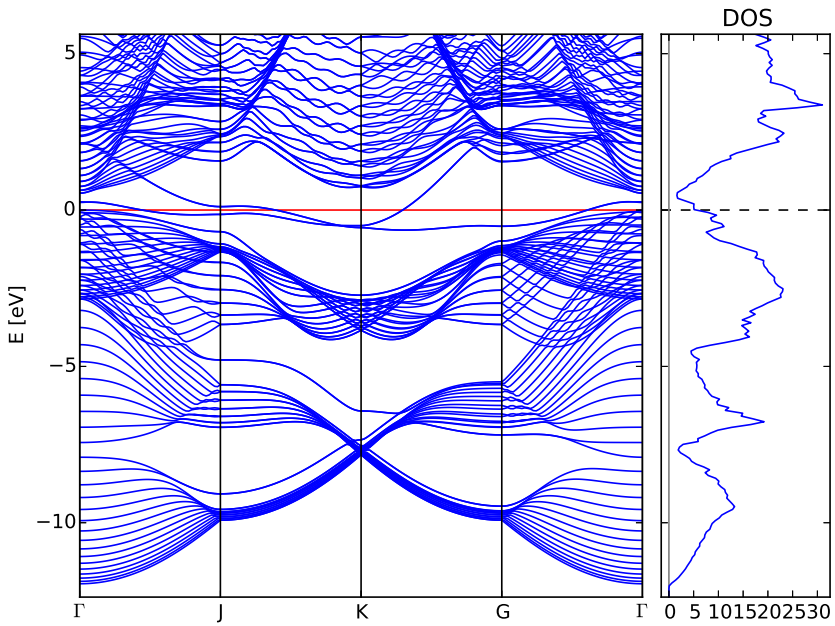


With H terminations.

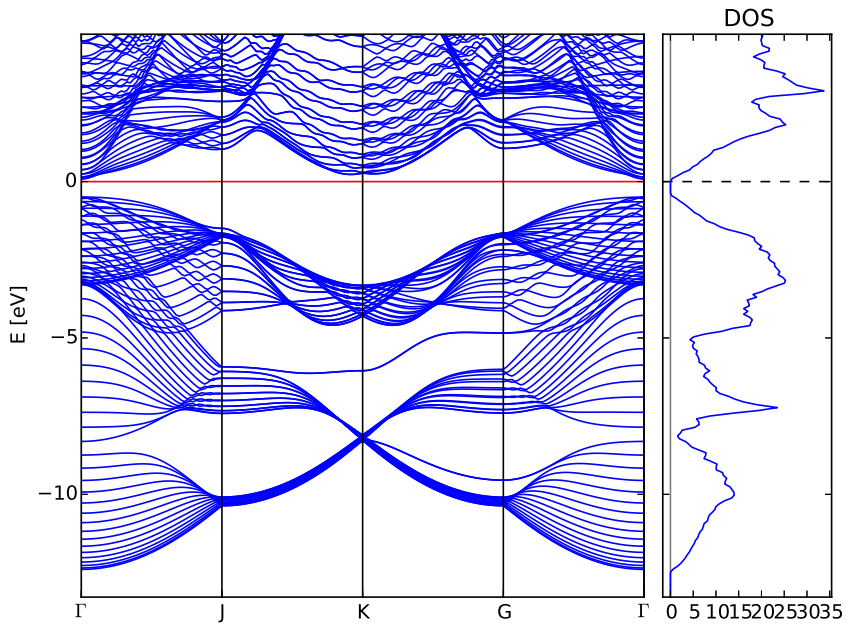


Latest results: $1 \times 1 \times 32$ atoms

Without H terminations.



With H terminations.



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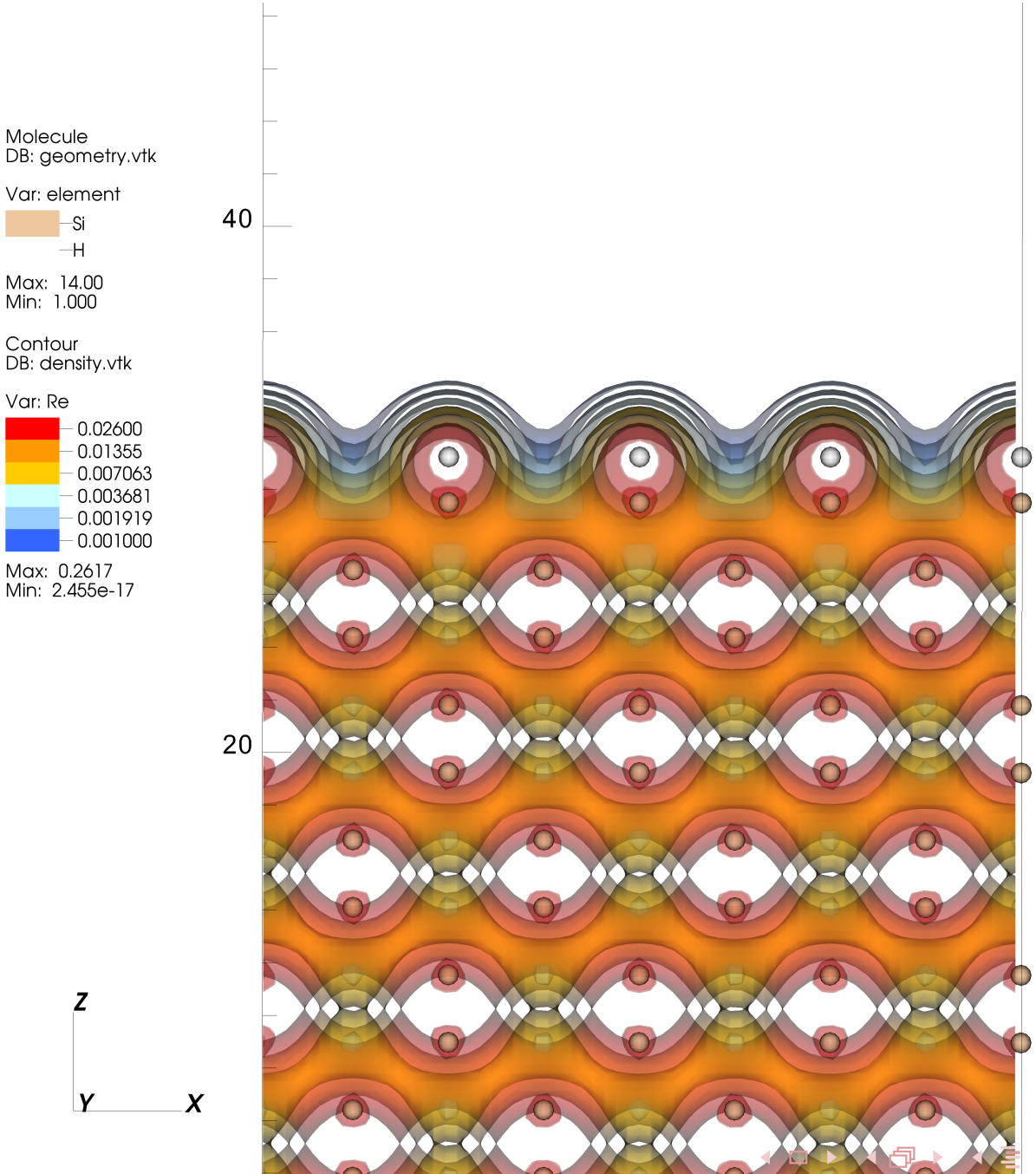
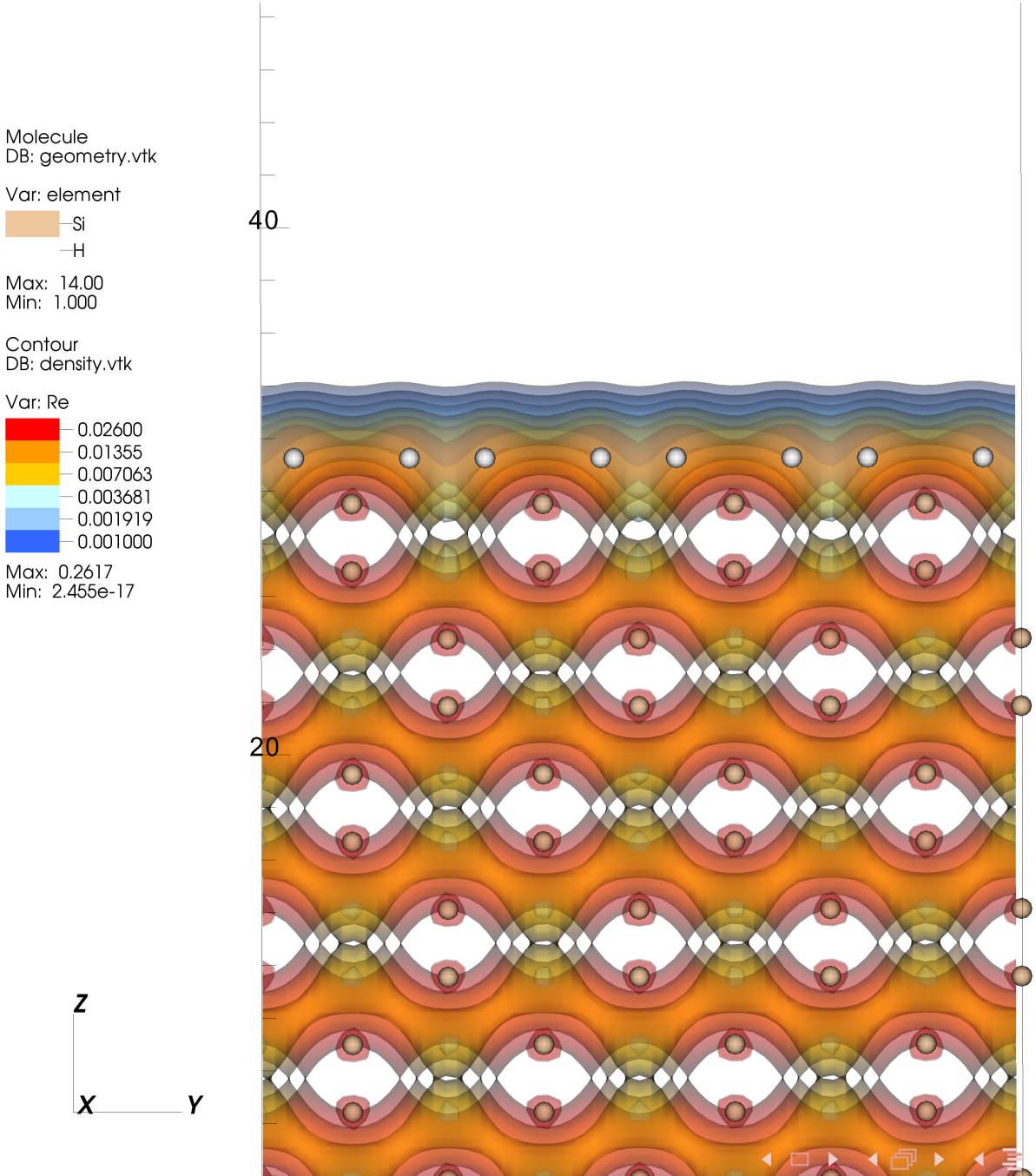
5. Conclusions

Ground-state density: $N_{GS}(xOz)$

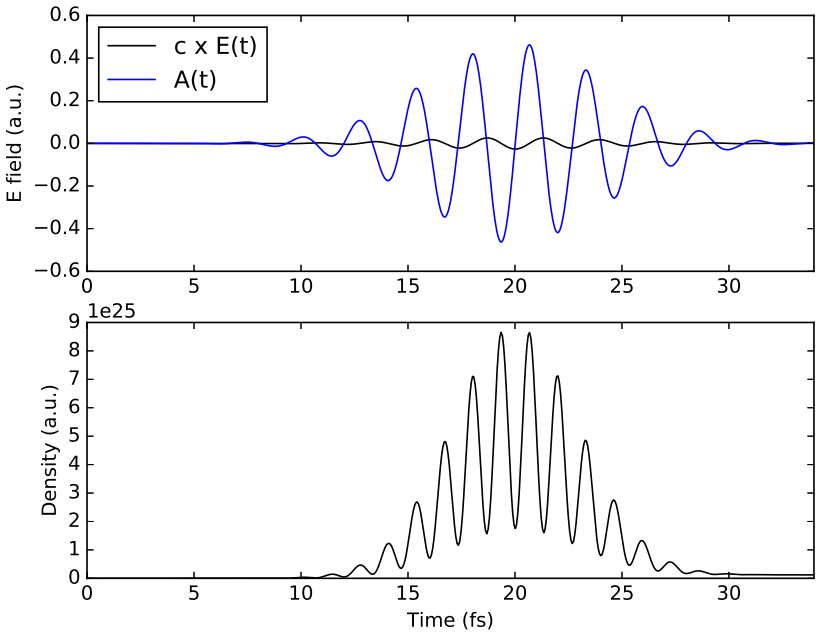


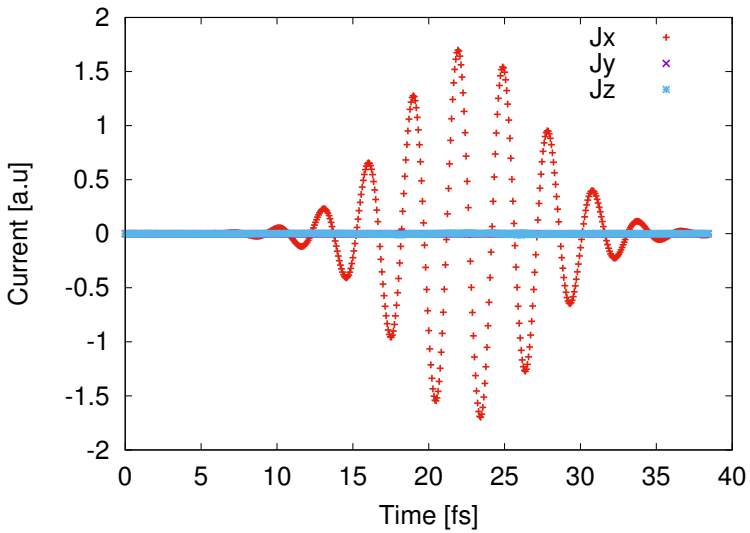
Ground-state density: $N_{GS}(yOz)$



Excited electrons N_{exc}

$\tau = 10$ fs, $\lambda = 800$ nm, $E = 1 \times 10^8$ V/m, $F \sim 0.013$ mJ/cm^2.



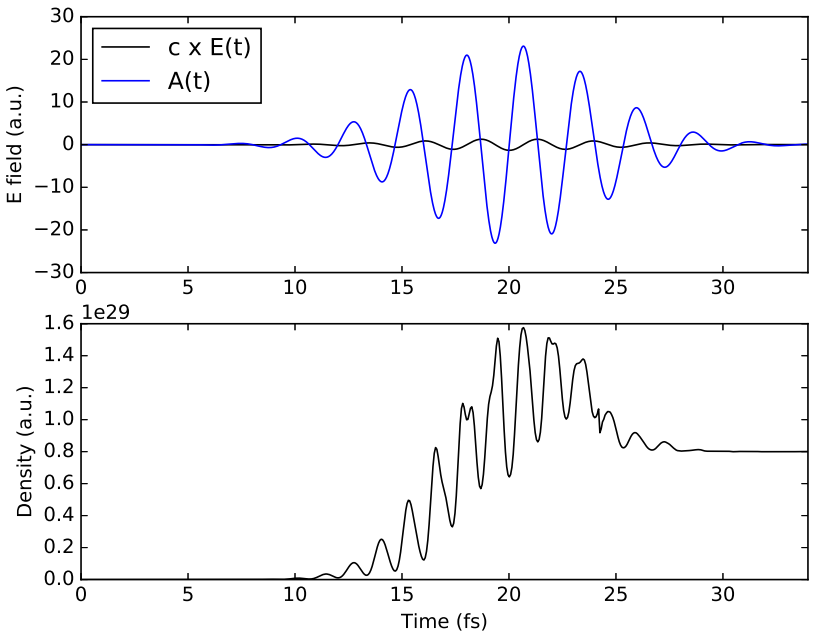


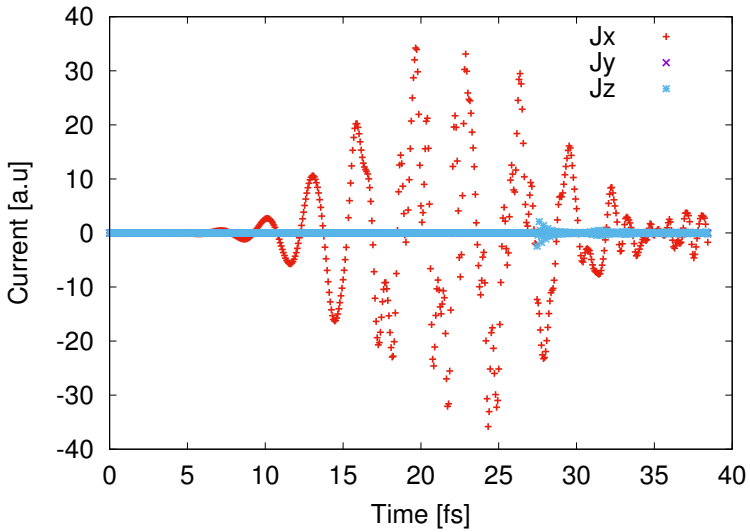
Interpretation

Current follows the electric field. Perturbative regime (then Drude model) are valid.

Excited electrons N_{exc}

$\tau = 10$ fs, $\lambda = 800$ nm, $E = 5 \times 10^9$ V/m $\sim F = 330$ mJ/cm^2 (close to damage threshold).



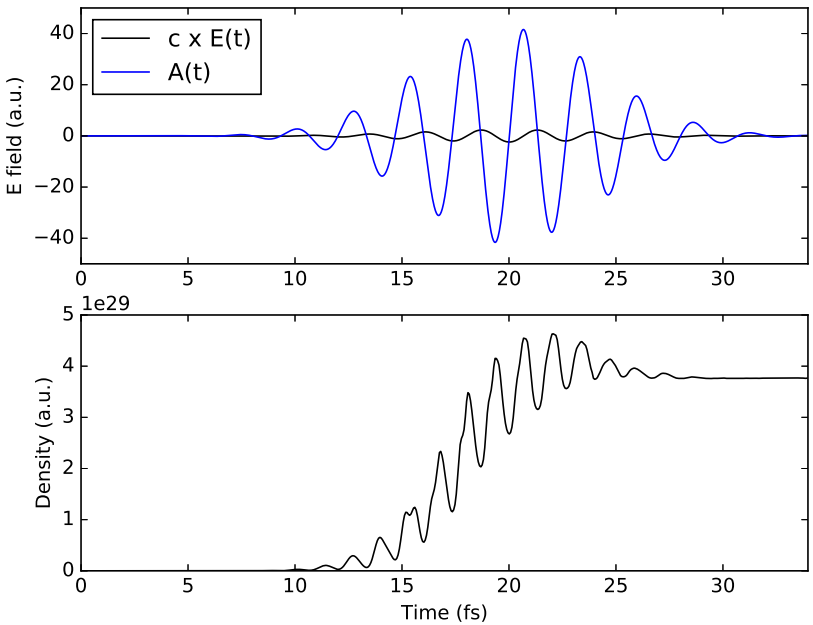


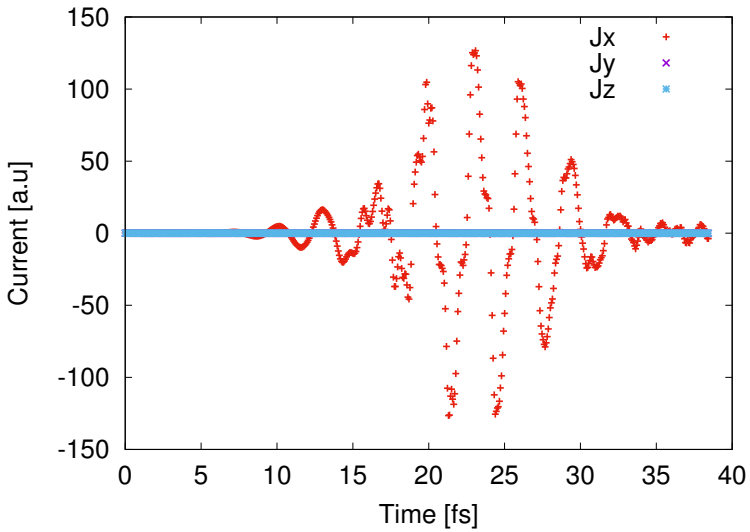
Observation

Generated current becomes highly non-linear. Perturbative approach and Drude model are no longer valid.

Excited electrons N_{exc}

$\tau = 10$ fs, $\lambda = 800$ nm, $E = 9 \times 10^9$ V/m, $F \sim 110$ mJ/cm^2.

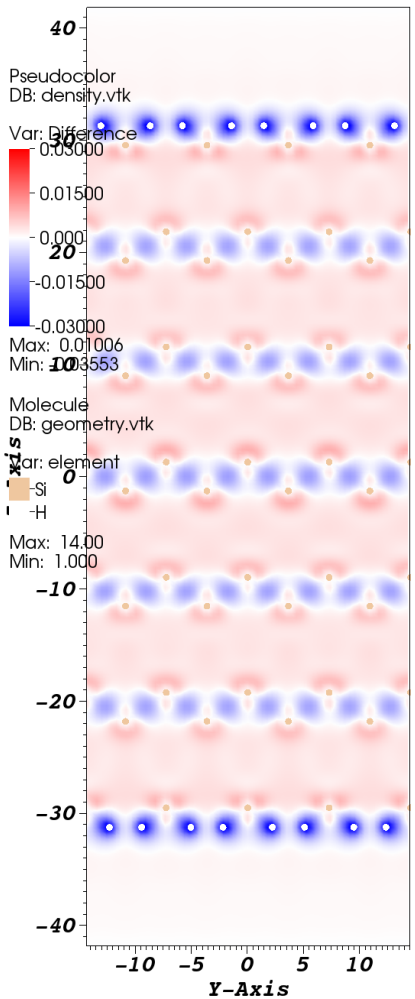


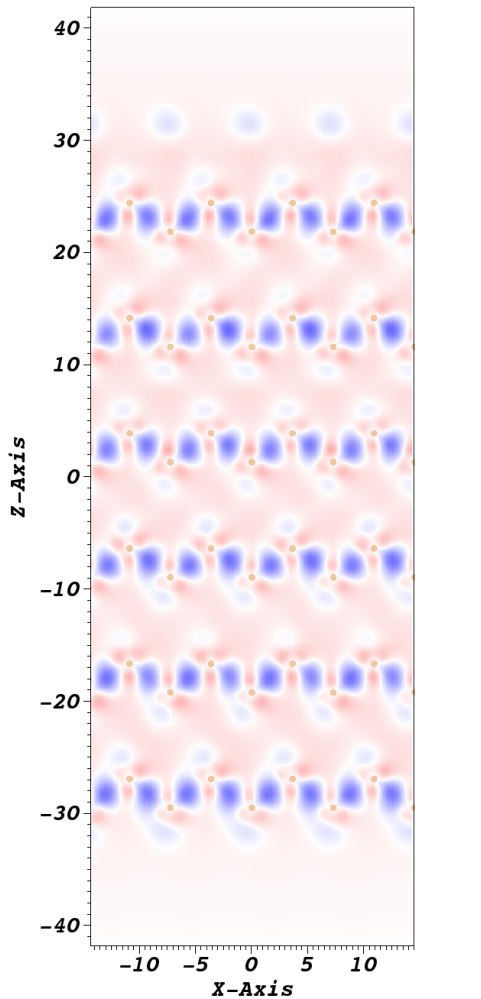


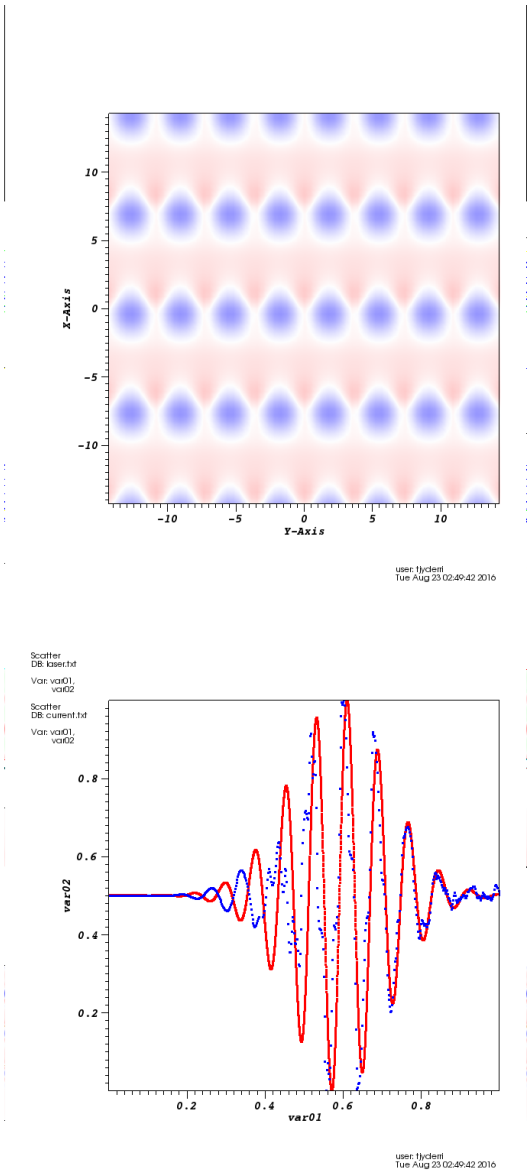
Observation

Peaks of A(t) correspond to maximum of excited electron density. Generated current is highly non-linear. Drude model not valid in this regime.

Electron density: $\Delta N_e = N_e(t) - N_{GS}$, $t = 23$ fs







Video

First observations

Striking points

Important	Using LDA, there is photo-emission: electrons are leaving the sample.
Observation	Local density is strongly affected by the laser oscillation.
Oscillation	Density appears to oscillate faster and faster $\Longrightarrow$high-harmonic generation ?

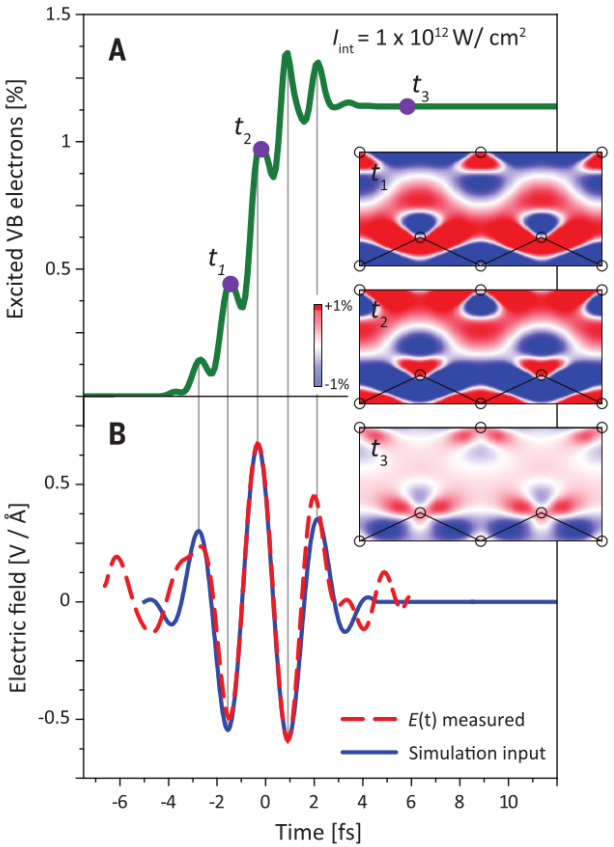
Notes

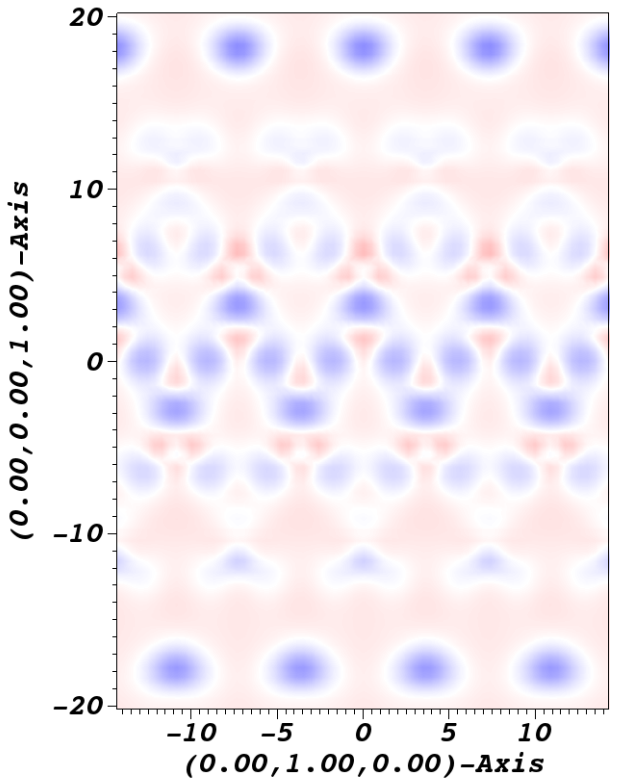
/!\	Sections are orthogonal, i.e., not in the plane of Si chemical bonds.
Observation	Depletion of electrons on the surface H: surface charging?
Observation	Electron response is aligned with the electric field.
Interpretation	Can we consider blue zones as holes, and red zones as free electrons?

Comparison with literature

Schultze, M.; Sato, S. A.; Yabana, K. et al,
Attosecond band-gap dynamics in silicon,
Science **346**, 1348-1352 (2014)

$F = 0.73\ \mathrm{J/cm^2}, \tau = 7\ \mathrm{fs}, \lambda = 800\ \mathrm{nm}.$



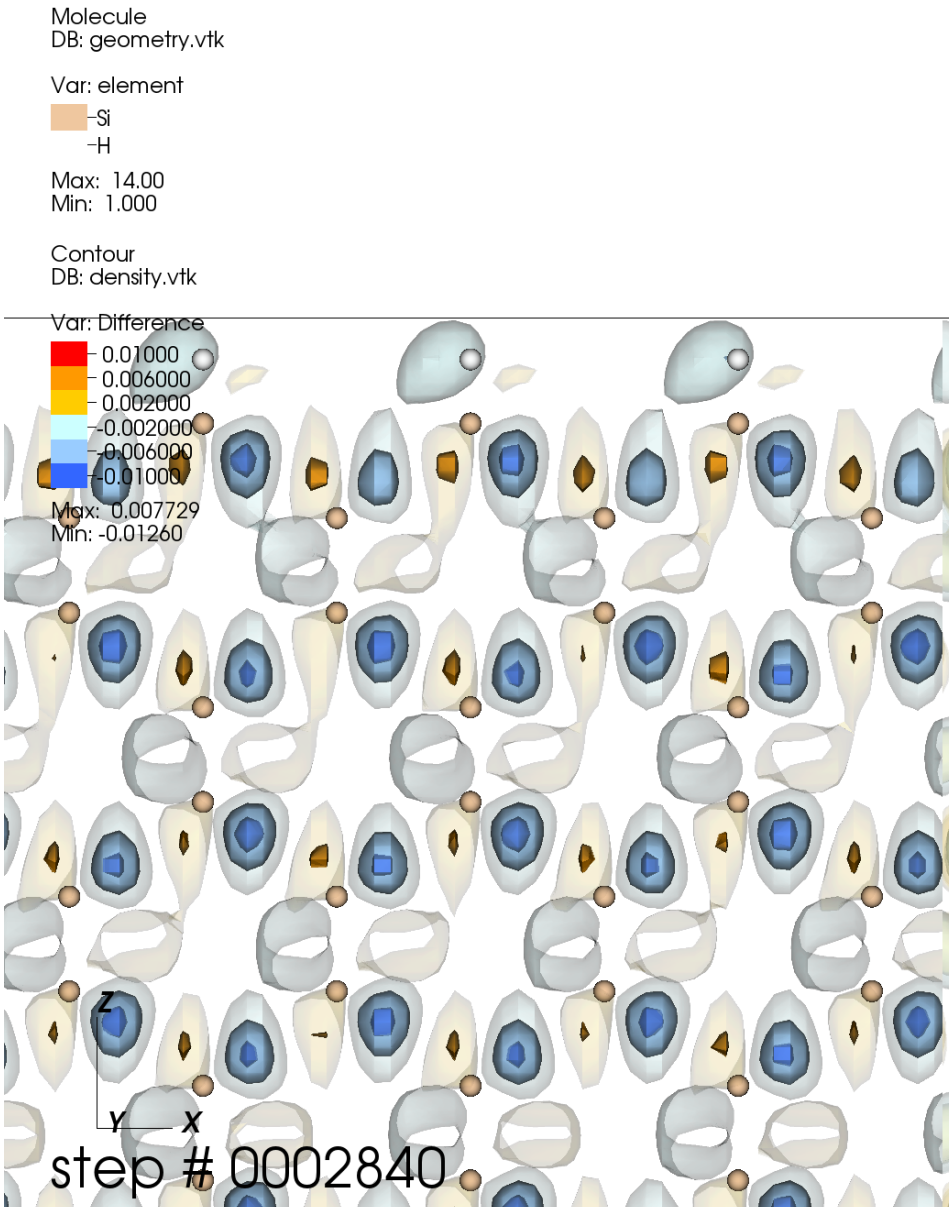


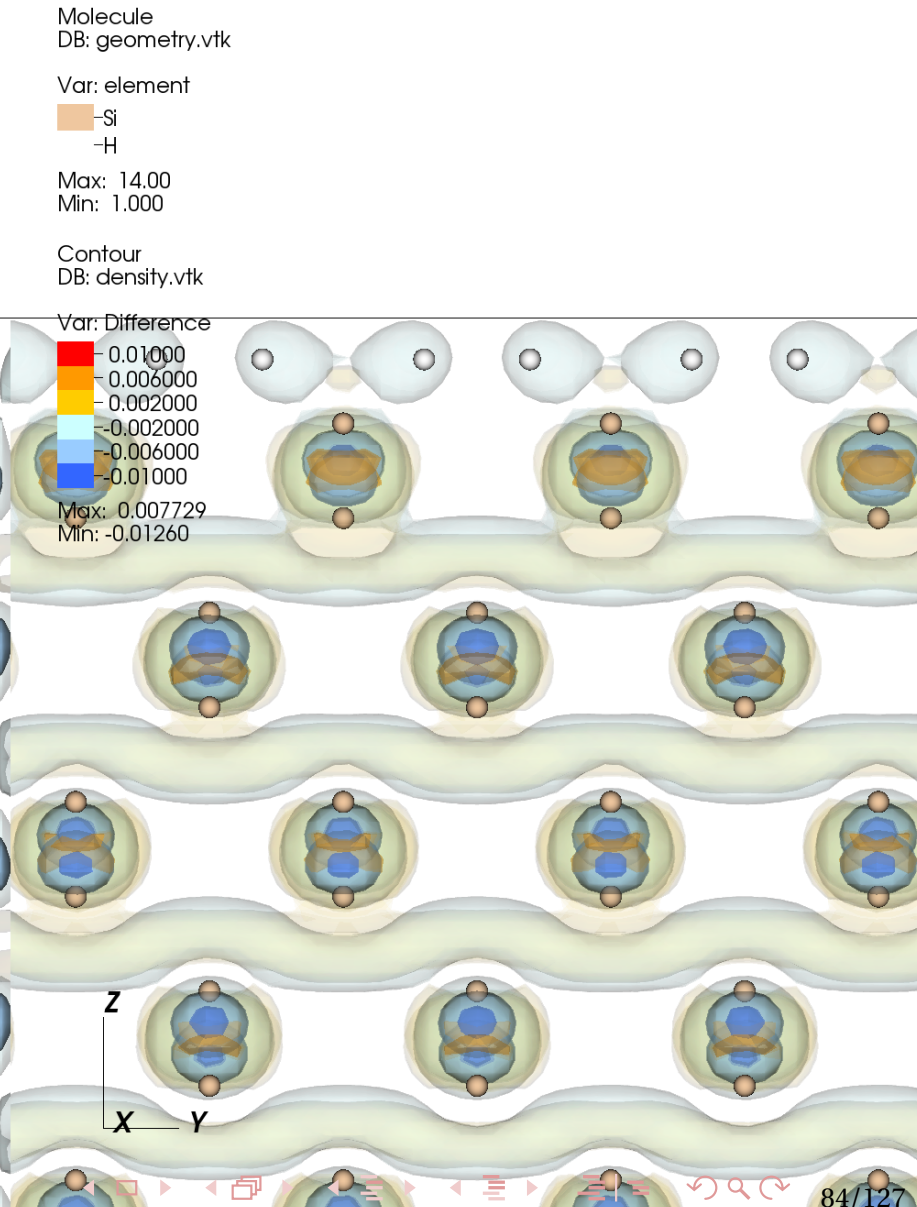
Video

To do

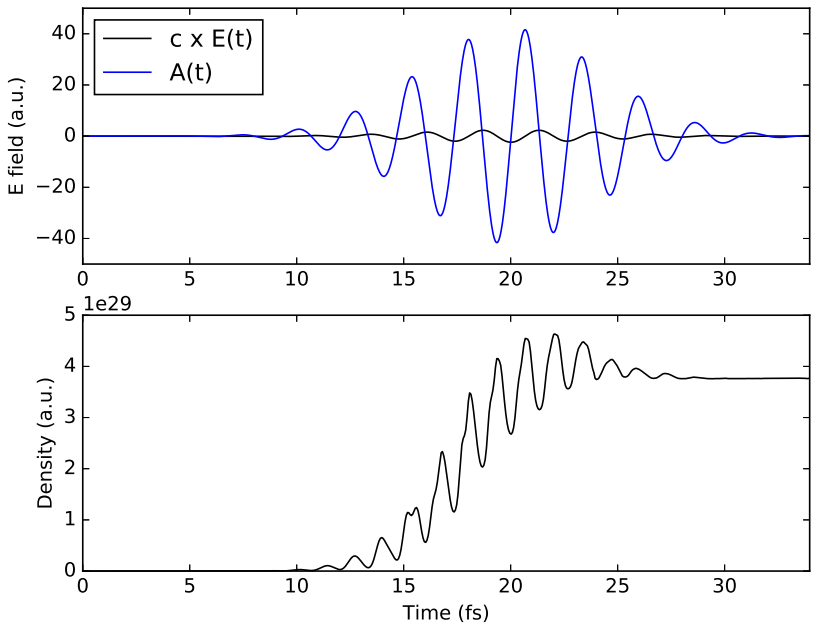
Still work remains to interpret the results.

3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 17.1$ fs

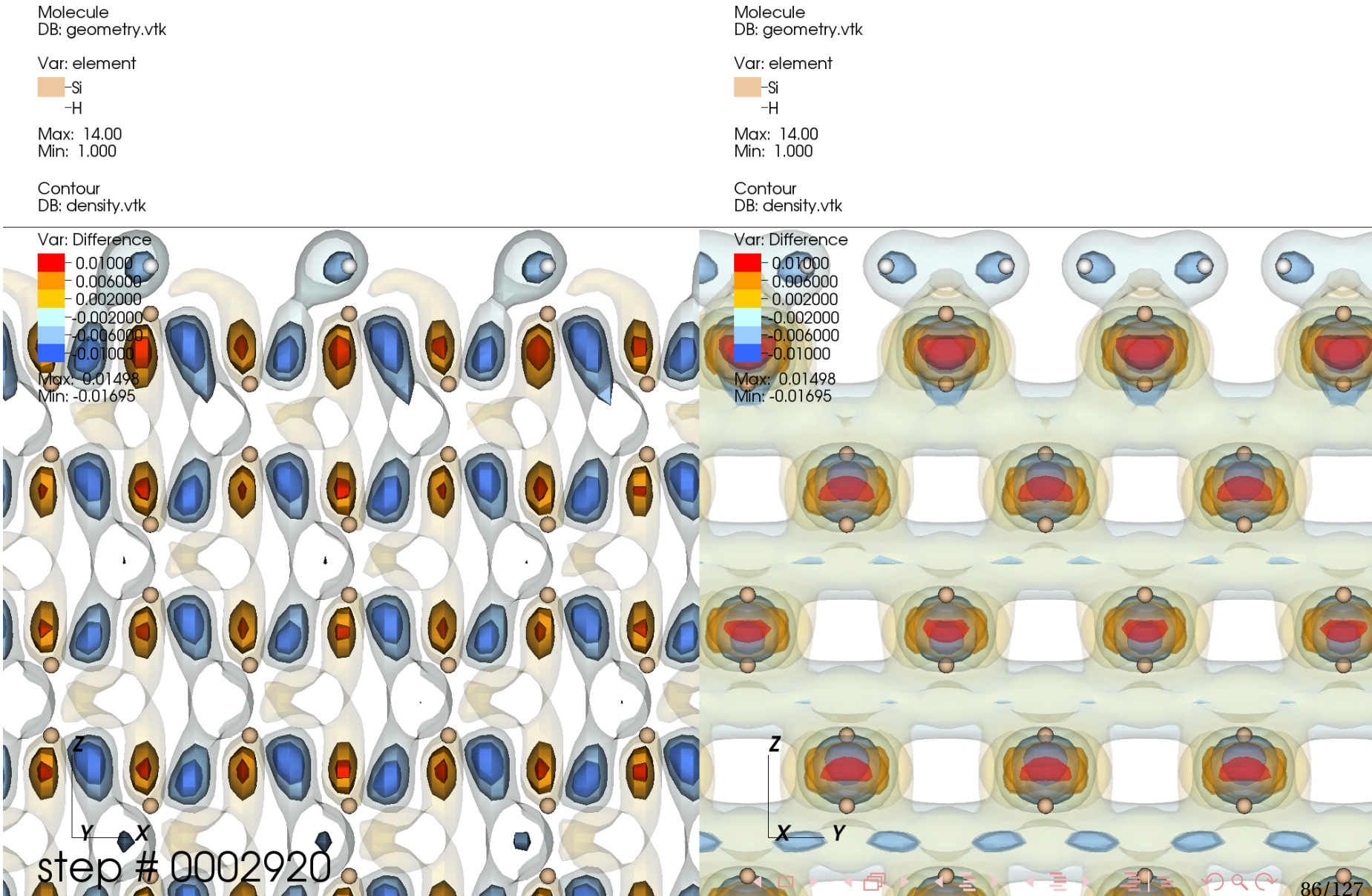




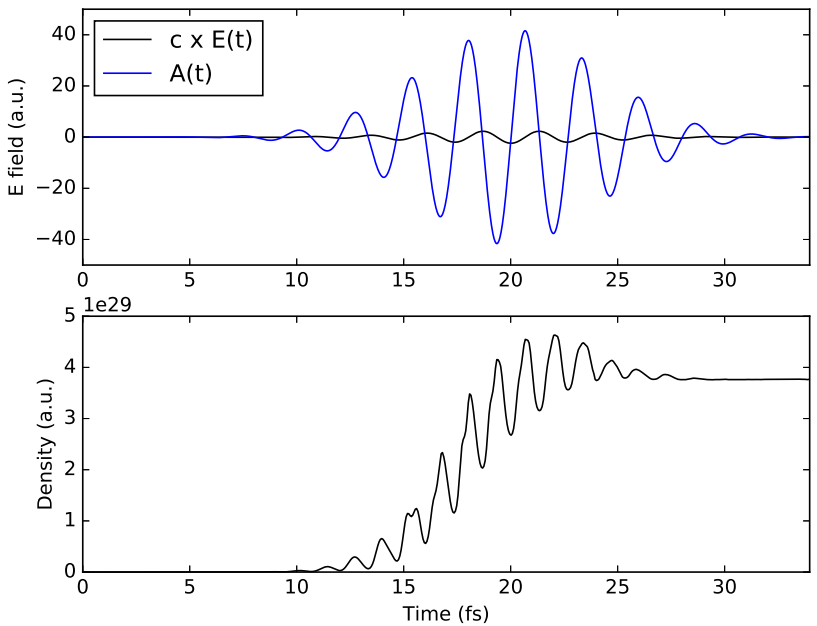
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 17.1$ fs



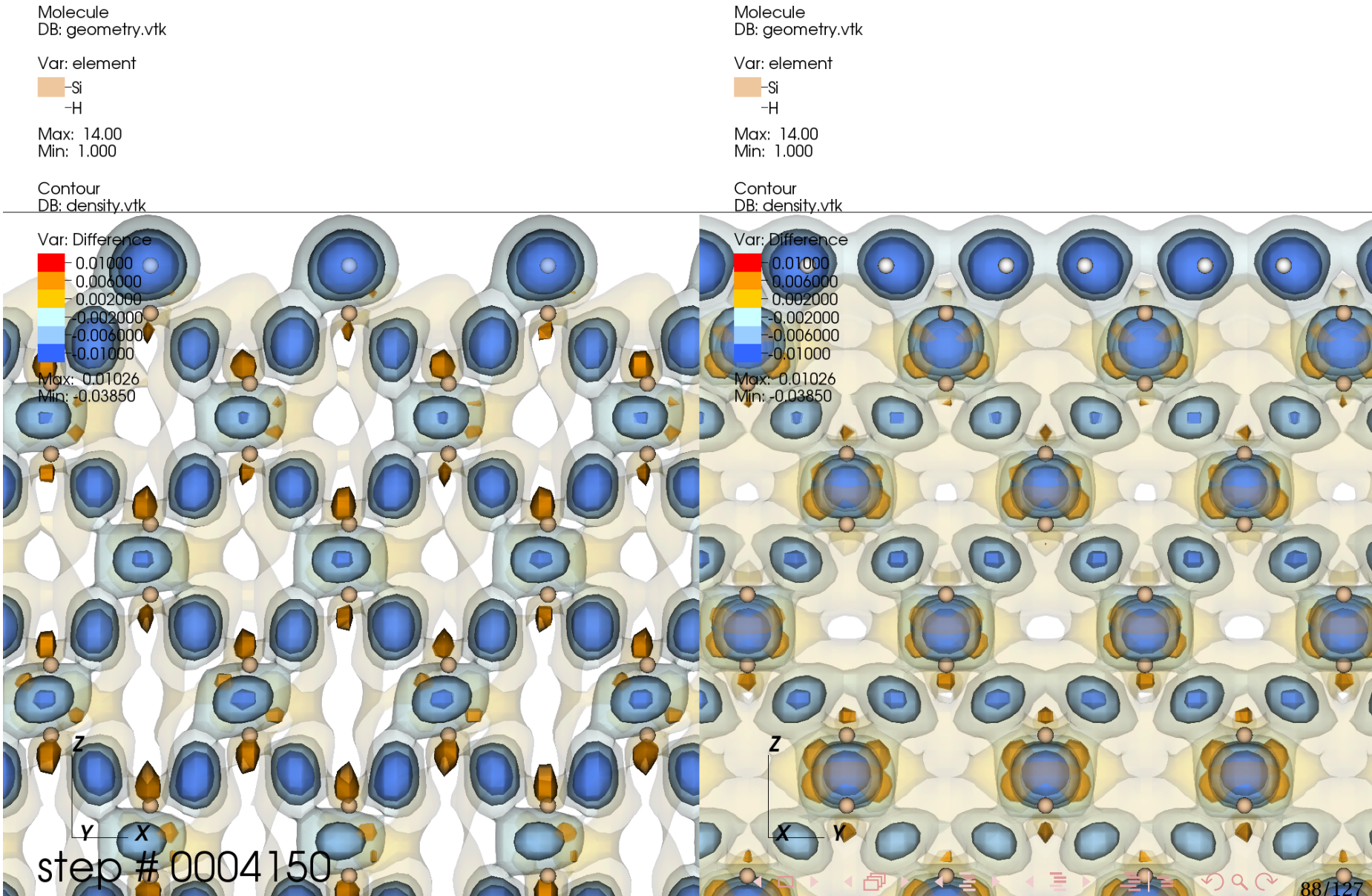
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 17.65$ fs



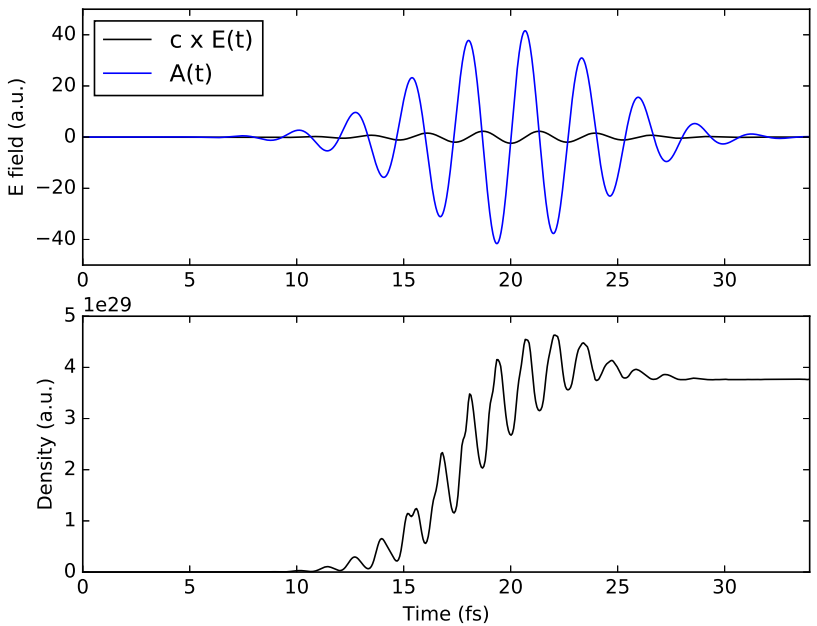
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 17.65$ fs



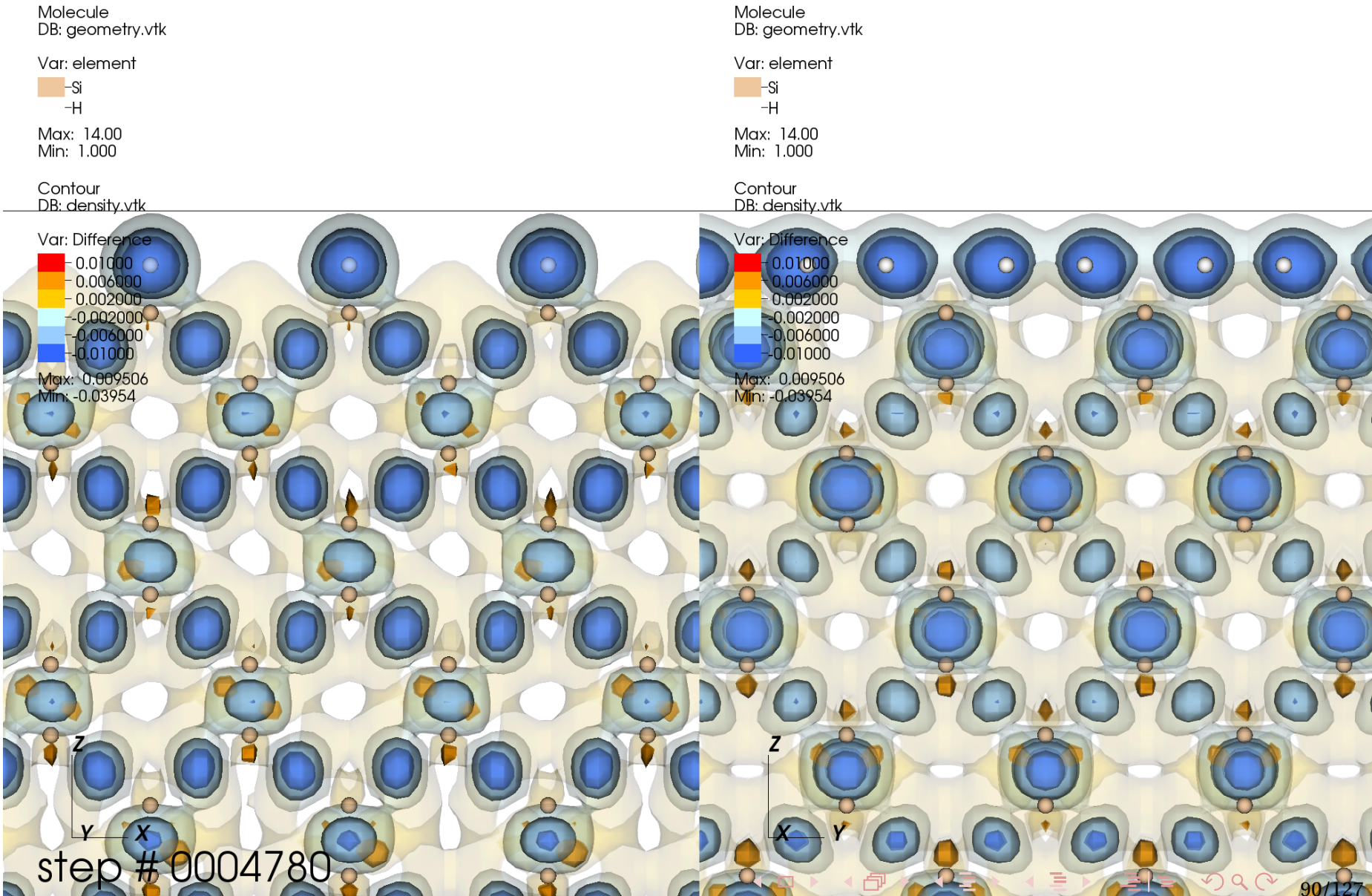
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 25.1$ fs



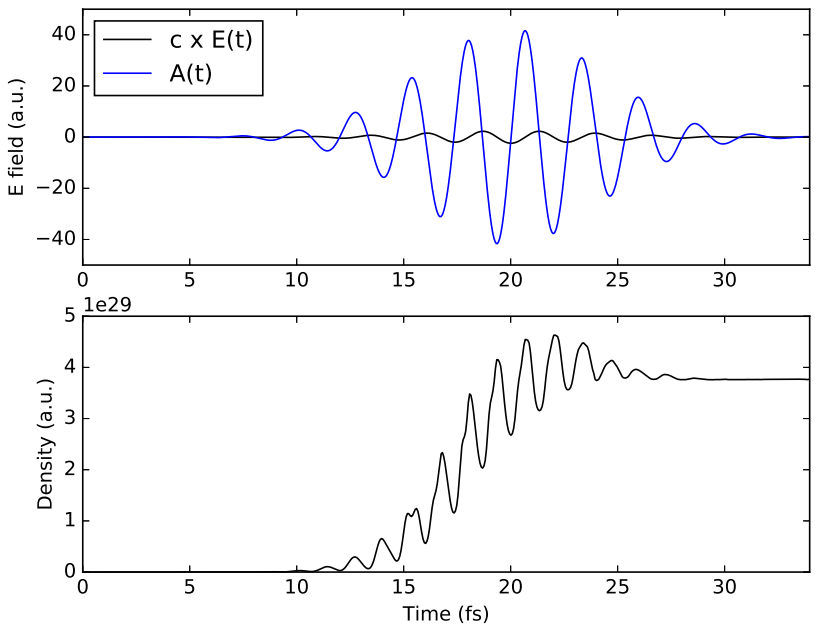
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 25.1$ fs



3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 28.9$ fs



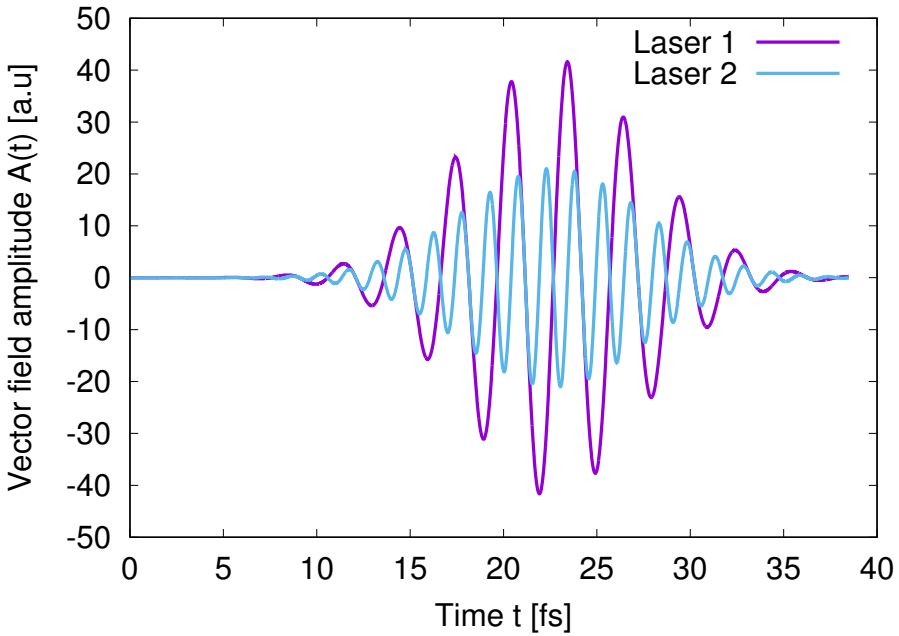
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 28.9$ fs



Outline

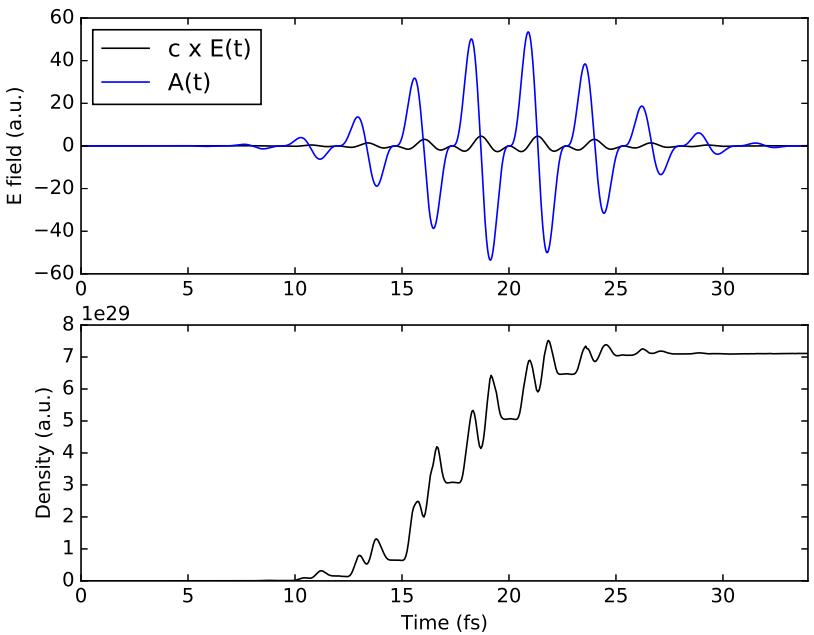
1. Context
   - Motivations
   - Open problems
   - How to address such problems?
   - Towards quantum-classical theory bridging
2. What to do with TD-DFT ?
   - What TD-DFT does not do.
   - Concept of TD-DFT
   - Current limits of TD-DFT
   - Calculation of linear absorption
   - Calculation of non-linear absorption
3. Preparation of TD-DFT experiments
   - Overview
   - Technical issues
   - Construction of the laser pulses
   - Construction of the Si samples
4. Some results
   - Band gap structure
   - Single wavelength simulations
   - Commensurable wavelength: $\lambda_1 = 800$ nm, $\lambda_2 = 400$ nm
   - Incommensurable wavelengths: $\lambda_1 = 1030$ nm, $\lambda_2 = 800$ nm
5. Conclusions

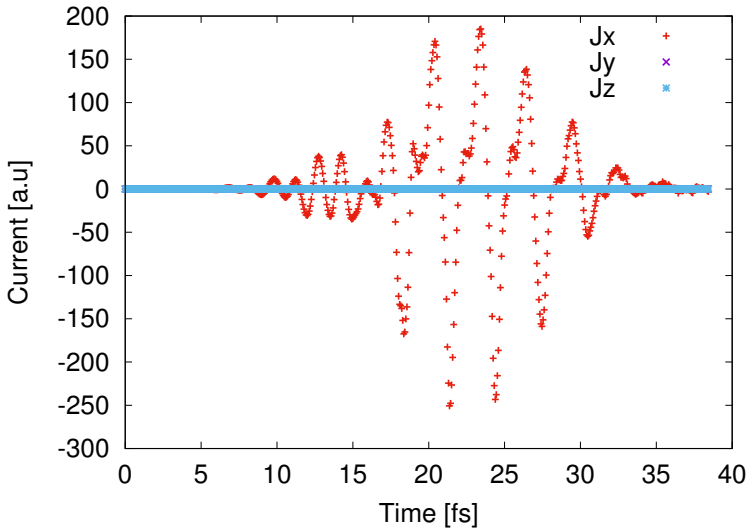
Laser field


Laser 1
Laser 2
Vector field amplitude A(t) [a.u]
Time t [fs]
50
40
30
20
10
0
-10
-20
-30
-40
-50
0
5
10
15
20
25
30
35
40

Excited electron density N_{ex}

- $\lambda_1 = 800\,\text{nm}, \tau = 10\,\text{fs}, E = 9 \times 10^9\,\text{V/m}, F \sim 110\,\text{mJ/cm}^2$.
- $\lambda_2 = 400\,\text{nm}, \tau = 10\,\text{fs}, E = 9 \times 10^9\,\text{V/m}, F \sim 110\,\text{mJ/cm}^2$.

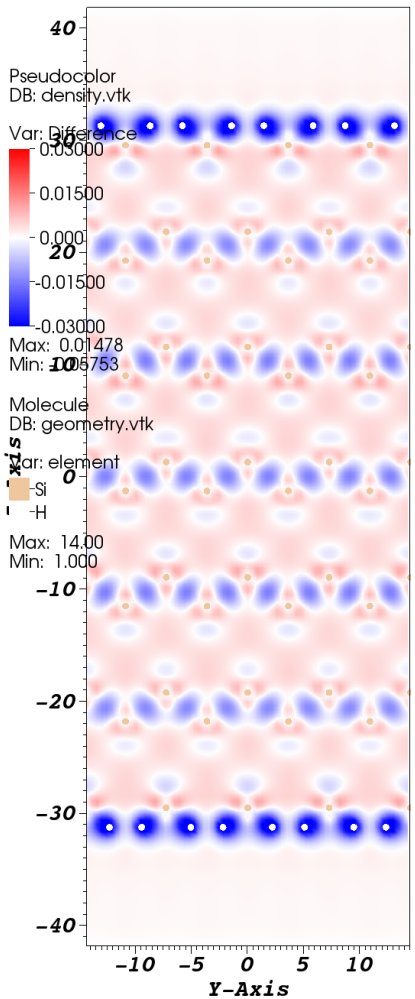


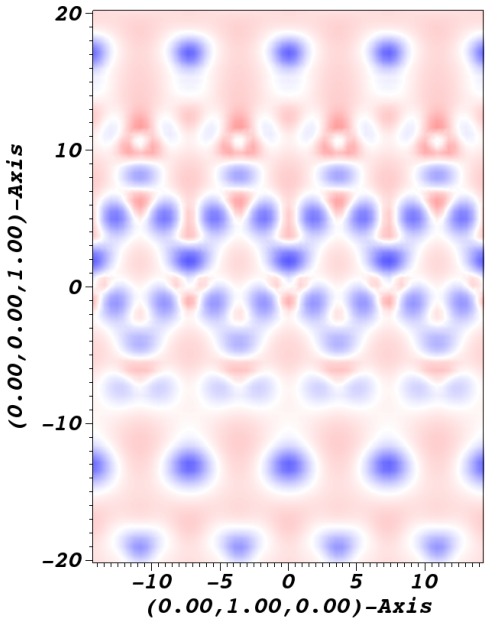


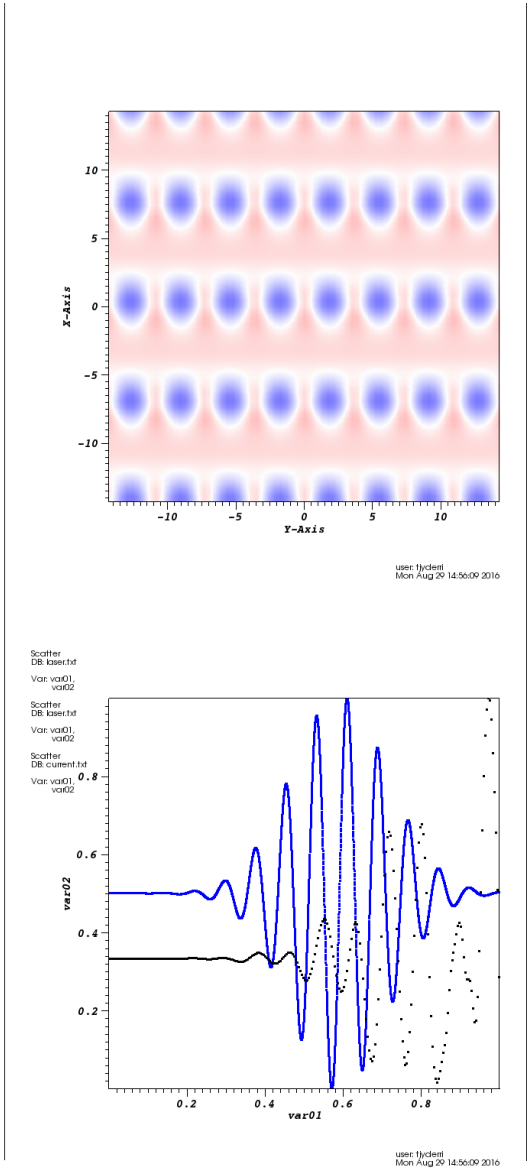
Observations

Generated current is now asymmetric (?)
Carrier excitation still follows the vector potential A(t).

Electron density: $\Delta N_e = N_e(t) - N_{GS}$

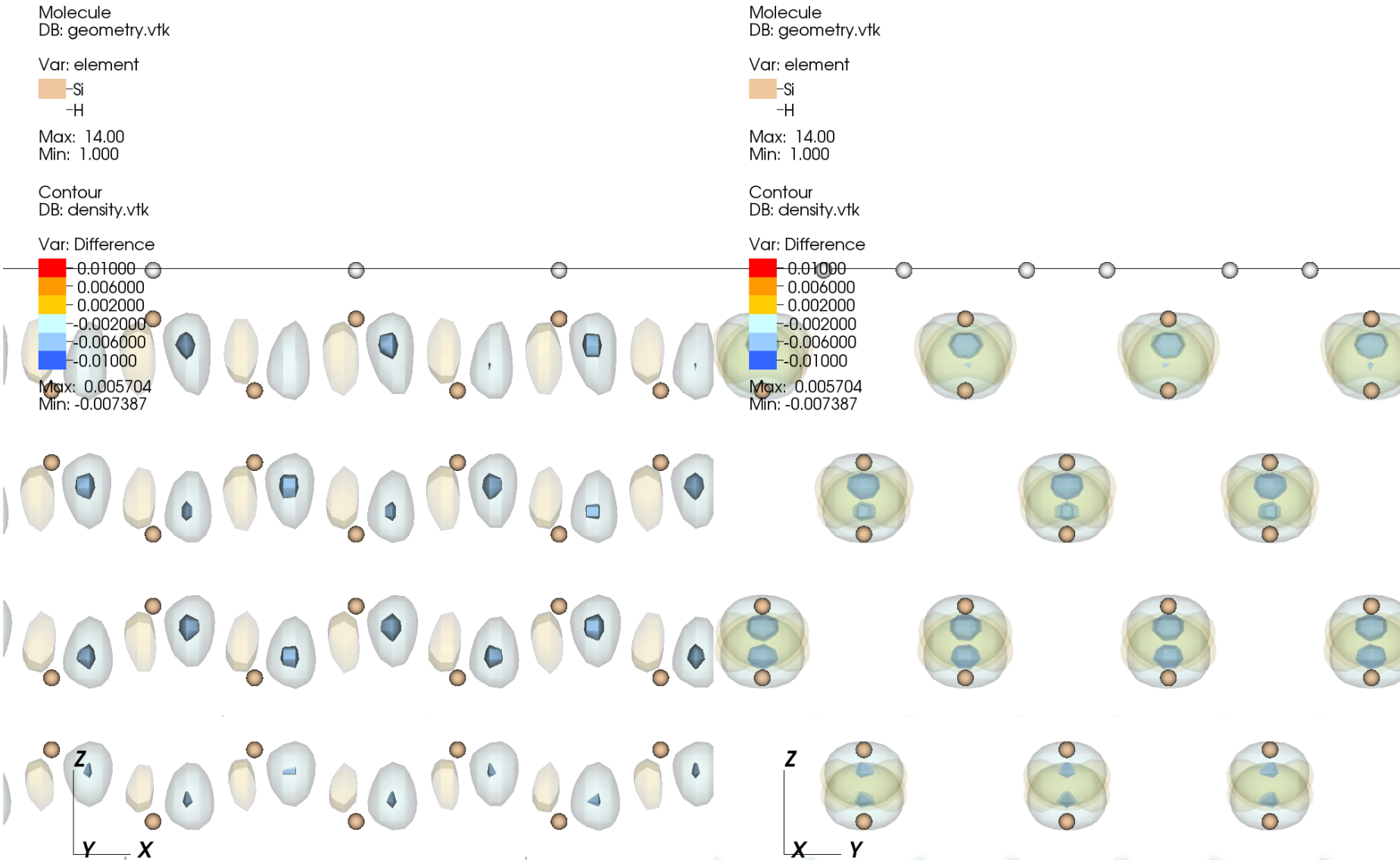




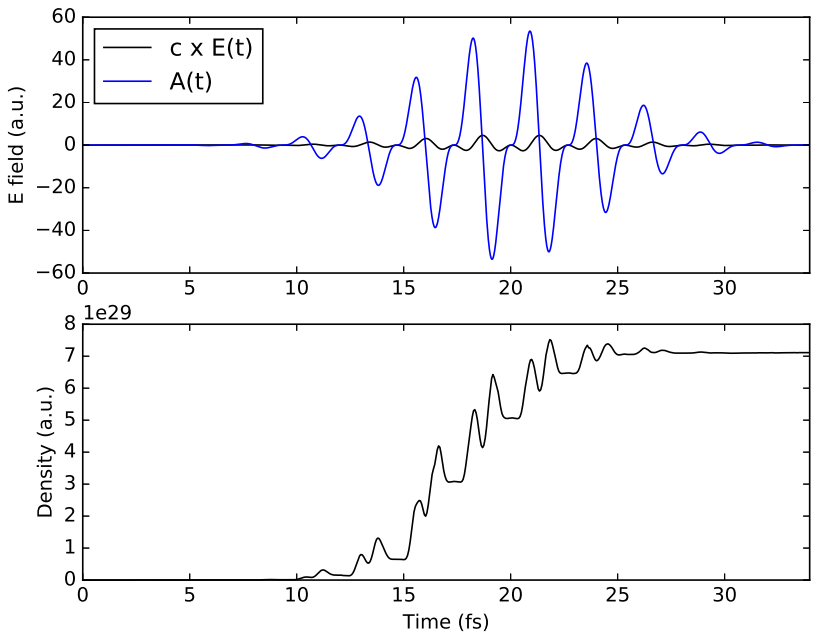


Video

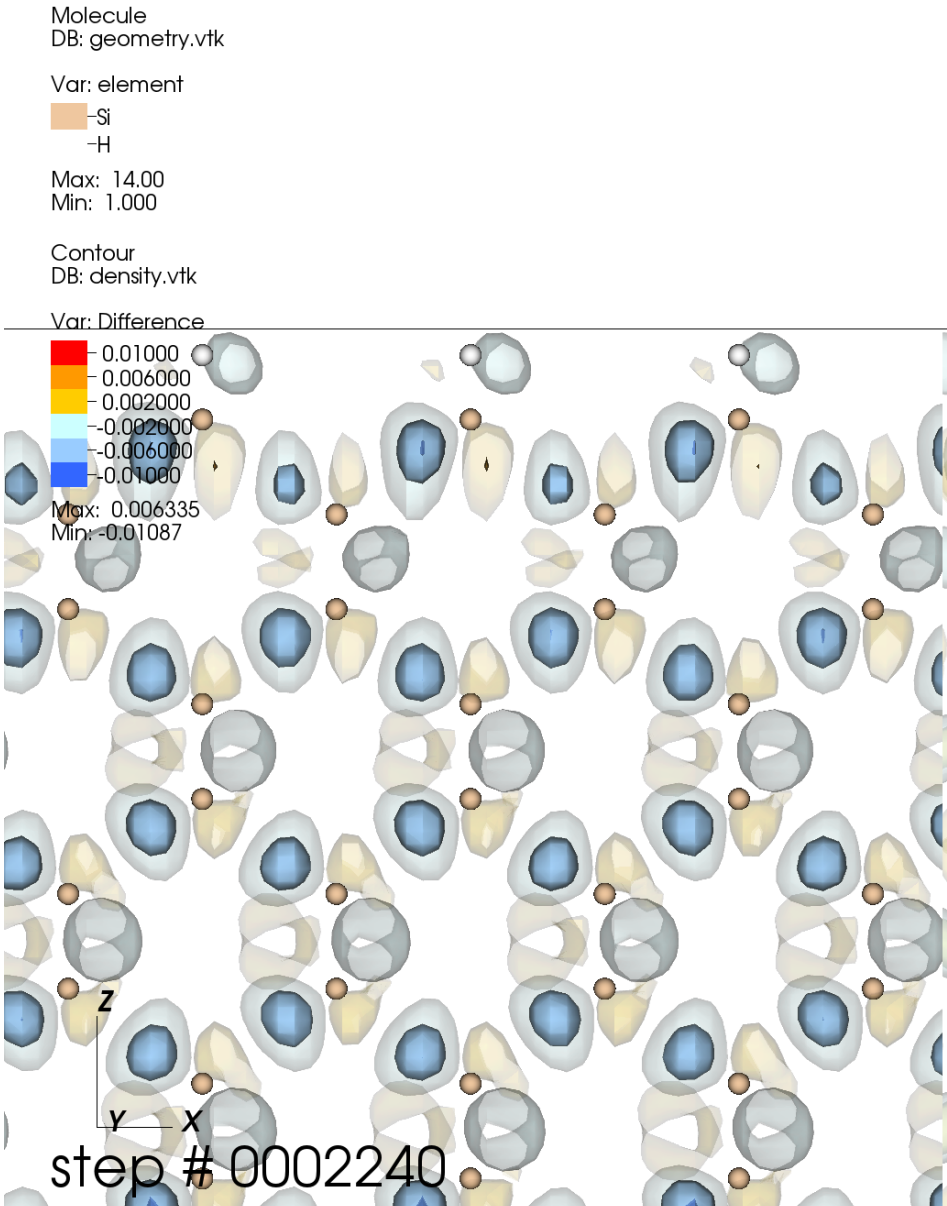
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 12.8$ fs

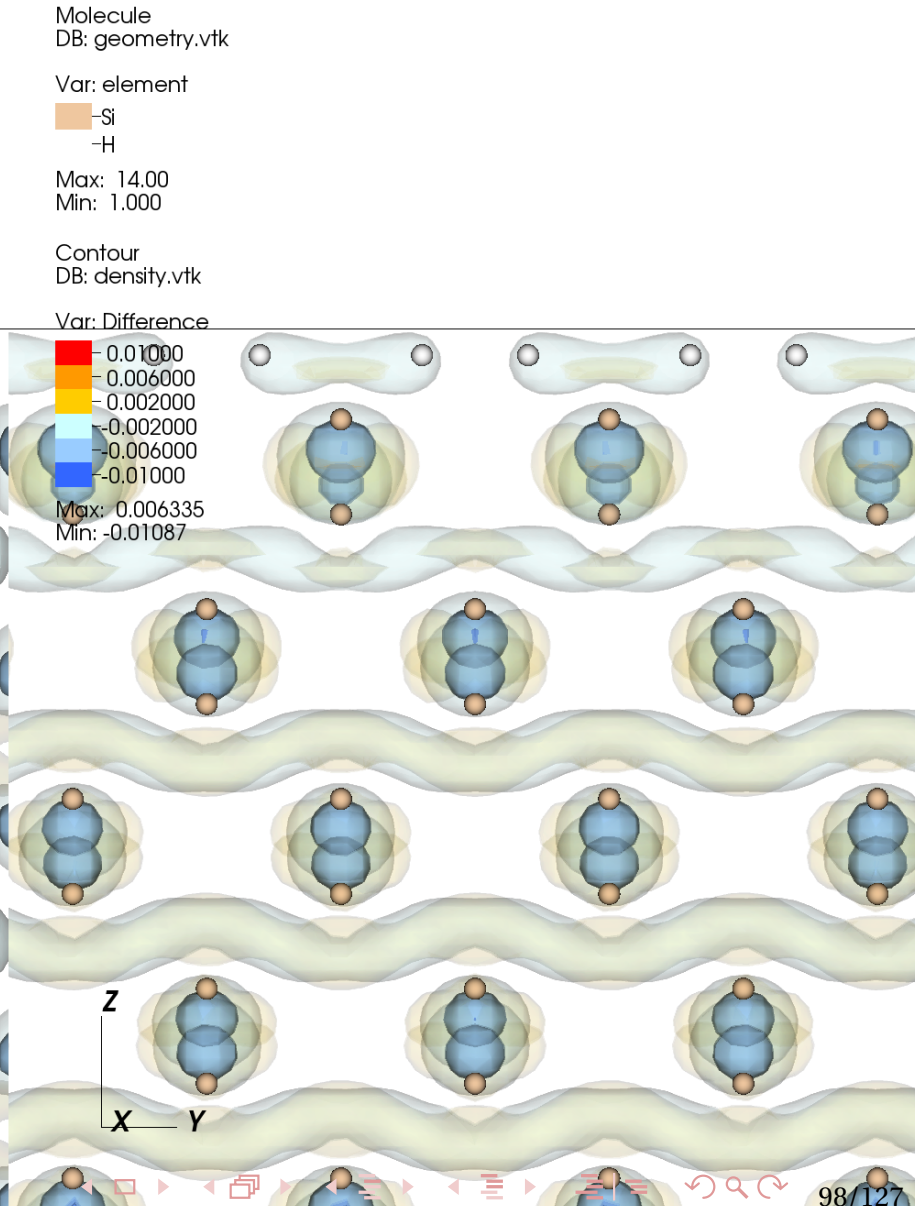


3d: $\Delta N_e(t) = N_e(t) - N_{GS}, t = 12.8$ fs

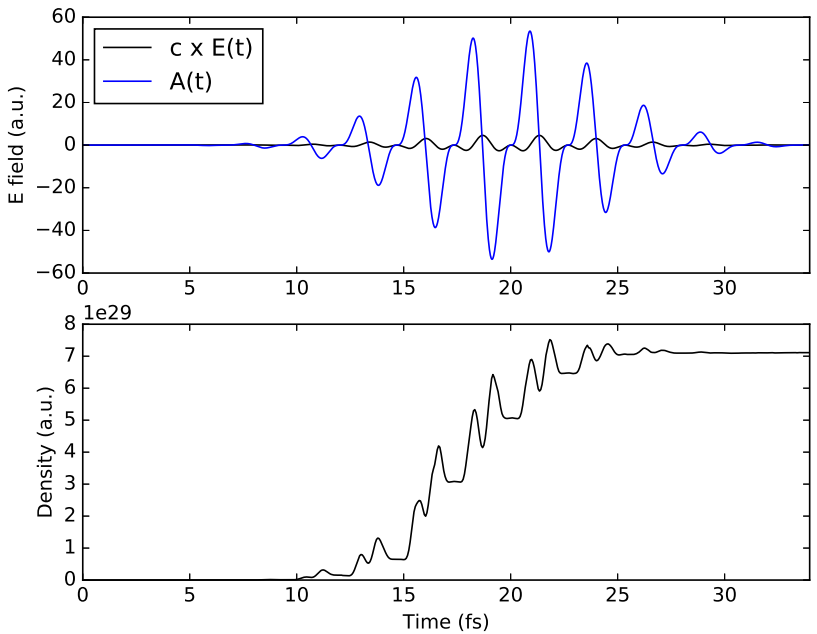


3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 13.5$ fs

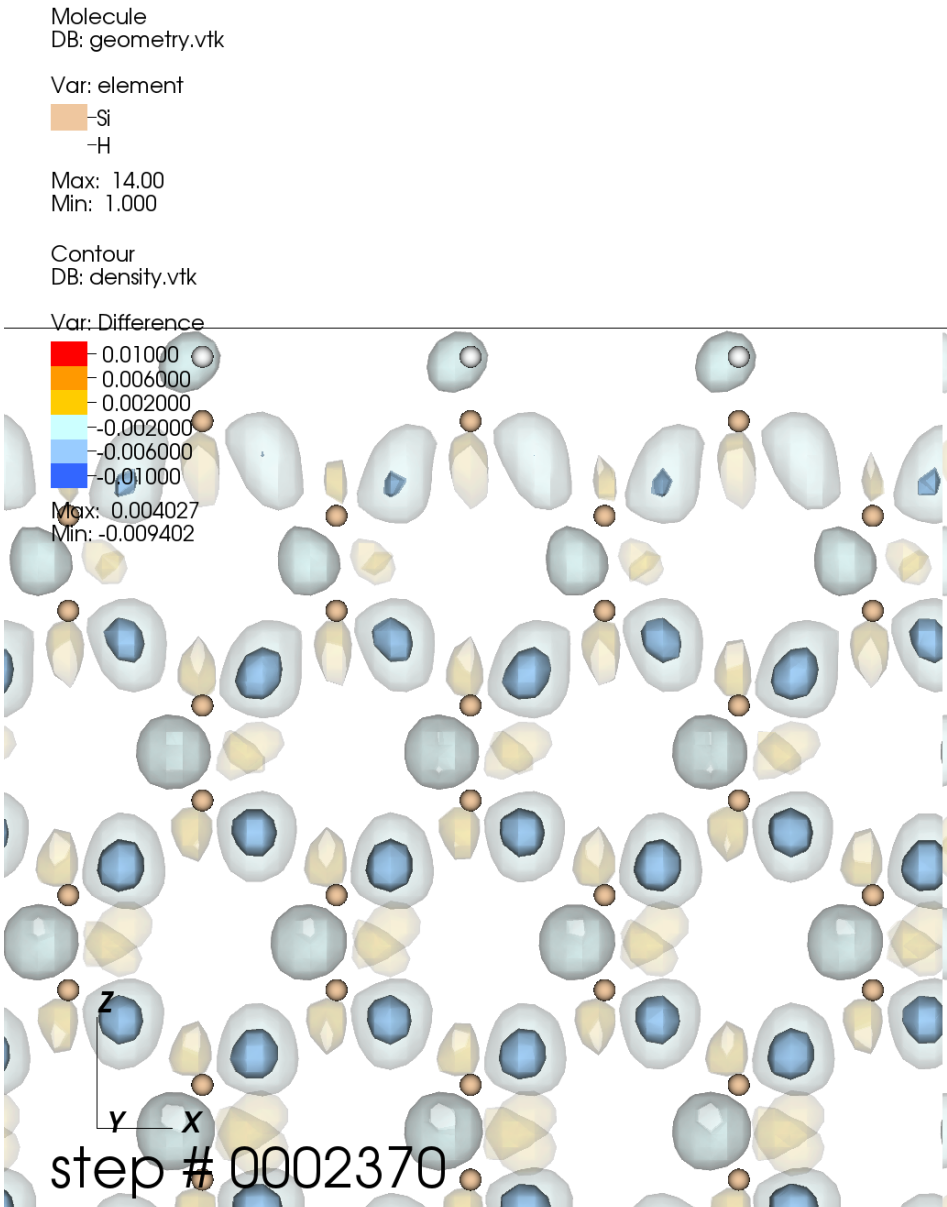


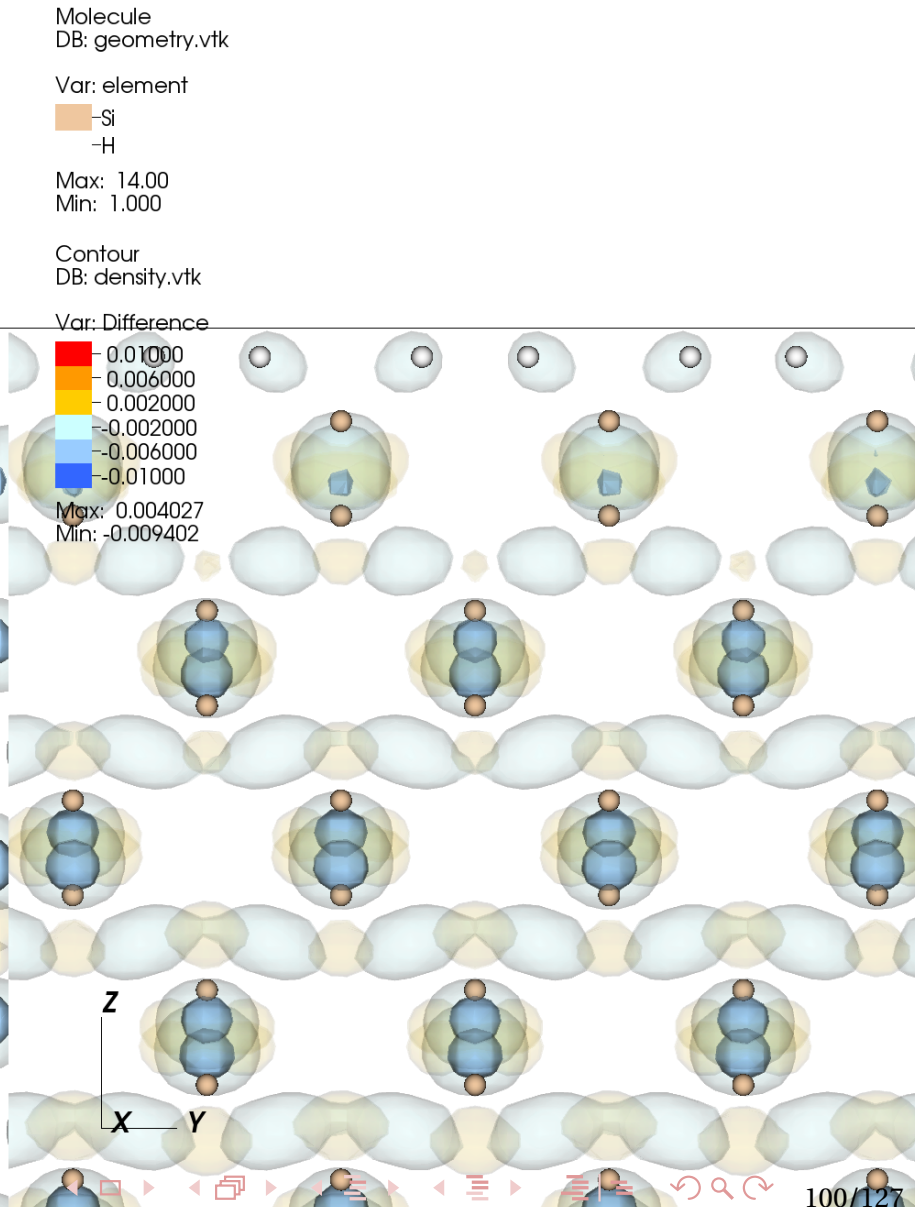


3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 13.5$ fs

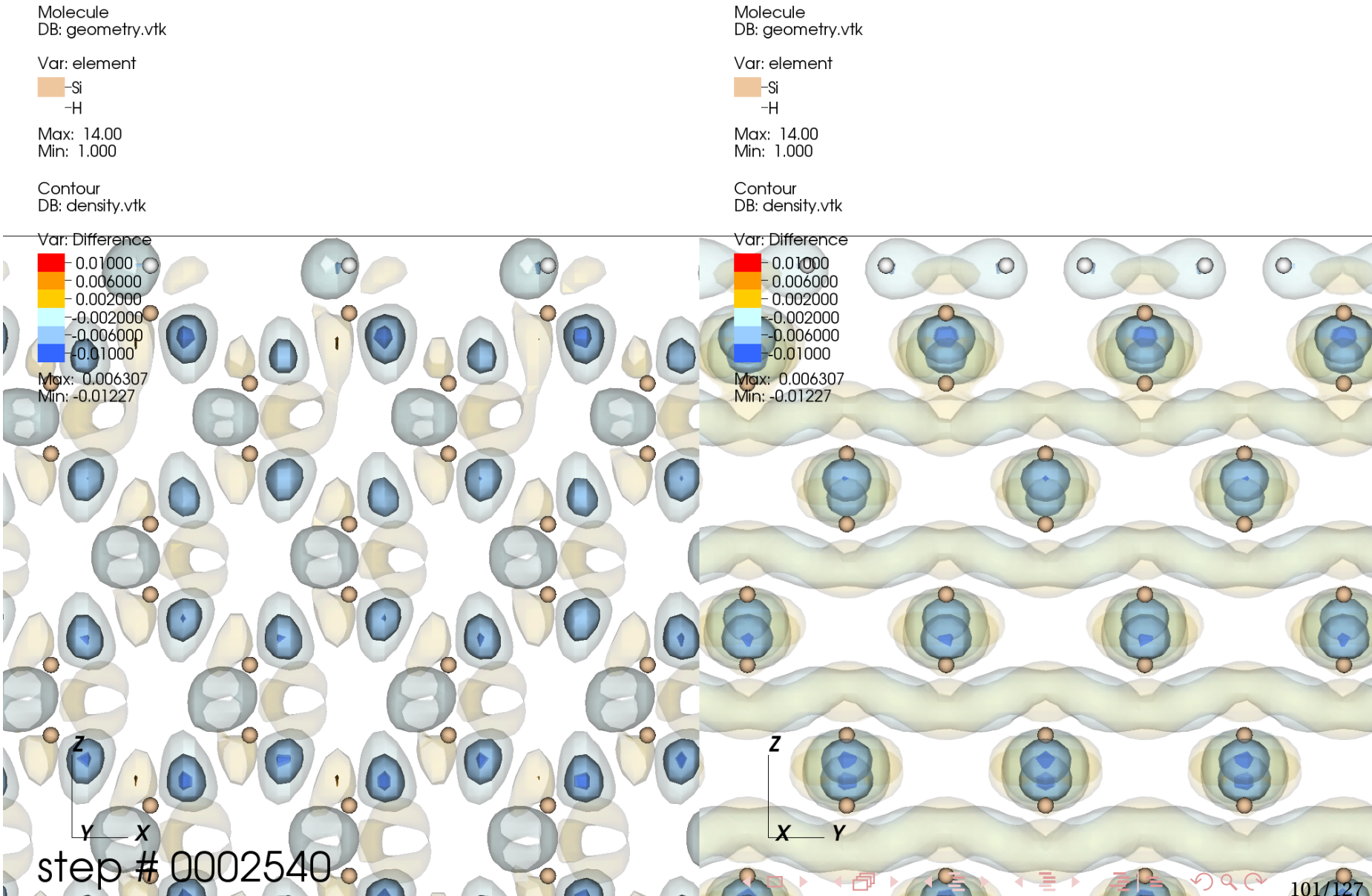


3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 14.3$ fs

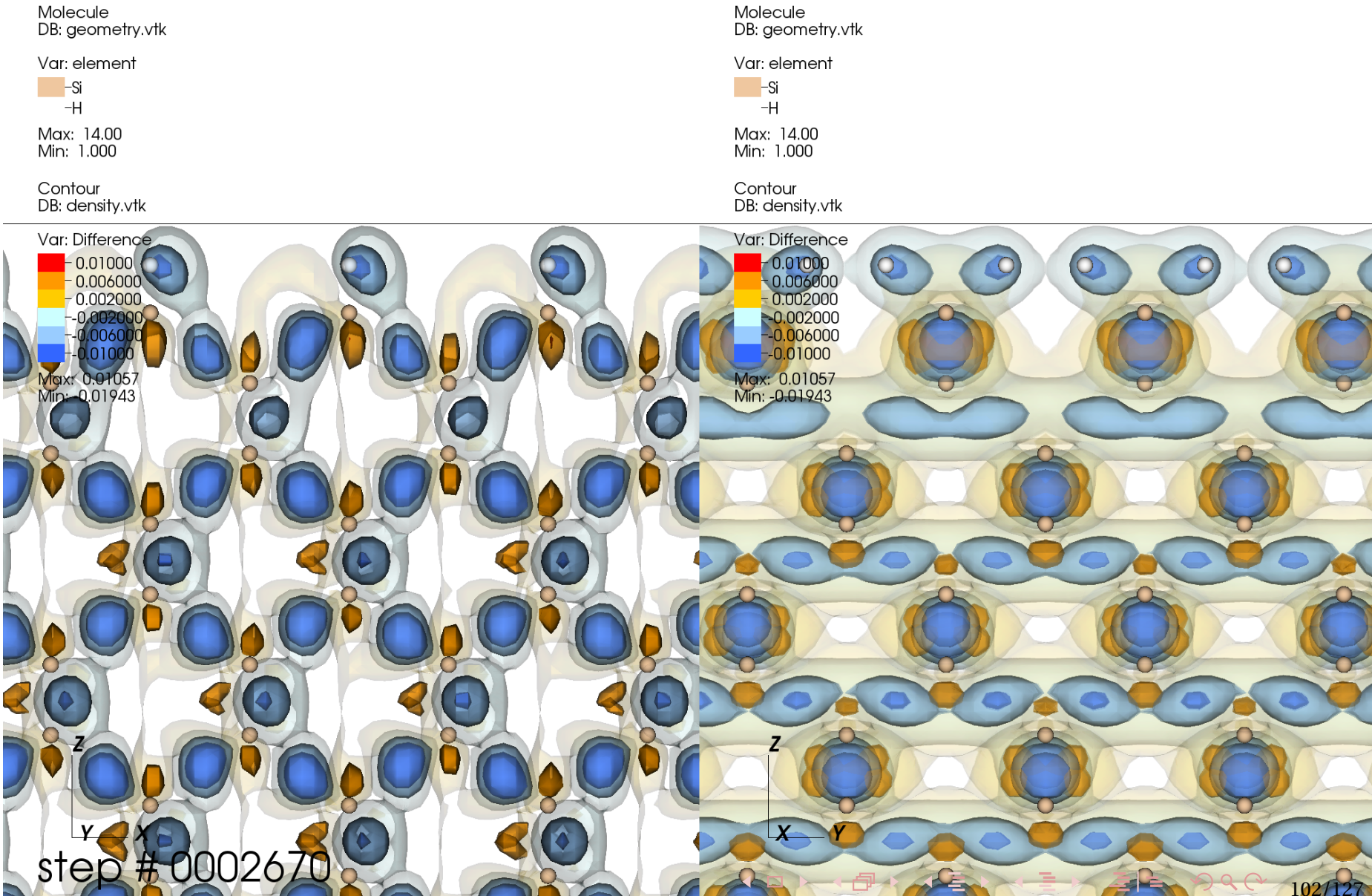




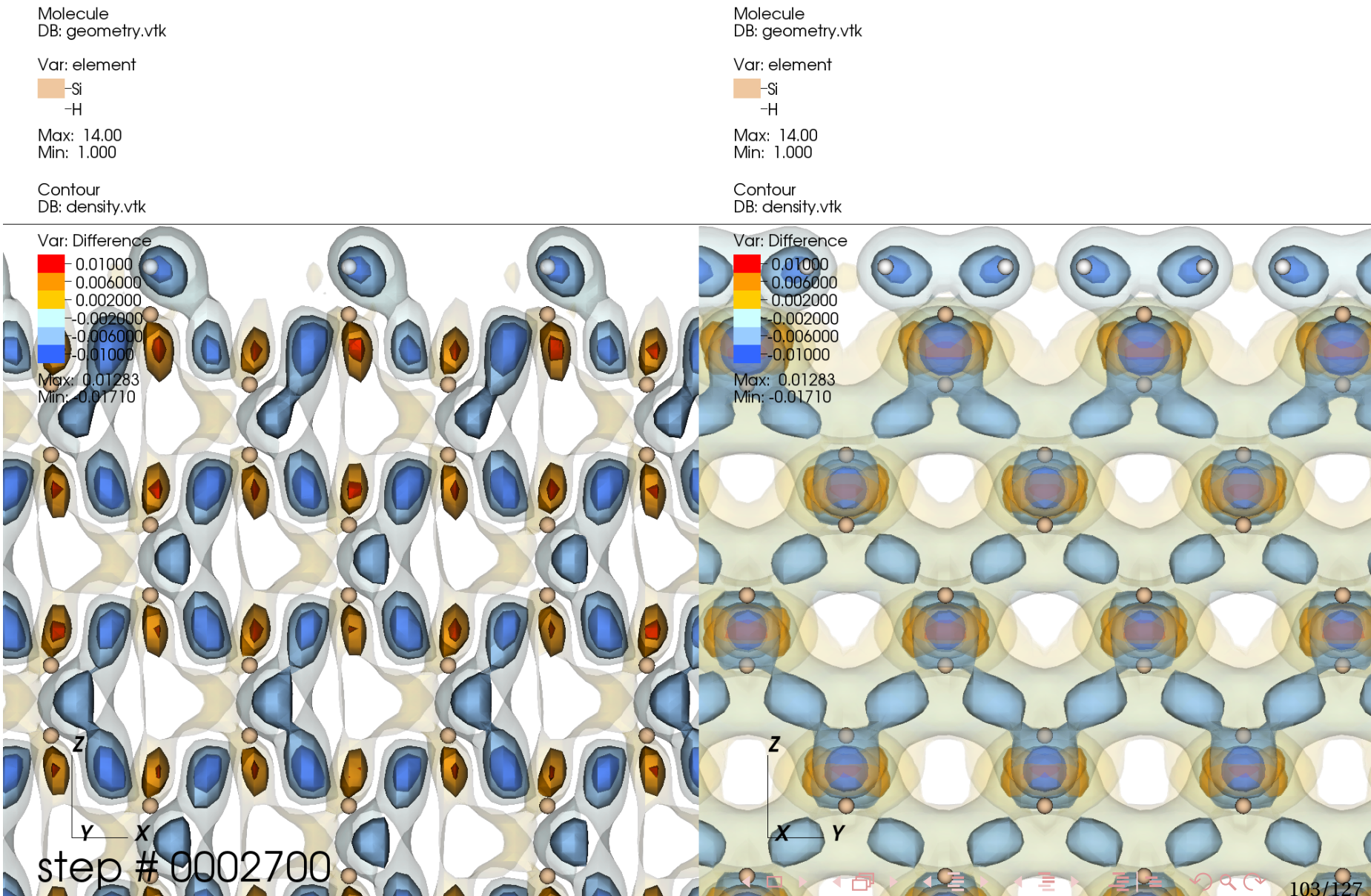
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 15.3$ fs



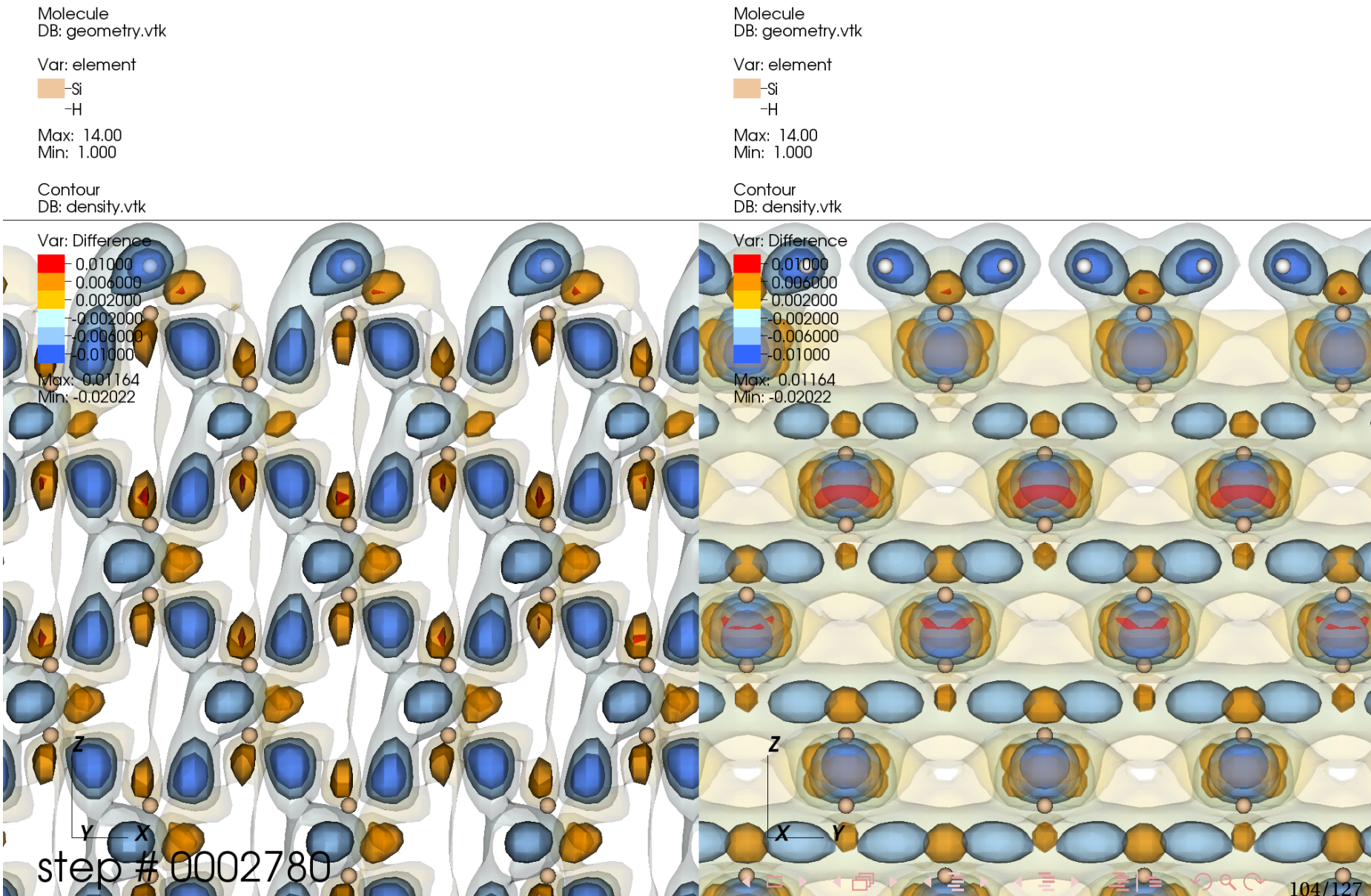
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 16.1$ fs



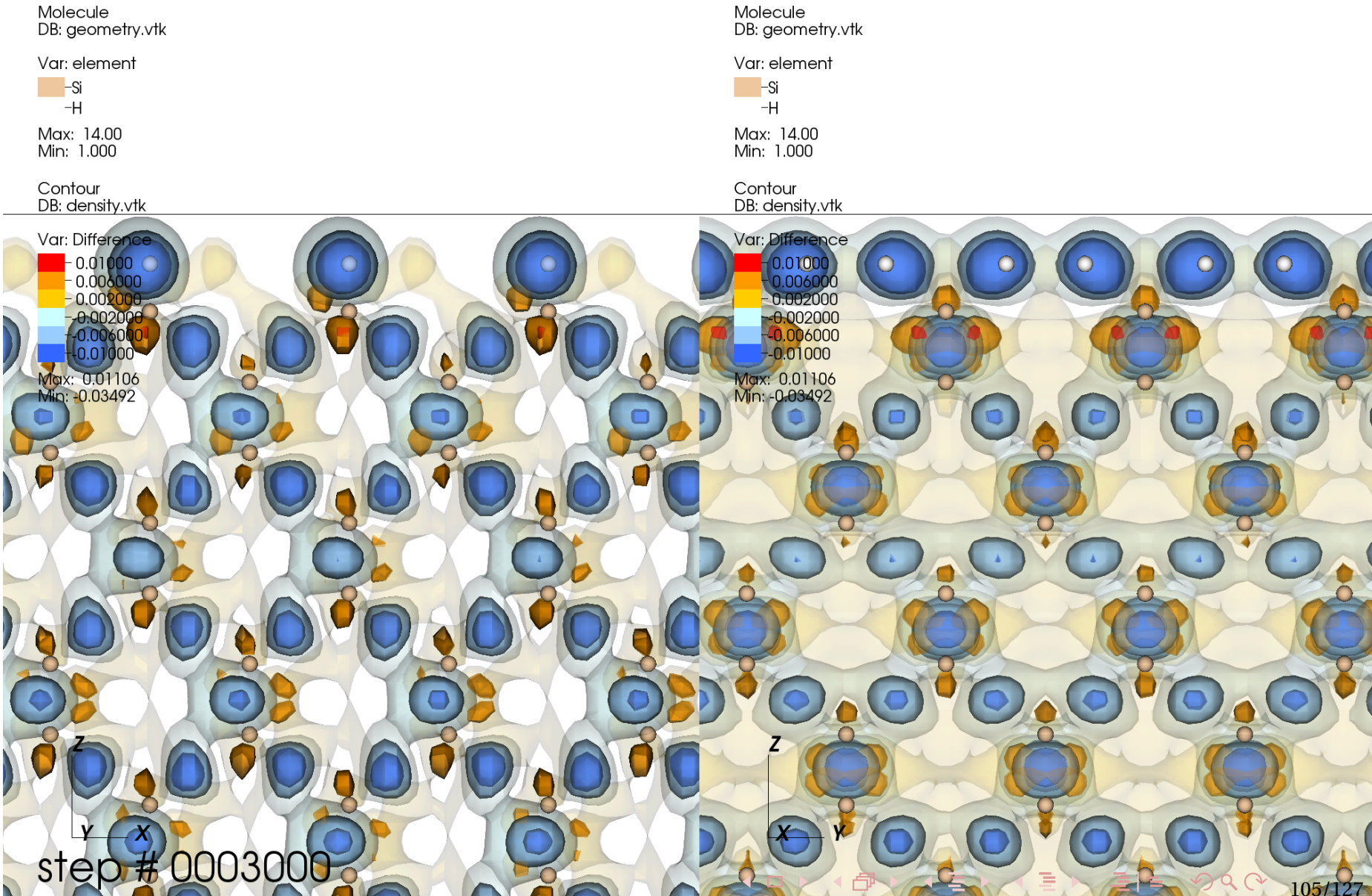
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 16.3$ fs



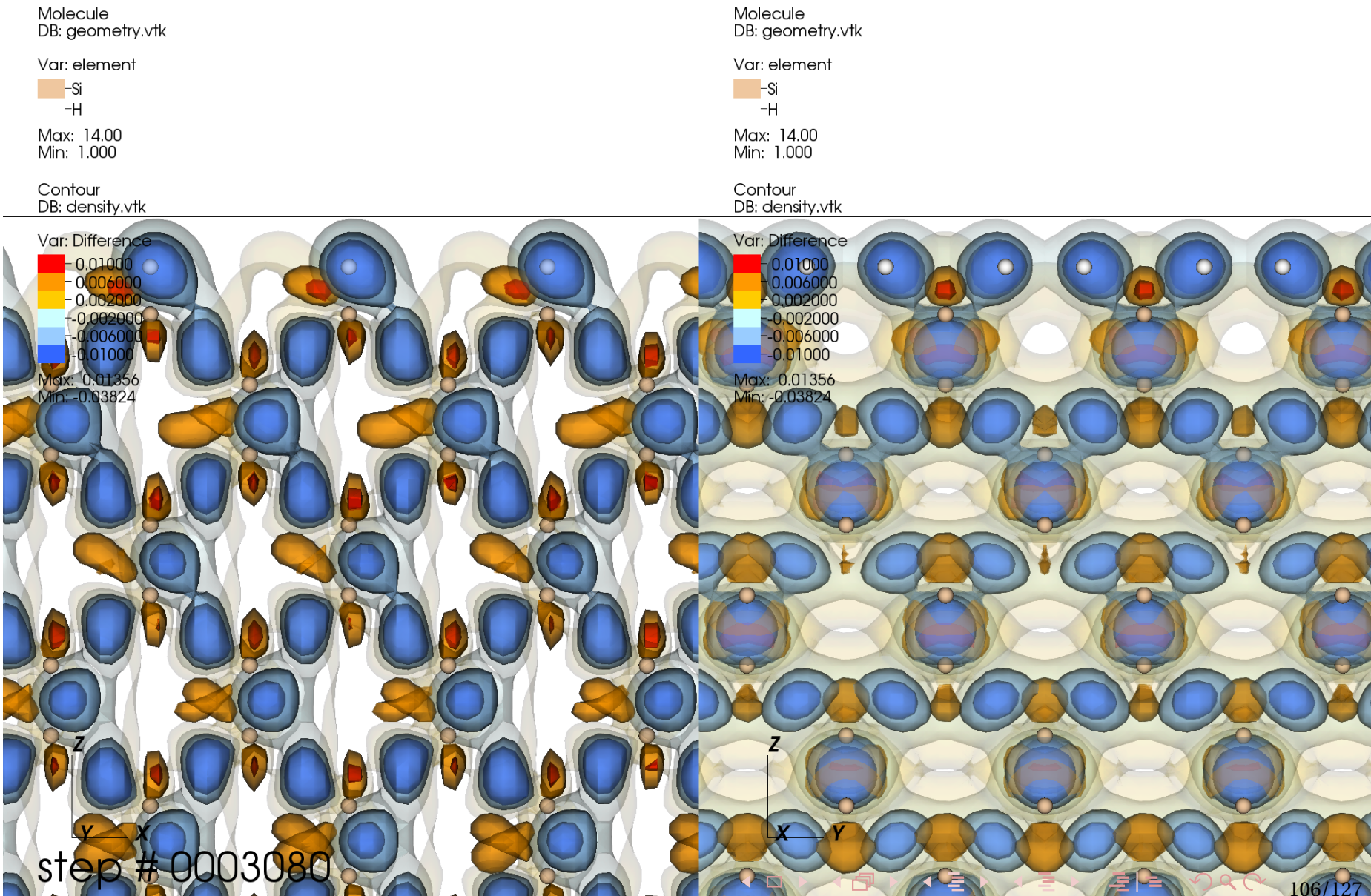
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 16.8$ fs



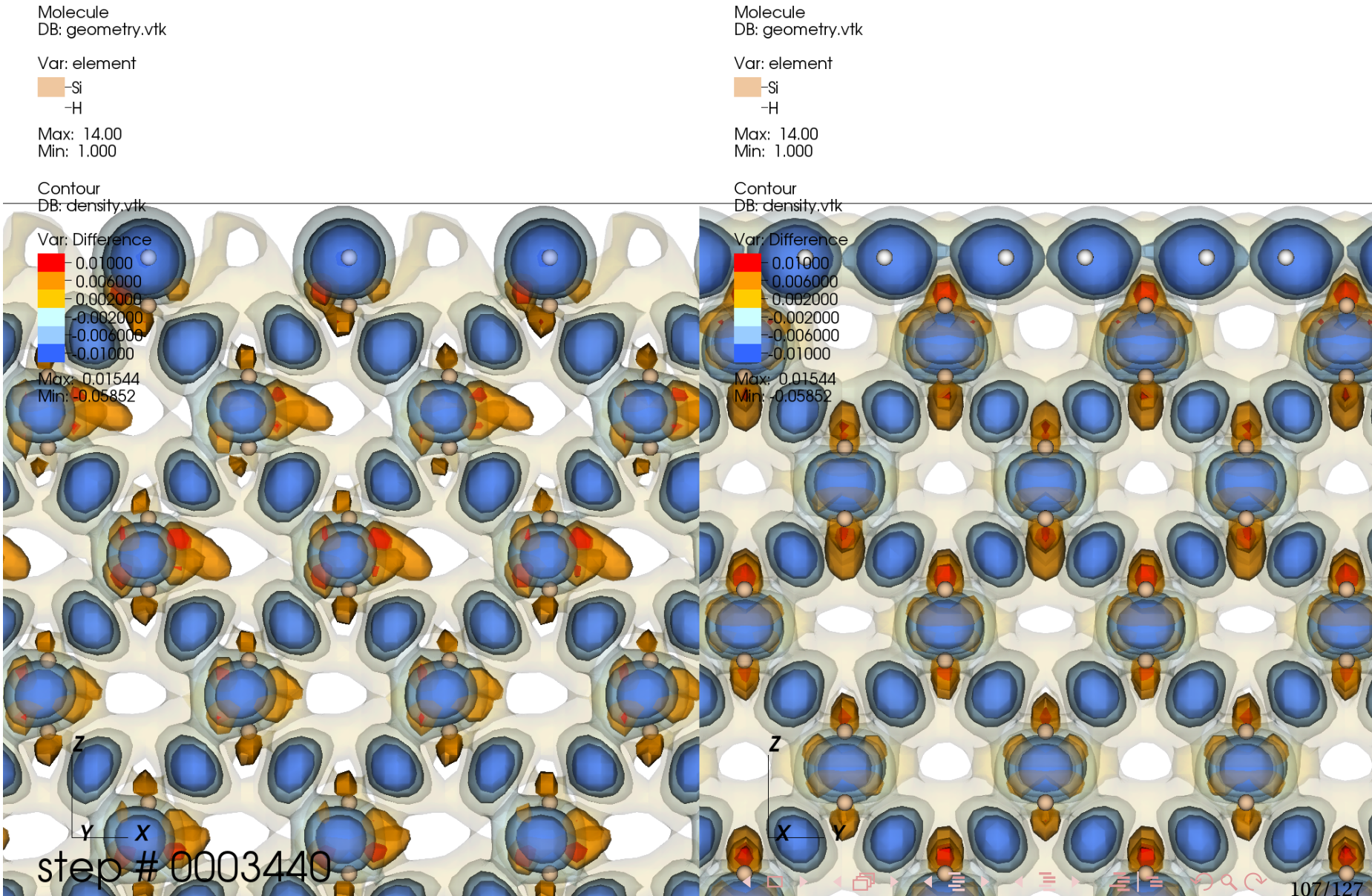
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 18.1$ fs



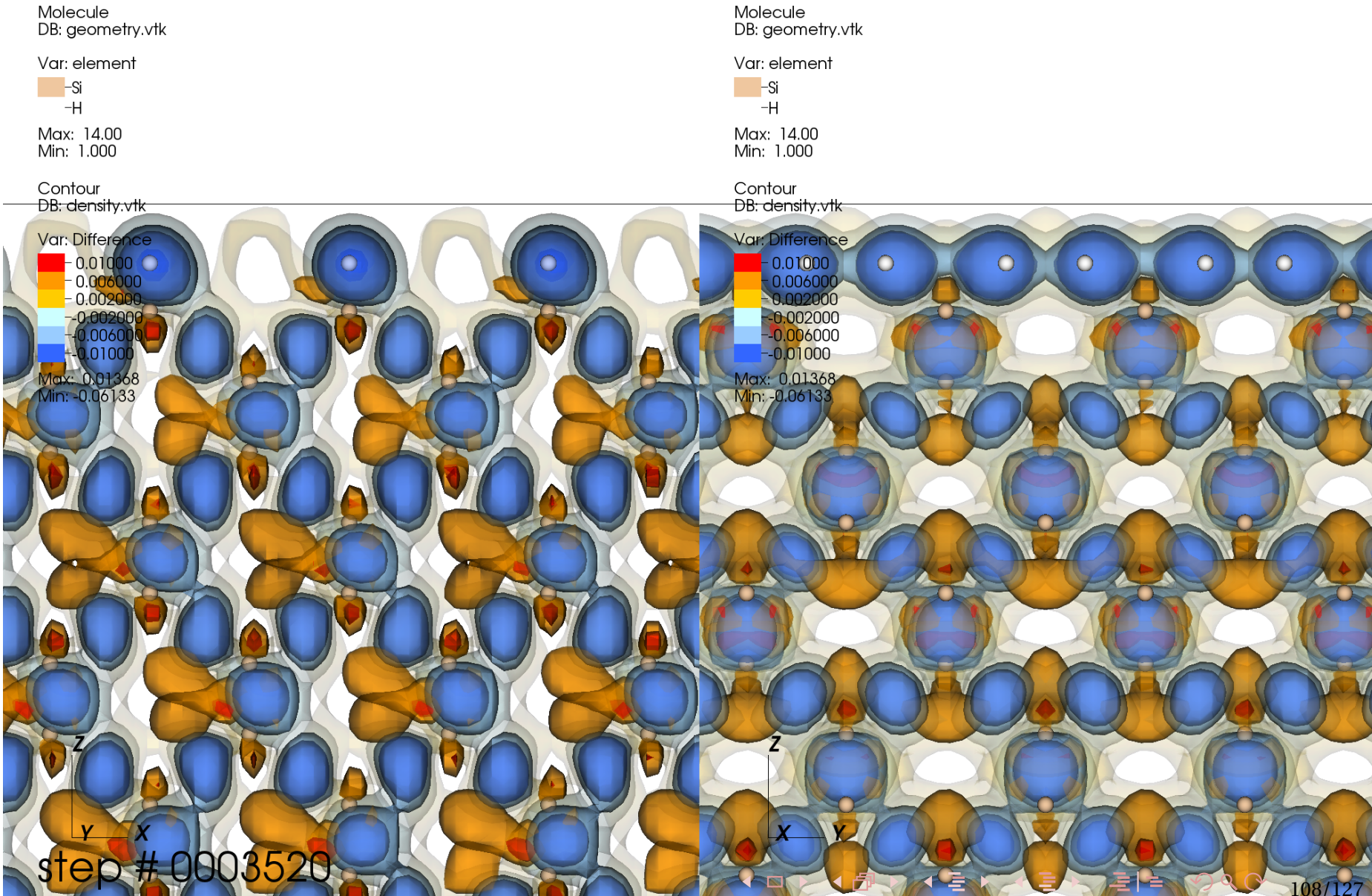
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 18.6$ fs



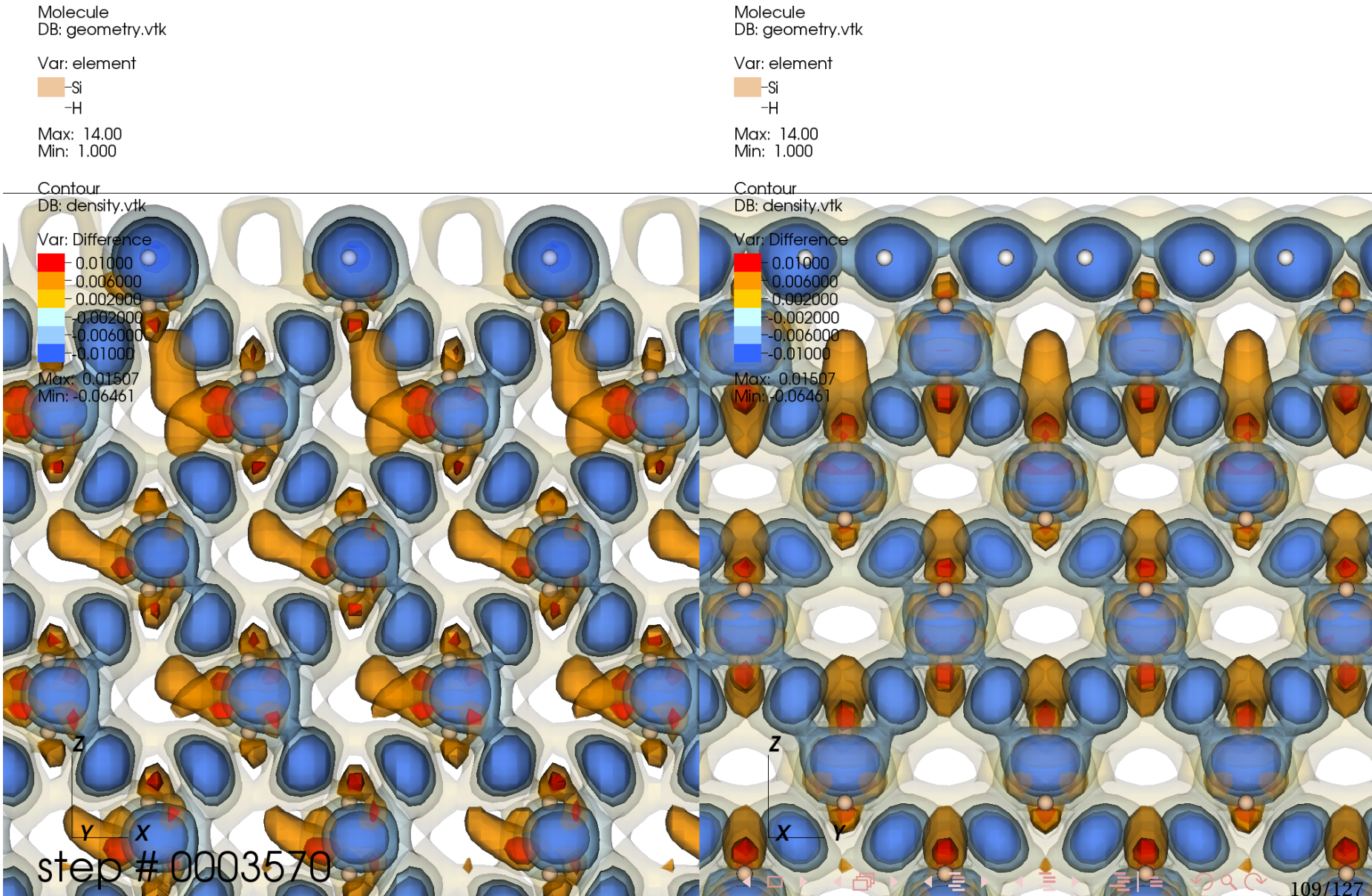
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 20.8$ fs



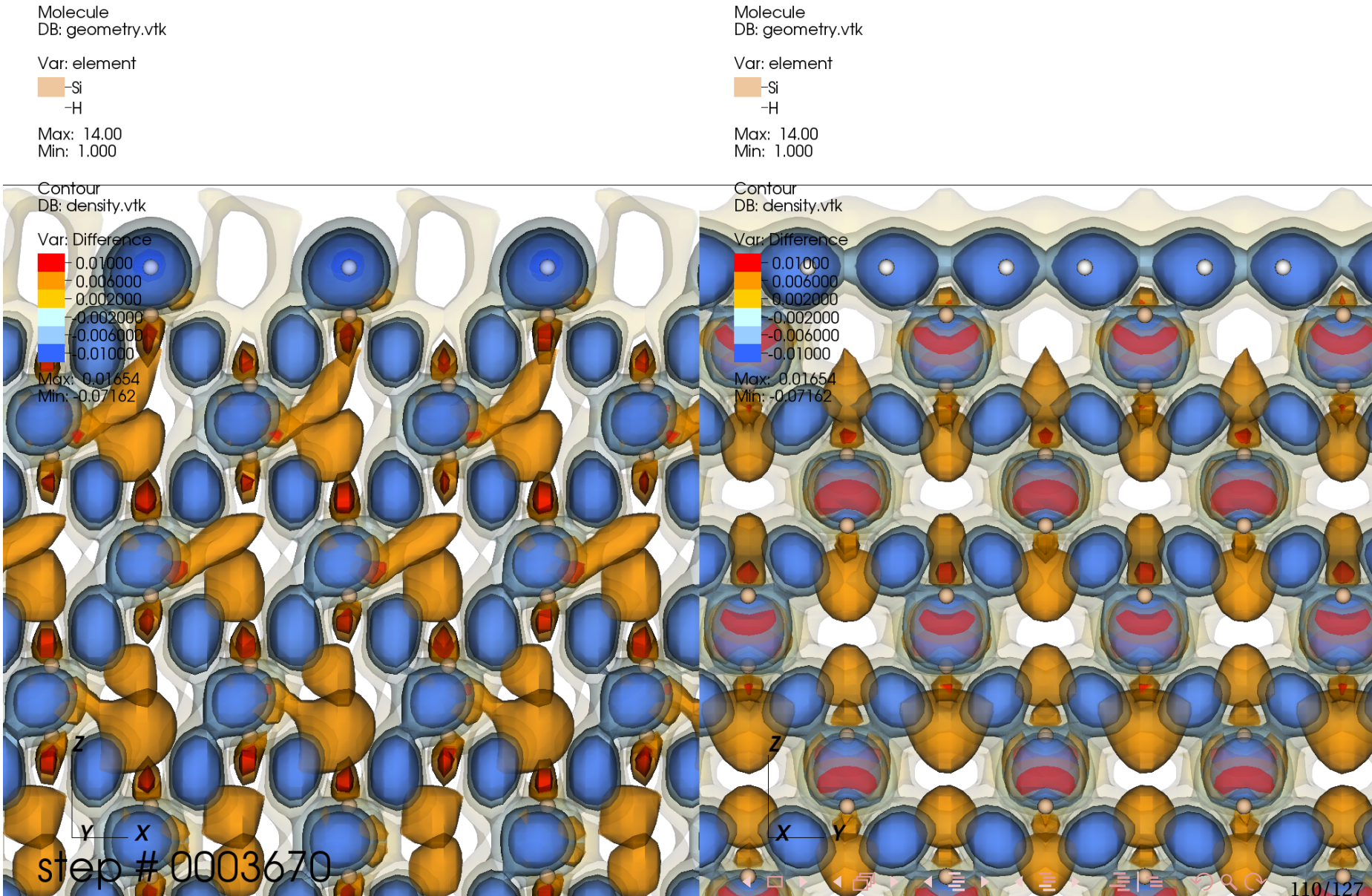
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 21.3$ fs



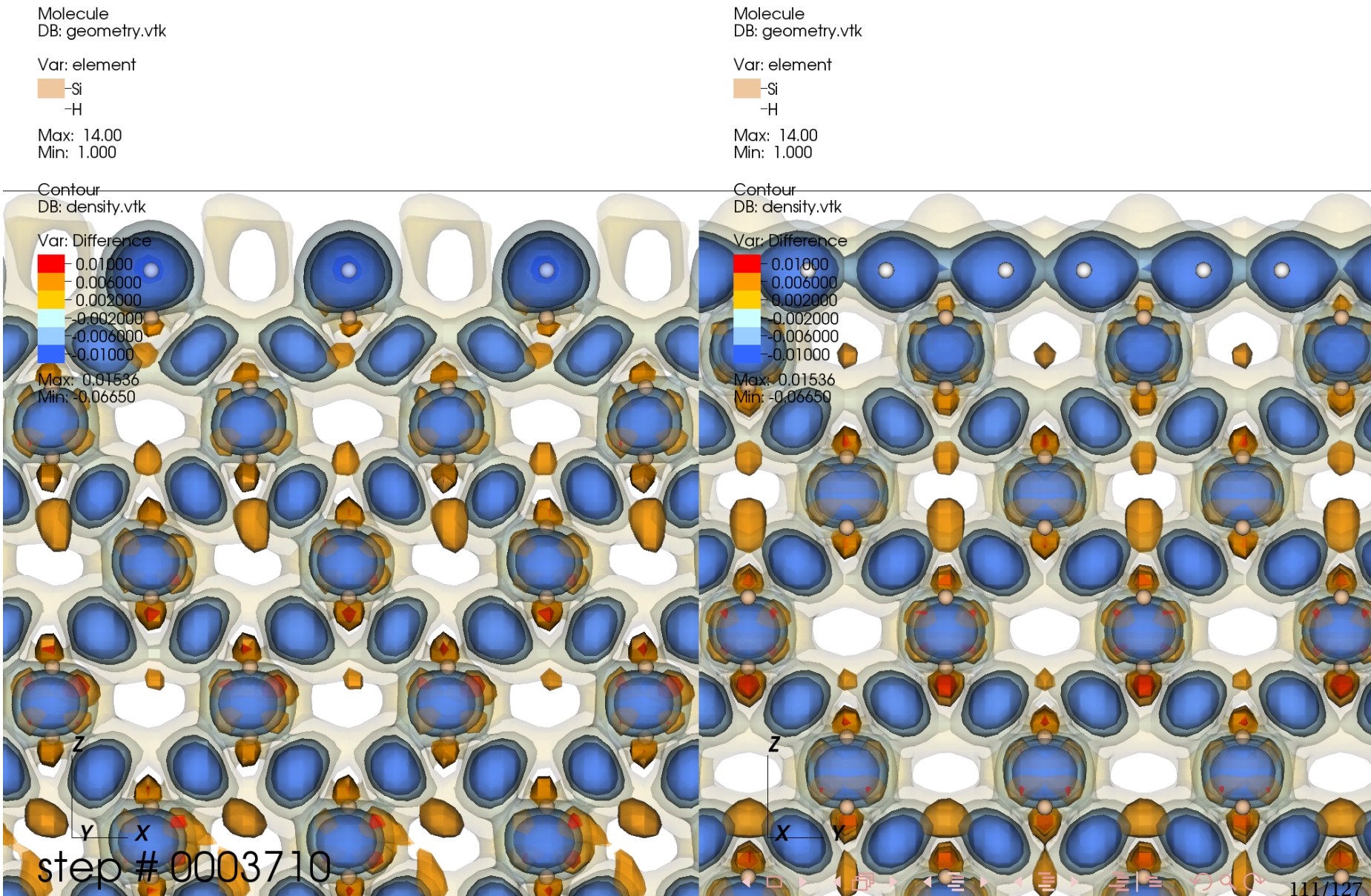
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 21.5$ fs



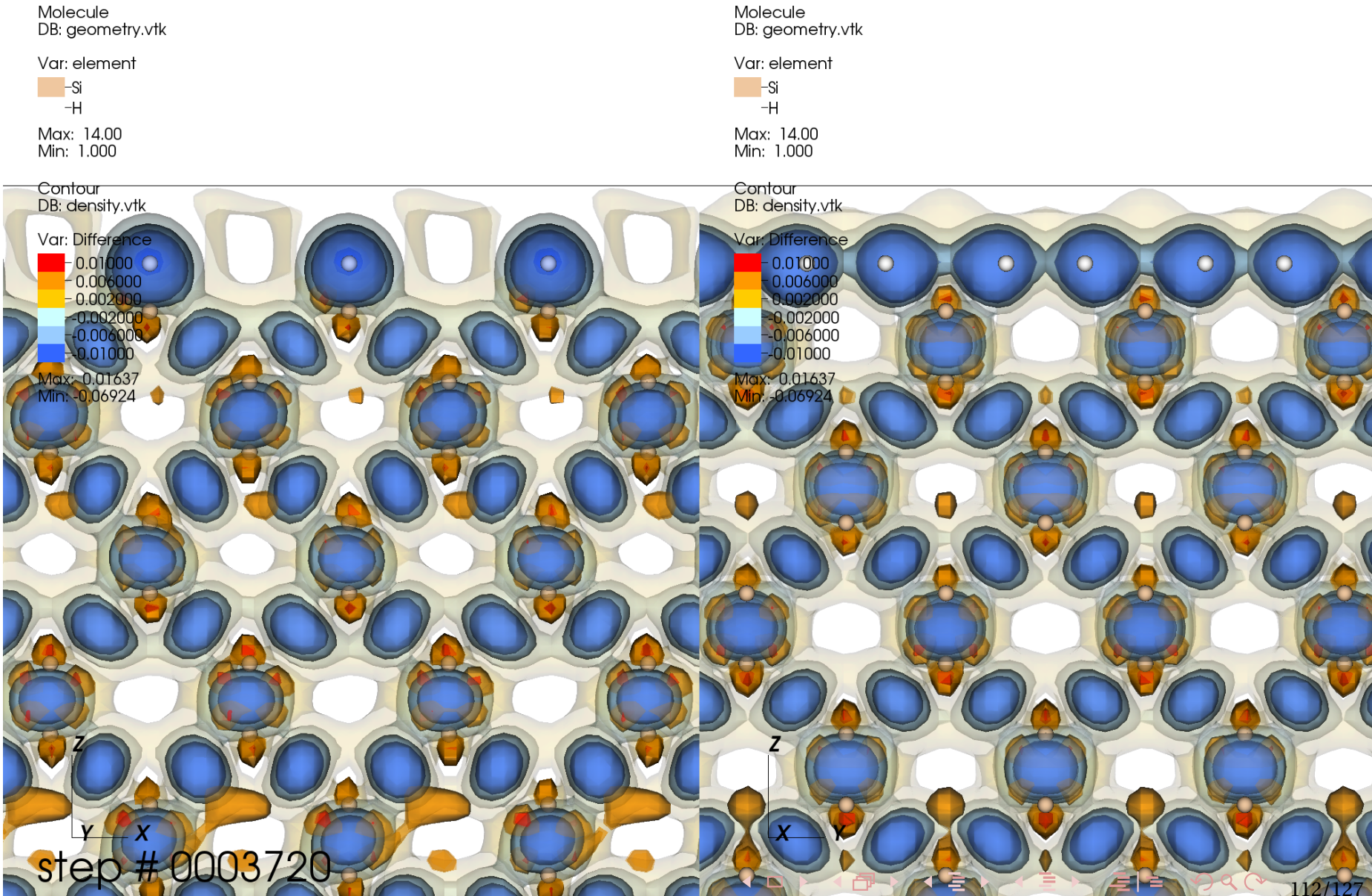
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 22.2$ fs



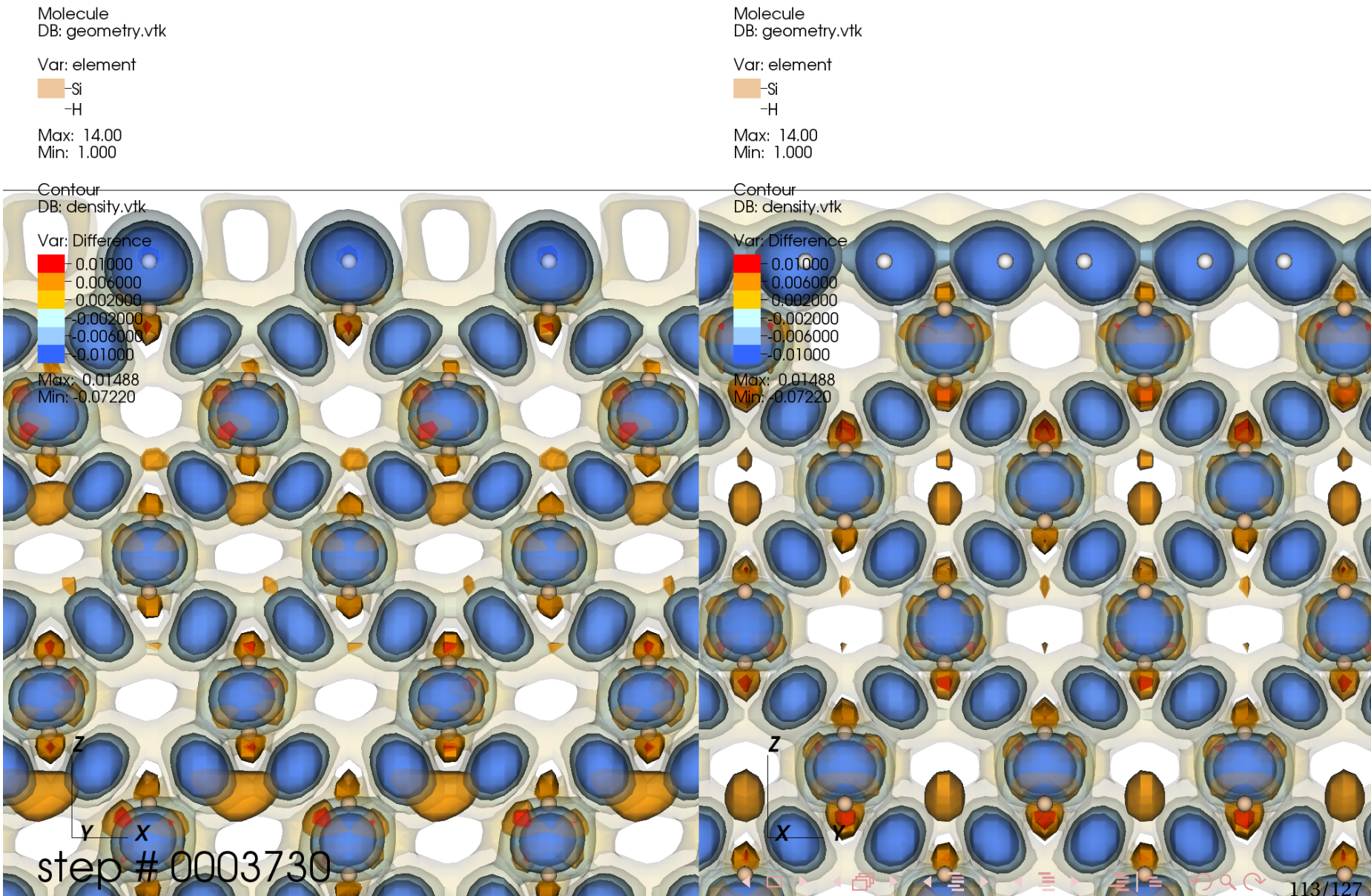
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 22.4$ fs



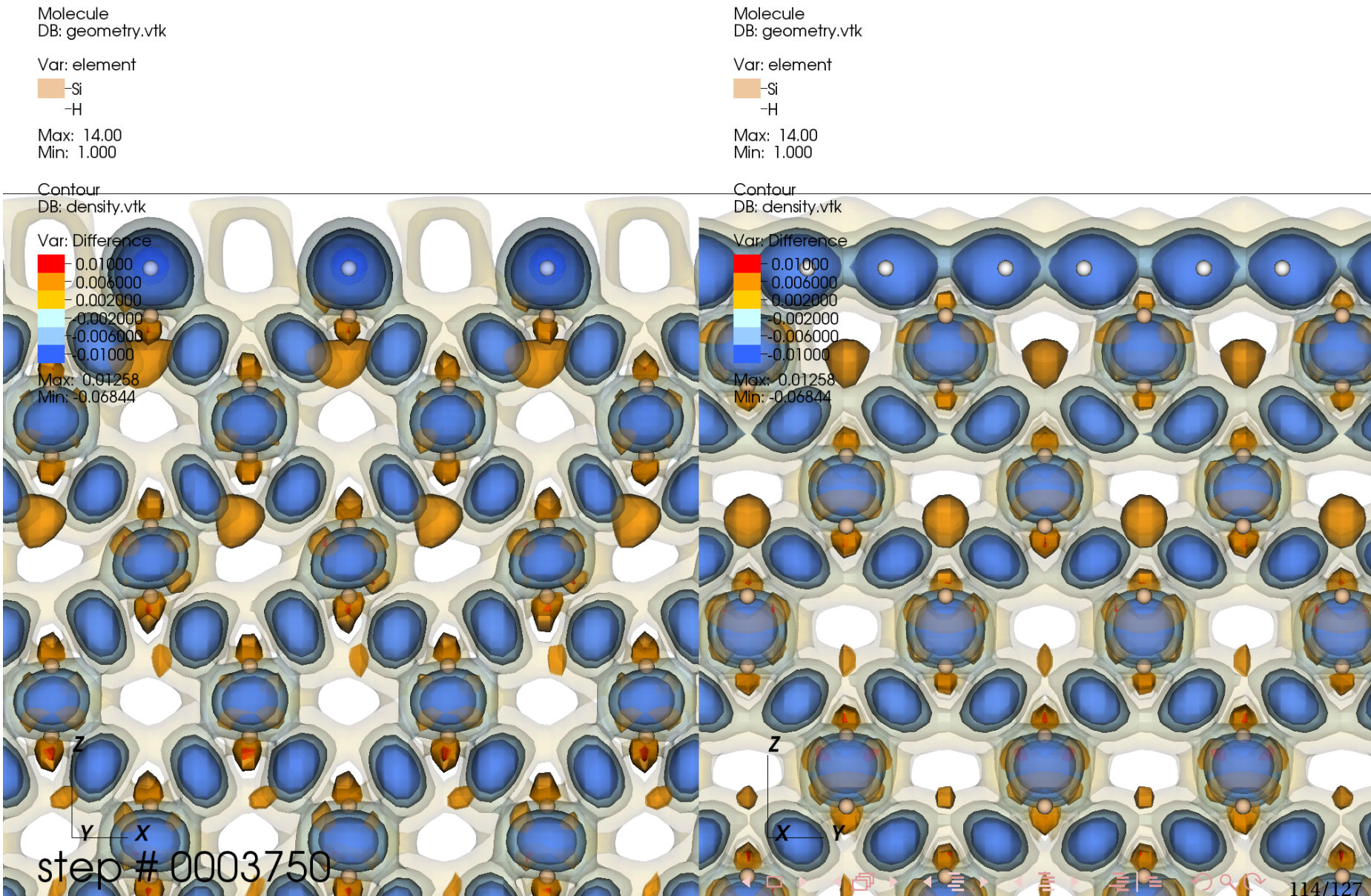
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 22.5$ fs



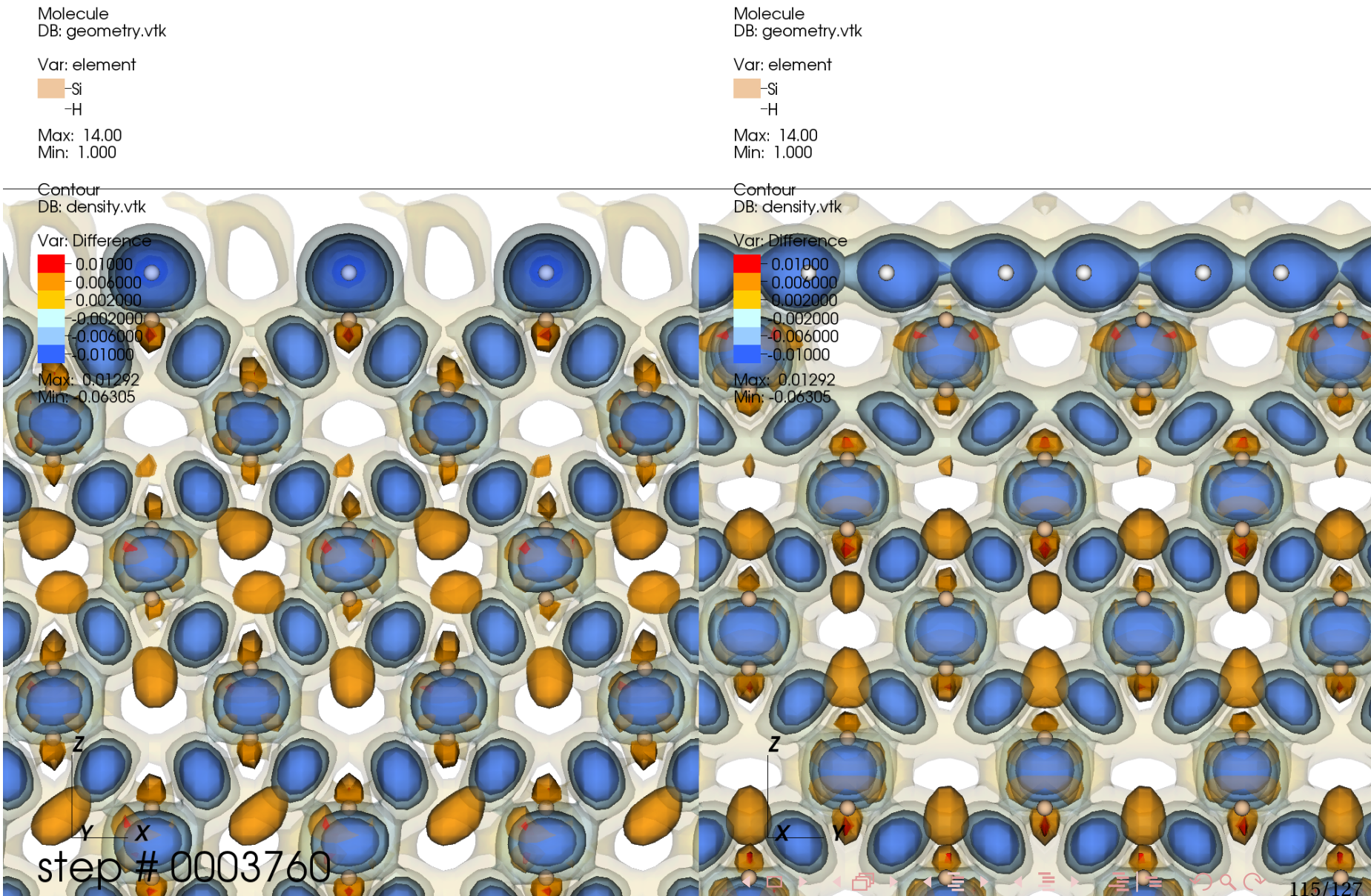
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 22.55$ fs



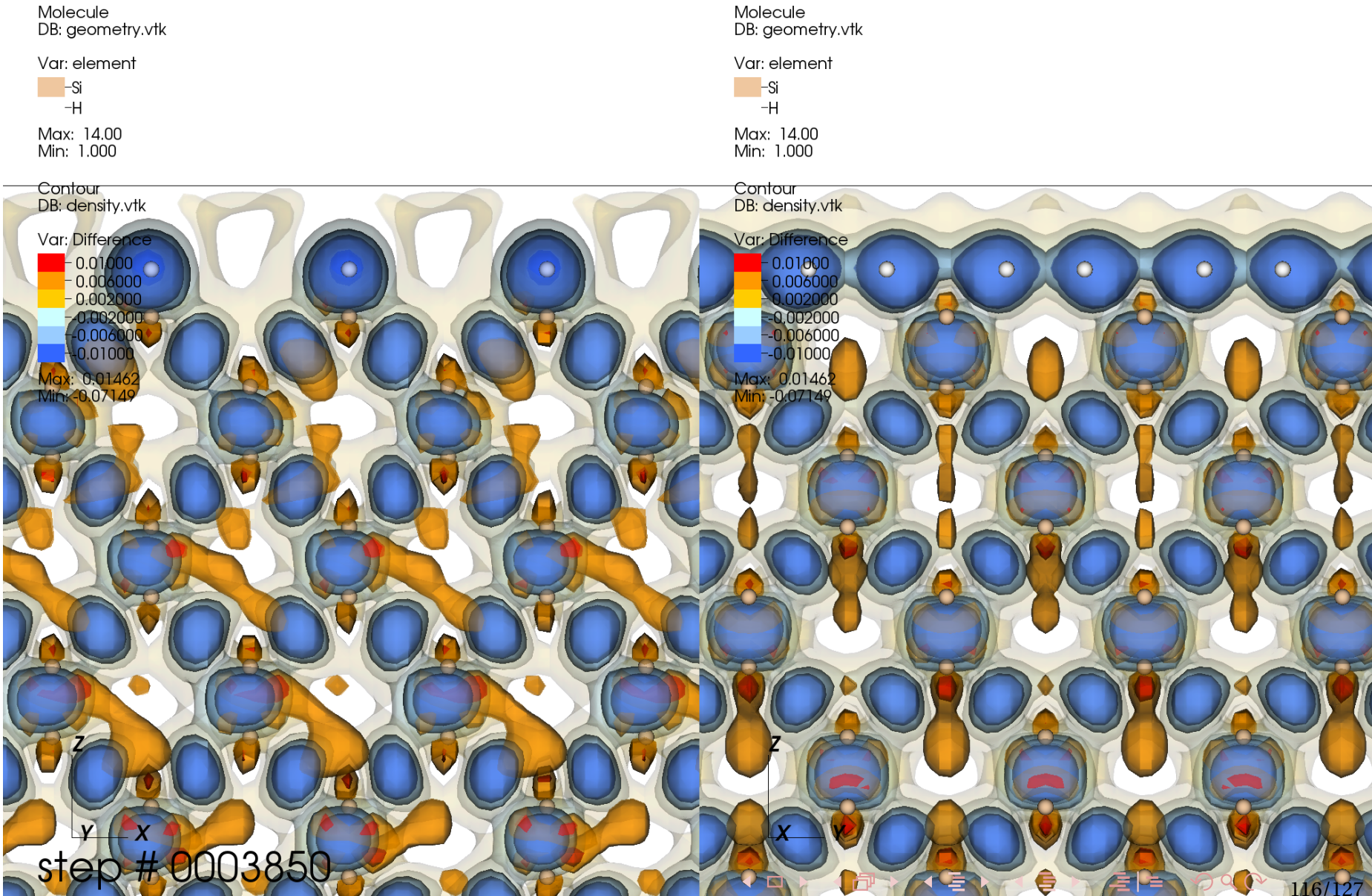
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 22.67$ fs



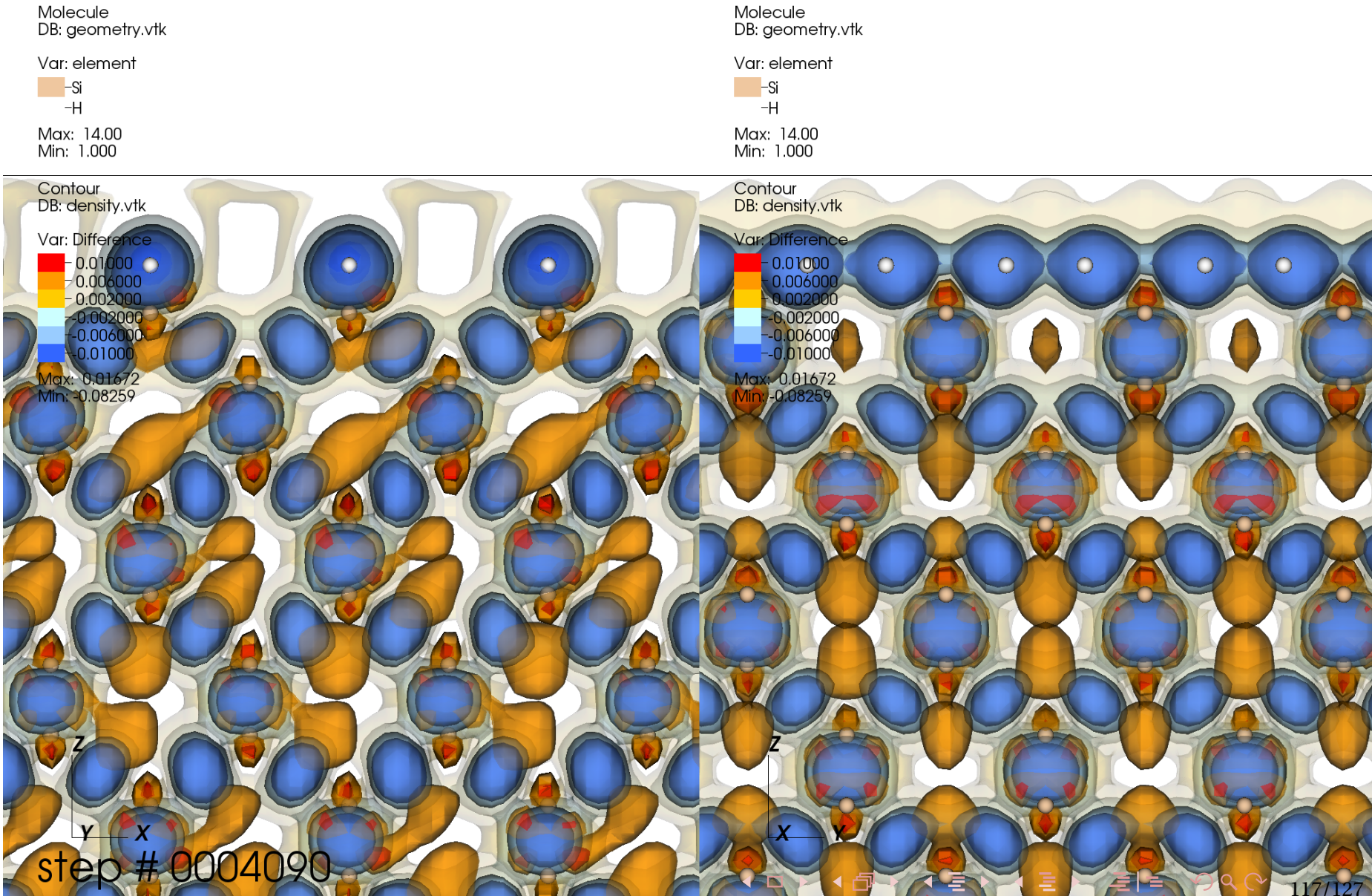
3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 22.73$ fs



3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 23.3$ fs



3d: $\Delta N_e(t) = N_e(t) - N_{GS}$, $t = 24.7$ fs



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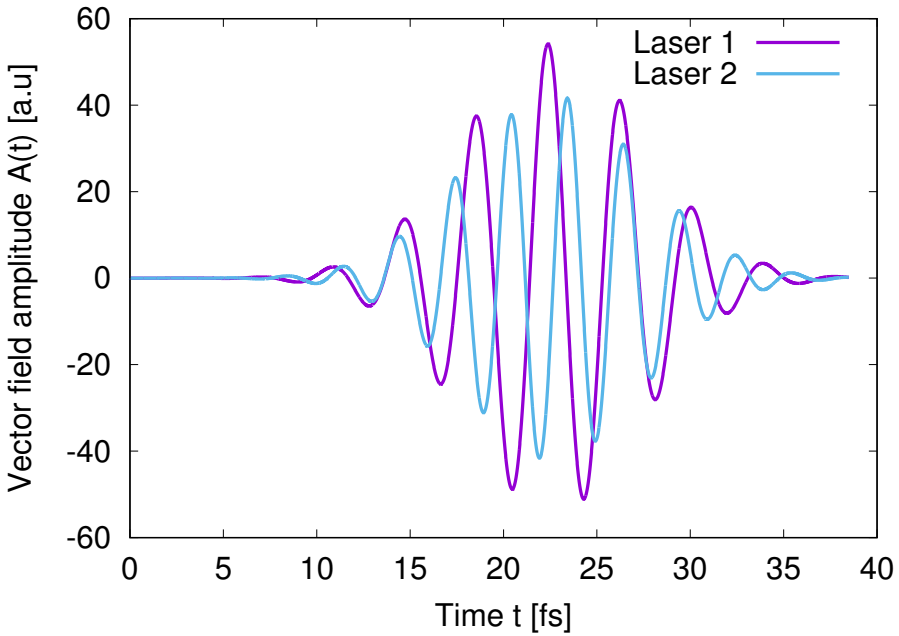
5. Conclusions

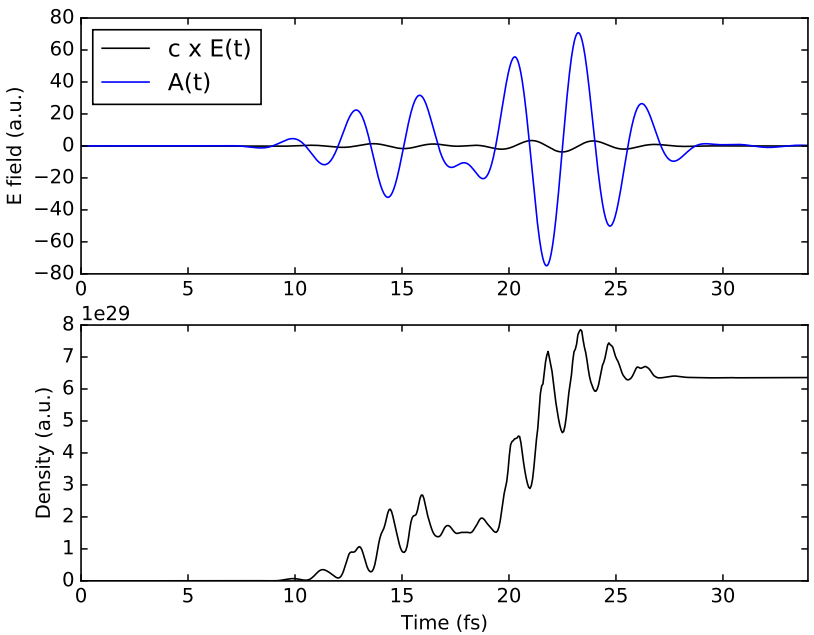
Laser field

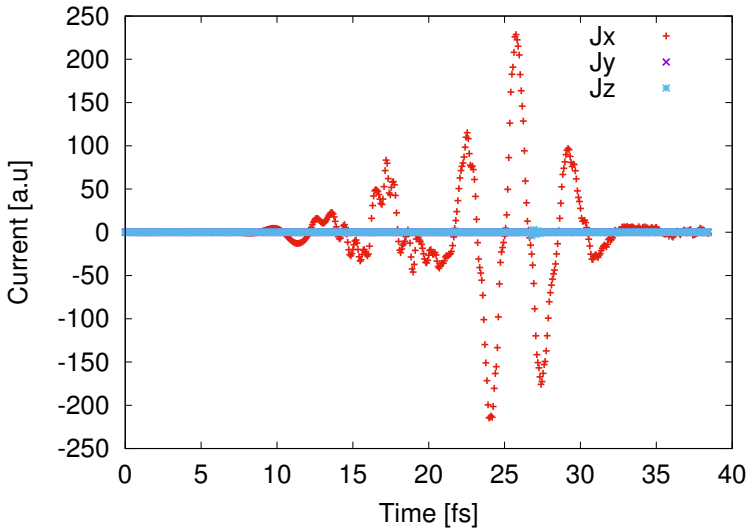
- $\lambda_1 = 1030\,\mathrm{nm}$, $\tau = 10\,\mathrm{fs}$, $E = 9 \times 10^9\,\mathrm{V/m}$, $F \sim 110\,\mathrm{mJ/cm^2}$.
- $\lambda_2 = 800\,\mathrm{nm}$, $\tau = 10\,\mathrm{fs}$, $E = 9 \times 10^9\,\mathrm{V/m}$, $F \sim 110\,\mathrm{mJ/cm^2}$.



Excited electron density

- $\lambda_1 = 1030\,\text{nm}, \tau = 10\,\text{fs}, E = 9 \times 10^9\,\text{V/m}, F \sim 110\,\text{mJ/cm}^2$.
- $\lambda_2 = 800\,\text{nm}, \tau = 10\,\text{fs}, E = 9 \times 10^9\,\text{V/m}, F \sim 110\,\text{mJ/cm}^2$.

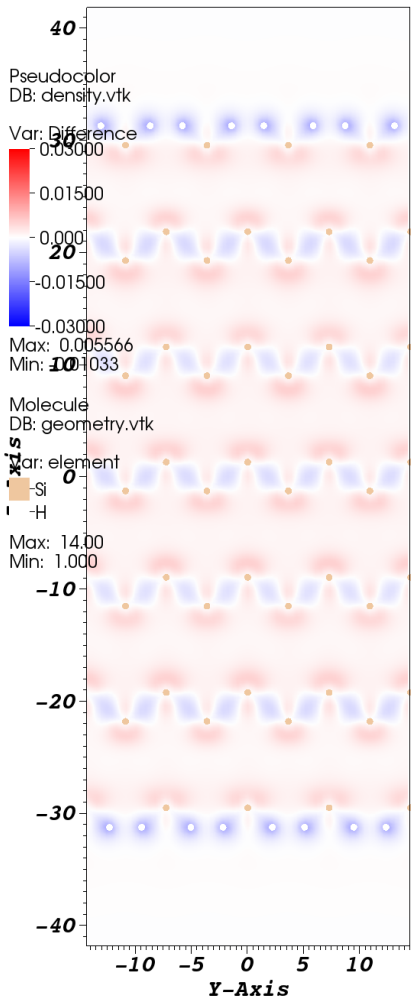


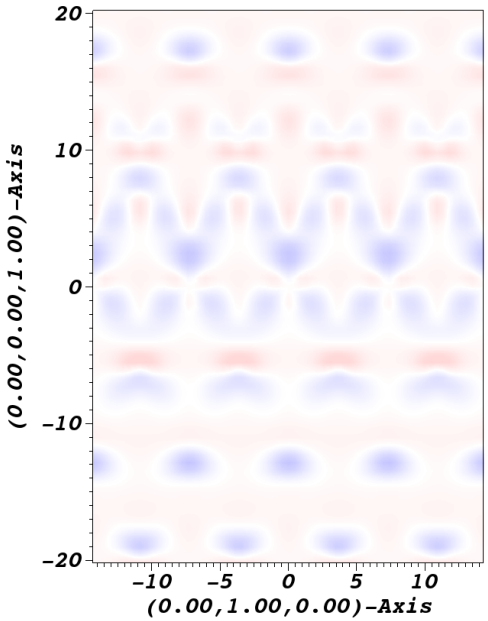


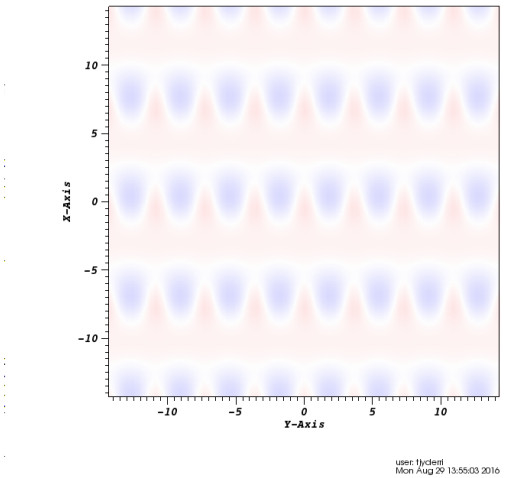
Observations

Generated current is asymmetric.
Carrier excitation follows the vector potential.

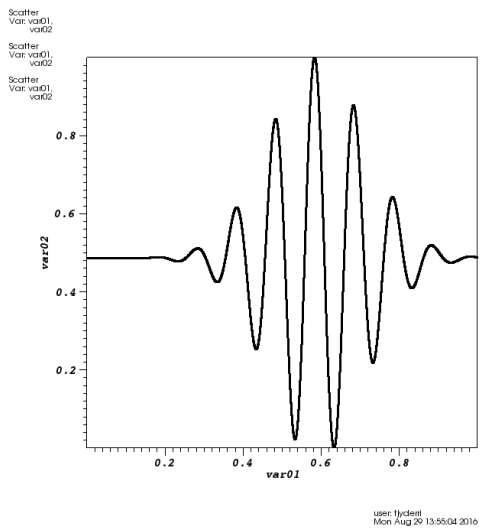
Density distribution







DB: laser.txt



Video

3d: $\Delta N_e = N_e(t) - N_{GS}$

3D video to come. Video

Next steps

For density calculation

1. Correct the band gap using better functionals (hybrid, meta-GGA).
2. Characterize the samples optically to check the functional, check differences between bulk and slab.
3. Perform density calculations for bulk (simpler, easier to interpret).
4. Compare with analytical single-color Keldysh theory (parabolic bands).

Possibly missing processes in continuum description (/!\ We use a very thin film, here).

1. High-harmonic generation: a simple FFT on J(t) will give the spectrum.
2. Photo-emission: count the number of electrons leaving the sample.

In general,

1. Remake the calculations using Ehrenfest Molecular Dynamics (may describe non-thermal melting or Coulomb explosion).
2. See the effect of longer pulses (same intensity).

Conclusions

- TD-DFT is a powerful method allowing to predict and address many effects in laser-matter interaction. It is formally exact, but requires approximations to perform numerical simulations. Exchange correlation potentials (”xc”) describe the local and non-local effects such as e-e, e-h collisions, memory etc.
- Single-color simulation of Si show: delocalization of the electrons during irradiation, non-linear current generation (Drude’s model failure with slabs)
- First bi-color excitation of Si simulations demonstrate the following:
  - Carrier-envelope phase must be controlled to obtain repeatable results.
  - In modification regime, non-linear current generation (high-harmonic generation!) modifies the excitation of conduction band electrons.
  - The distribution of density is substantially modified by bi-color pulses, may directly affecting the atomic bounds.
- Some problems remain to be solved to close this work (better xc, photo-emission mask): more visits in MPSD to come. First: 2nd November for 2 weeks.
- Future data analysis are expected to reveal effects on high-harmonic generation, photo-emission, and non-thermal melting.

Thank you!!

- Many thanks to Heiko Appel and his wife for eating in a Mexican restaurant in Kreuzberg (Berlin) in 2014 and not running away from my table.
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Missing blocks of Octopus

Dream to be implemented in octopus

No	Otobe/Yabana's definition of excited density.
Todo	Electron Localization Function output for solids
Test	Ehrenfest Dynamics for mixed boundary conditions ("MoveIons=Yes")
Running	Time-dependent convergence in vacuum size.

Current manuscripts

`Git:thgit2.rz-berlin.mpg.de:/pub/QuantumLaP.git`

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